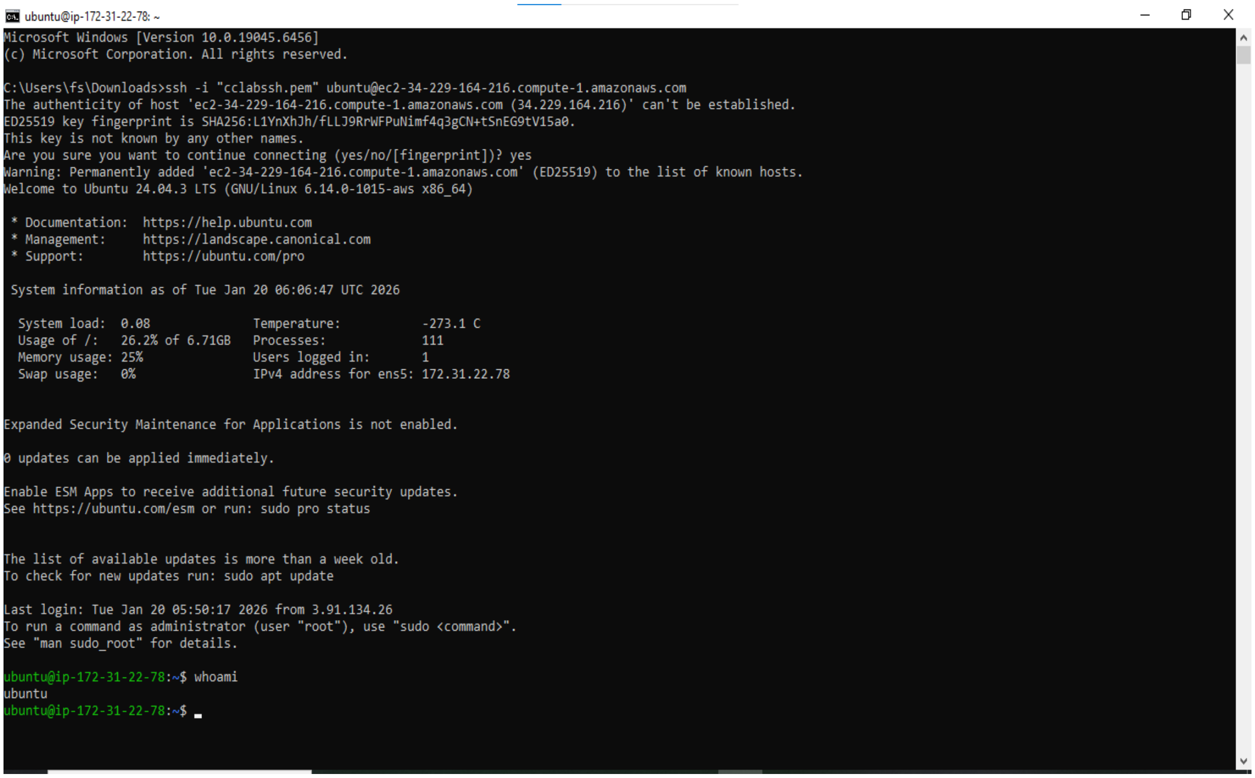
WEEK 1

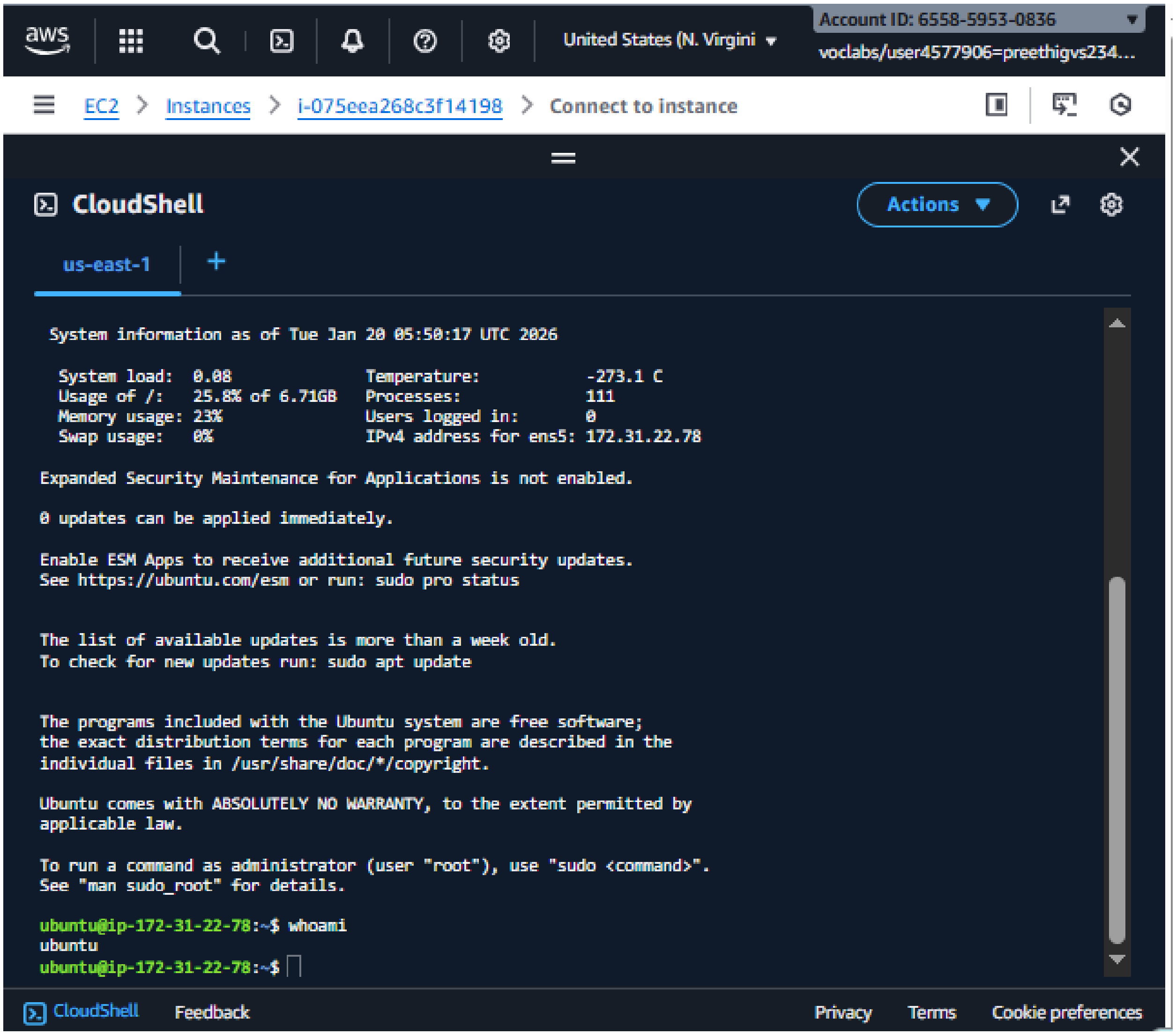
CCLAB TASK 1

1) Connect EC2 Using SSH on local machine

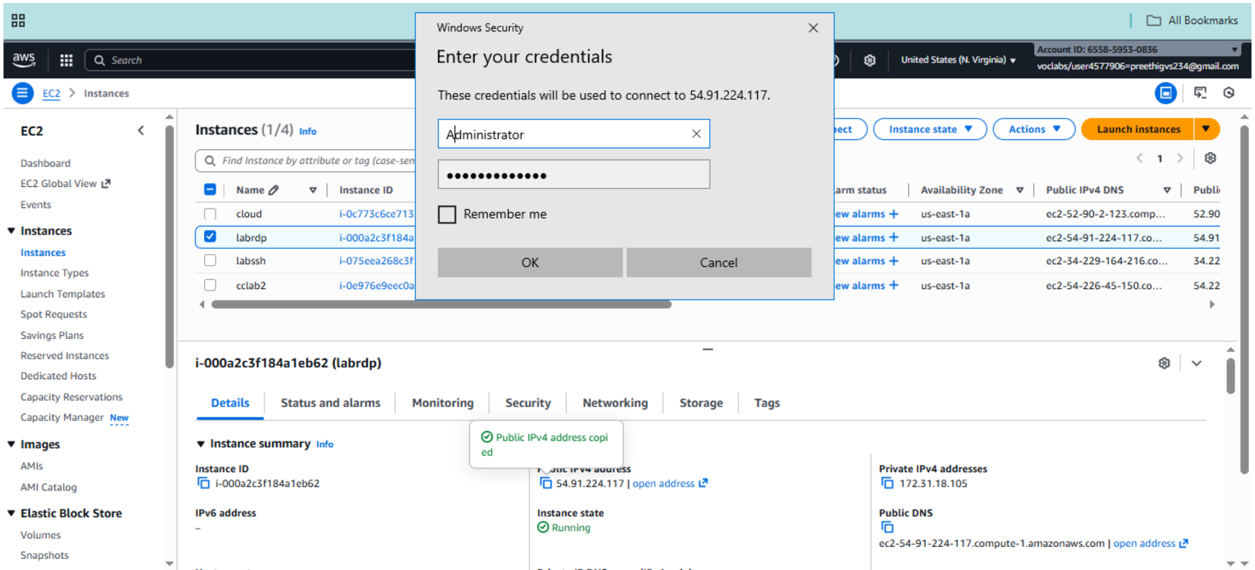


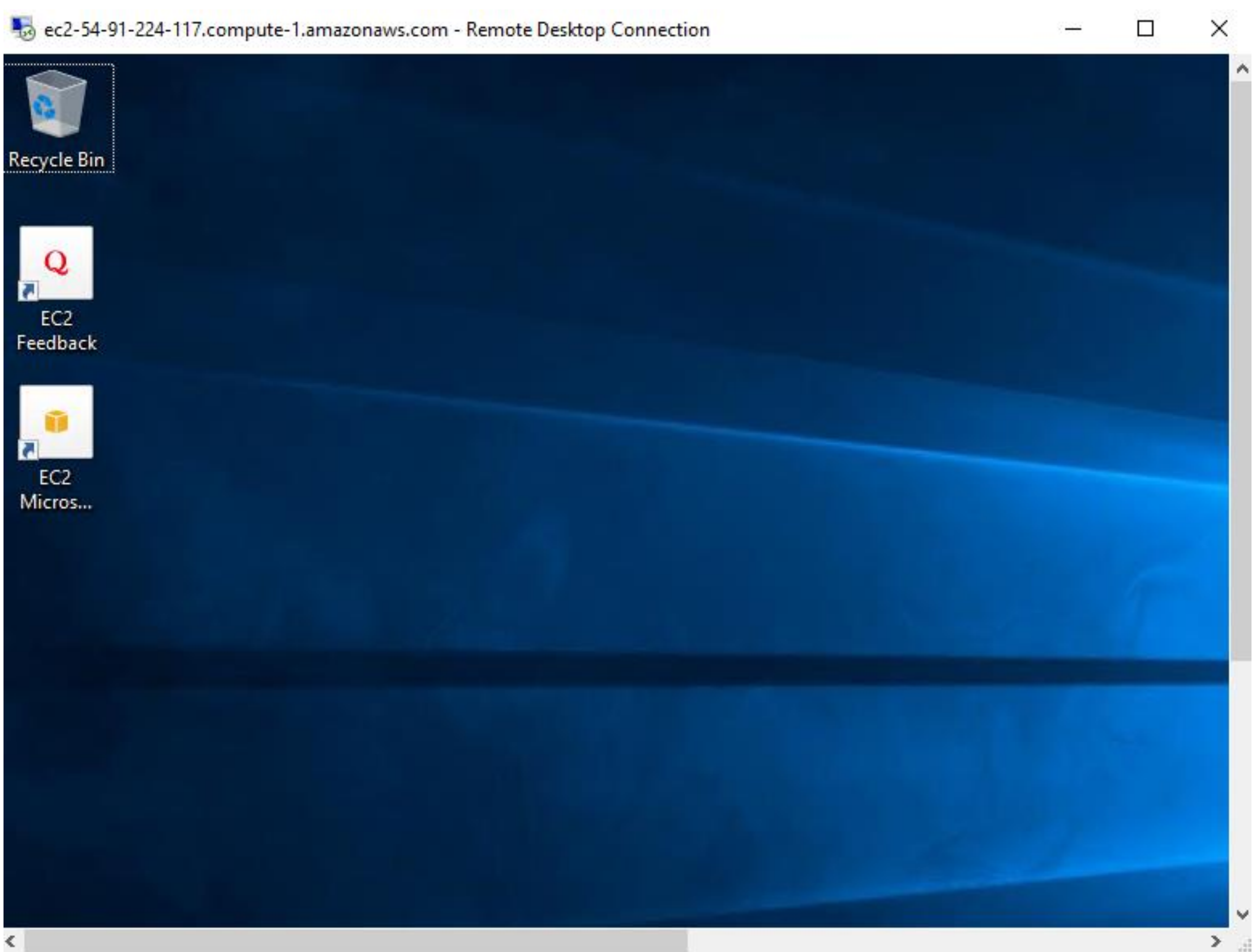
```
ubuntu@ip-172-31-22-78: ~
Microsoft Windows [Version 10.0.19045.6456]
(c) Microsoft Corporation. All rights reserved.

C:\Users\fs\Downloads>ssh -i "cclabssh.pem" ubuntu@ec2-34-229-164-216.compute-1.amazonaws.com
The authenticity of host 'ec2-34-229-164-216.compute-1.amazonaws.com (34.229.164.216)' can't be established.
ED25519 key fingerprint is SHA256:L1YnXhJh/fLLJ9RrWFPuNimf4q3gCN+tSnEG9tV15a0.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-34-229-164-216.compute-1.amazonaws.com' (ED25519) to the list of known hosts.
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1015-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Tue Jan 20 06:06:47 UTC 2026

  System load:  0.08              Temperature:           -273.1 C
  Usage of /:   26.2% of 6.71GB   Processes:             111
  Memory usage: 25%               Users logged in:       1
  Swap usage:   0%                IPv4 address for ens5: 172.31.22.78


Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


The list of available updates is more than a week old.
To check for new updates run: sudo apt update

Last login: Tue Jan 20 05:50:17 2026 from 3.91.134.26
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-22-78:~$ whoami
ubuntu
ubuntu@ip-172-31-22-78:~$
```

2) Connect EC2 using cloudshell

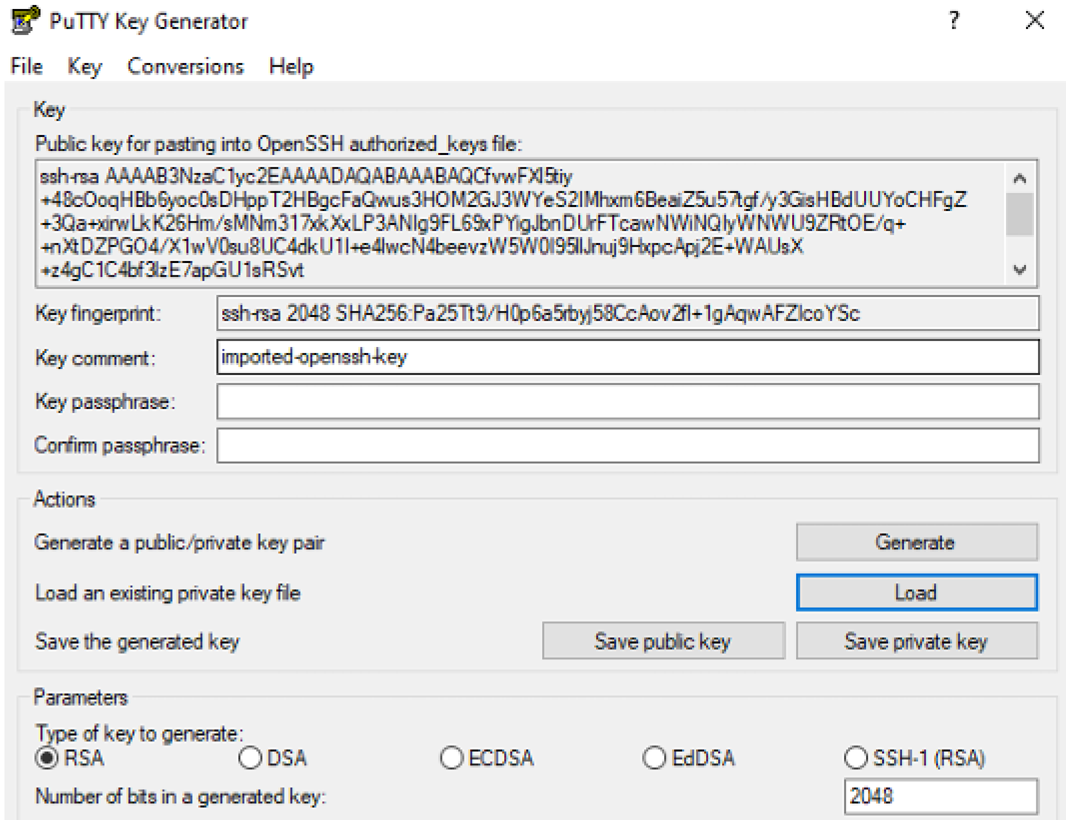


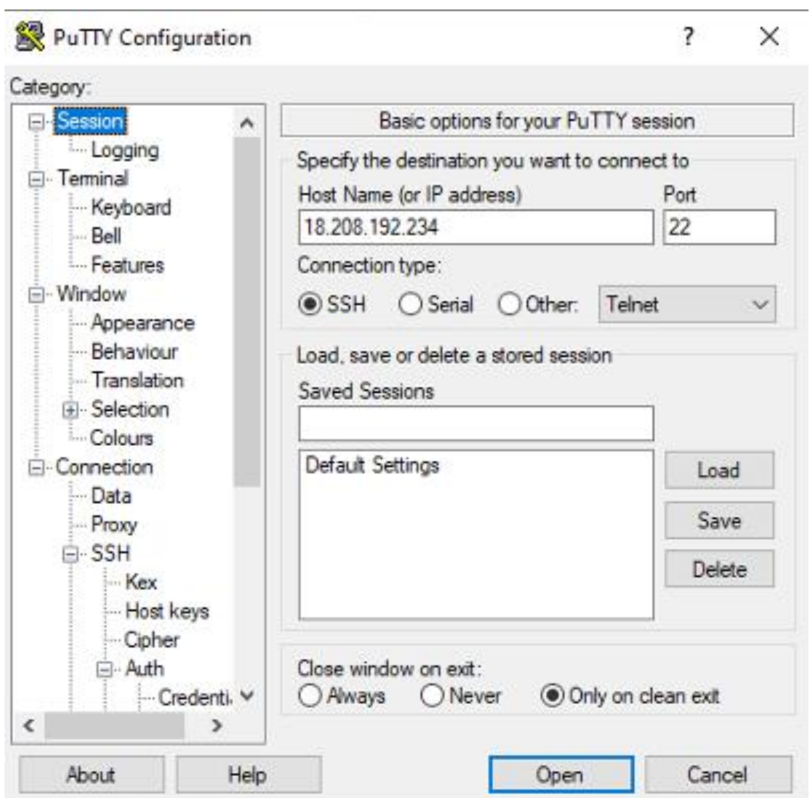
3)EC2 instance connecting using RDP

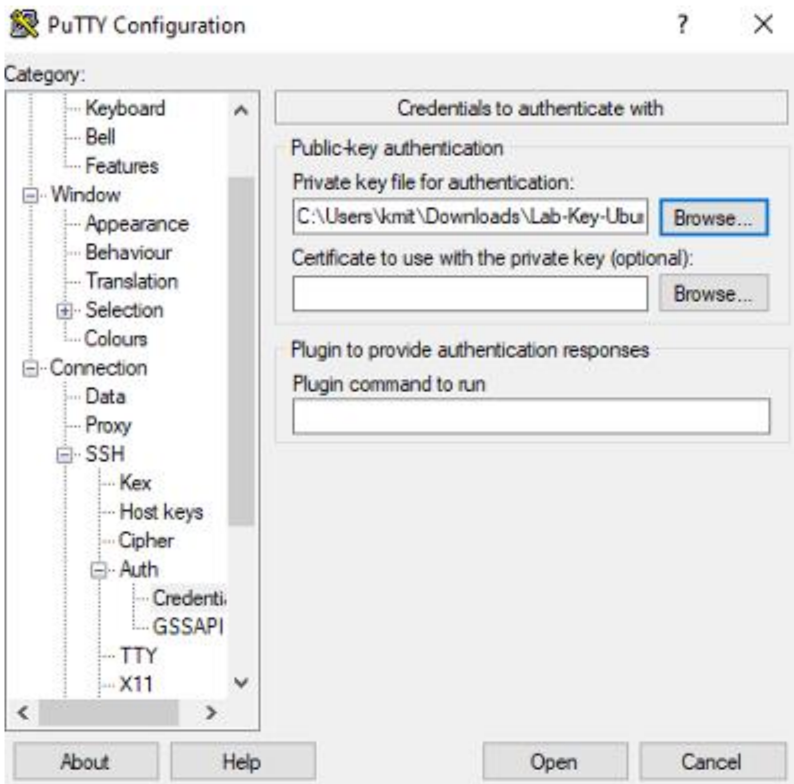


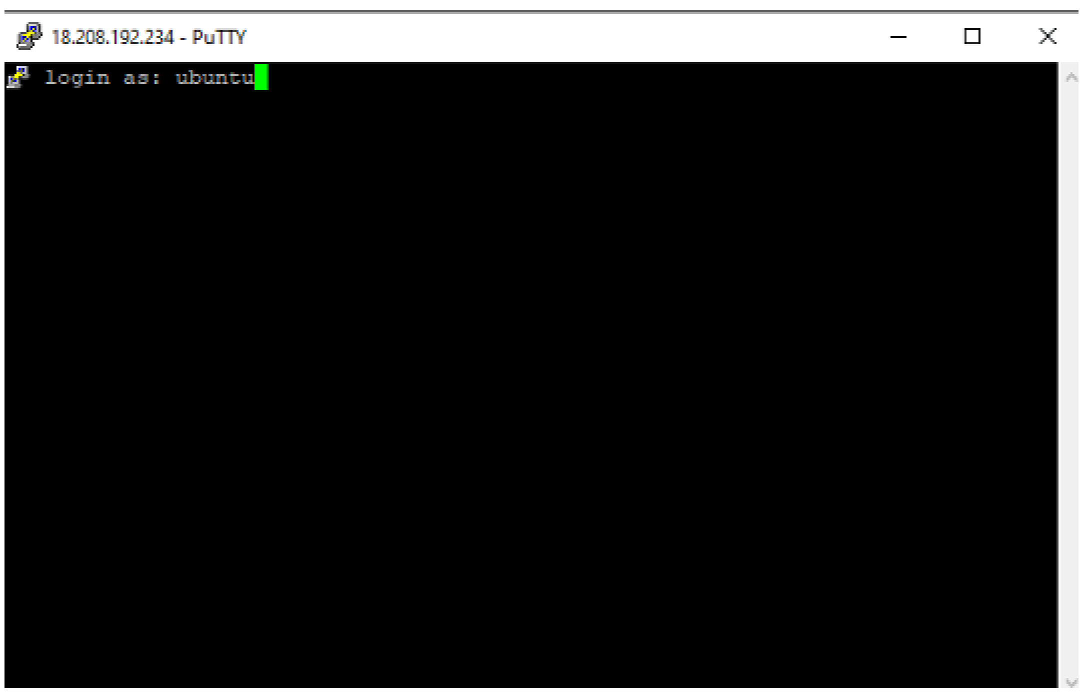


4) EC2 connection using PuTTY (pem to puTTY)

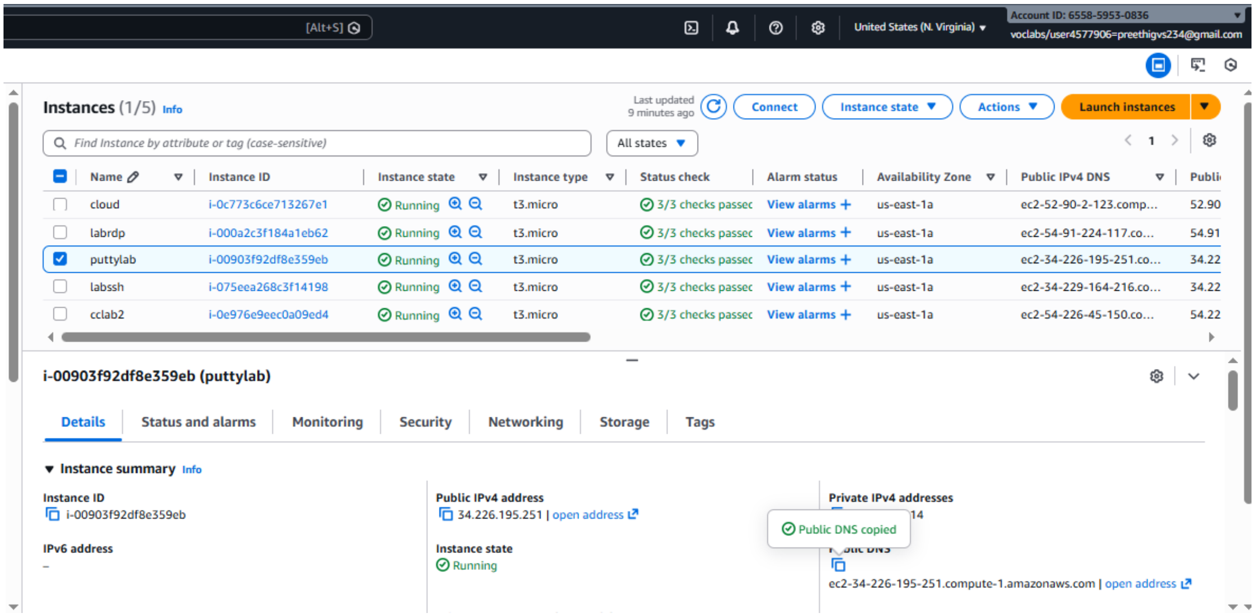


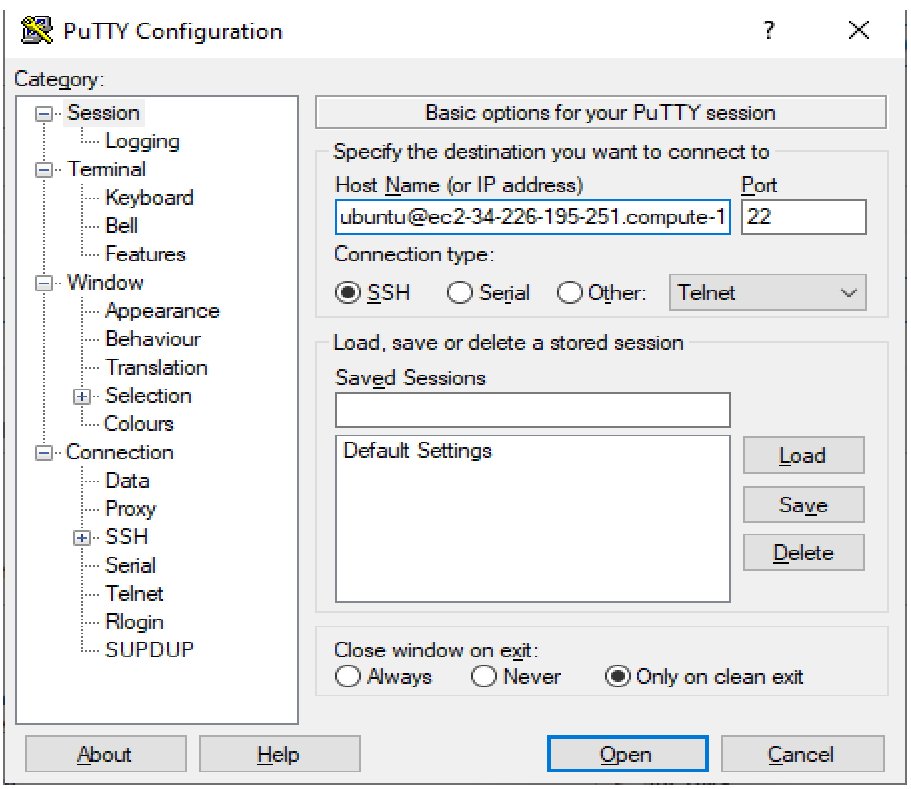



PuTTY Configuration
Category:
Keyboard
Bell
Features
Window
Appearance
Behaviour
Translation
Selection
Colours
Connection
Data
Proxy
SSH
Kex
Host keys
Cipher
Auth
Credenti
GSSAPI
TTY
X11
Credentials to authenticate with
Public-key authentication
Private key file for authentication:
C:\Users\kmit\Downloads\Lab-Key-Ubu
Browse...
Certificate to use with the private key (optional):
Browse...
Plugin to provide authentication responses
Plugin command to run
About
Help
Open
Cancel


18.208.192.234 - PuTTY
login as: ubuntu

5) EC2 using puTTY (.ppk file)





ubuntu@ip-172-31-21-14: ~

```
0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


The list of available updates is more than a week old.
To check for new updates run: sudo apt update


The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-21-14:~$ whoami
ubuntu
ubuntu@ip-172-31-21-14:~$
```

Week-2

1. **a) Create and attach an EBS volume to an EC2 instance, and**
   b) scale the instance by increasing CPU and RAM by changing the instance type.

2.Attach and permanently mount an EBS volume to a Linux EC2 instance to ensure data persistence across reboots.

3. Create a snapshot of the attached EBS volume and use it to create and attach a new volume to an EC2 instance in another AWS region.

---

1. **a) Create and attach an EBS volume to an EC2 instance, and**
   b) scale the instance by increasing CPU and RAM by changing the instance type.

a) Creating and attaching an EBS volume to an EC2 instance

Step 1: Create a New EBS Volume

Open the EC2 Dashboard.
From the left-hand menu, select Volumes under Elastic Block Store (EBS).
Click Create Volume.
Choose:
Volume type (gp3/gp2)
Size of the volume
Availability Zone (must be the same as the EC2 instance)
Click Create Volume.

Step 2: Attach the EBS Volume to the Existing EC2 Instance

In the Volumes section, select the newly created volume.
Click Actions → Attach Volume.
Select the target EC2 instance from the list.
Specify the device name (example: /dev/xvdf).
Click Attach.

Step 3: Verify the Volume on the EC2 Instance

Connect to the EC2 instance using SSH.
Check the attached disks:
lsblk
Identify the new volume (e.g., /dev/xvdf).

Note: Ensure that the EBS volume and the EC2 instance belong to the **same Availability Zone**, as cross-AZ attachment is not supported.

b)Procedure: increasing/decreasing of CPU and RAM

Step 1. - Navigate to the EC2 dashboard.
Step 2. - Stop the EC2 instance associated with the EBS volume.
Step 3. - Click "Actions" and then—instance settings—change the instance type"Modify Volume".
Step 4. -Increase/ decrease the size or change the volume type to a higher performance specification.
Step 5. - Click "Modify" to apply the changes.
Step 6: Now the start the machine you see with increase /decrease of size and performance

---

2.Attach and permanently mount an EBS volume to a Linux EC2 instance to ensure data persistence across reboots.

Objective

Attach an additional EBS volume (example: 20 GB) to a Linux EC2 instance and mount it using the **system-default device names** that appear on modern EC2 (Nitro) instances.

Note: Although you select a device name like /dev/xvdf in the AWS Console, the OS typically shows the disk as **NVMe** devices such as /dev/nvme1n1 and partitions as /dev/nvme1n1p1.

Step 1: Create an EBS Volume

1. Log in to **AWS Management Console**
2. Go to **EC2 → Volumes**
3. Click **Create volume**
4. Choose:
   - **Volume type:** gp3 (recommended)
   - **Size:** 20 GB (example)
   - **Availability Zone:** Same AZ as the EC2 instance
5. Click **Create volume**

Step 2: Attach the EBS Volume to EC2

1. Select the newly created EBS volume
2. Click **Actions → Attach volume**
3. Choose:
   - **Instance:** Your Linux EC2 instance
   - **Device name (console level):** /dev/xvdf
4. Click **Attach**

Internally, Linux will expose this as an NVMe device (for example /dev/nvme1n1).

Step 3: Connect to the EC2 Instance

Connect using SSH:

ssh -i key.pem ec2-user@<public-ip>

Step 4: Verify the Attached Disk

List all block devices:

lsblk

You will typically see:

- Root volume: /dev/nvme0n1 (with partitions like nvme0n1p1)
- New EBS volume: /dev/nvme1n1 (**no partition yet**)

Check filesystem information:

lsblk -fs

Step 5: Create a Partition on the New Volume

Create a partition on the **unpartitioned disk**:

fdisk /dev/nvme1n1

Inside fdisk:

- Press m → Help menu
- Press n → New partition
- Press p → Primary partition
- Press **Enter** → Default partition number
- Press **Enter** → Default first sector
- Press **Enter** → Default last sector (use full disk)
- Press w → Write changes and exit

Notify the kernel:

partprobe

Step 6: Verify the New Partition

lsblk -fs

You should now see:

- /dev/nvme1n1p1

Step 7: Format the Partition

Format the partition with **XFS** filesystem:

mkfs.xfs /dev/nvme1n1p1

Verify:

lsblk -fs

Step 8: Create a Mount Directory

mkdir /mnt/archana

Step 9: Mount the Volume

mount /dev/nvme1n1p1 /mnt/archana

Verify:

df -h

Step 10: Make the Mount Persistent (fstab)

Get the UUID:

blkid /dev/nvme1n1p1

Example output:

UUID="f1e294e6-7657-4761-bba5-bb2d0eab3936"

Edit /etc/fstab:

nano /etc/fstab

Add this line at the end:

UUID=f1e294e6-7657-4761-bba5-bb2d0eab3936   /mnt/archana   xfs   defaults,nofail   0   0

Save and exit.

Test the entry:

mount -a

navigate to cd /mnt/archana

create the files here

touch f1

Step 11: Verify Persistence

1. Reboot the instance:

reboot

2. Reconnect and check:

df -h

→ The volume should mount automatically.

Step 12: Detach and Reattach Volume (Data Recovery Scenario)

Detach the Volume

- **EC2 → Volumes → Select volume**
- **Actions → Detach volume**

Attach to Another EC2 Instance (Same AZ)

1. Attach the volume to a new instance
2. Log in to the new instance
3. Create mount directory:

mkdir /mnt/archana

4. Mount the existing partition:

mount /dev/nvme1n1p1 /mnt/archana

✅ All previously stored data will be available.

Summary Flow

Create Volume → Attach → lsblk → fdisk → mkfs → mkdir → mount → fstab → reboot

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3. Create a snapshot of the attached EBS volume and use it to create and attach a new volume to an EC2 instance in another AWS region.

Attaching Volumes across regions of EC2 using the snapshot

To create a snapshot of an EBS volume and attach it to an instance in another region in AWS, you'll need to follow these steps:

Step 1. Create a Snapshot:

a. Sign in to the AWS Management Console.
b. Navigate to the EC2 Dashboard.

c. Click on "Volumes" in the left-hand navigation pane.
d. Select the EBS volume you want to create a snapshot of.
e. Click on the "Actions" dropdown menu above the volume list and select "Create Snapshot."
f. Provide a name and description for the snapshot.
g. Click on the "Create Snapshot" button to initiate the snapshot creation process.

Step 2. Copy the Snapshot to Another Region:
a. Once the snapshot is created, go to the "Snapshots" section in the EC2 Dashboard.
b. Select the snapshot you just created.
c. Click on the "Actions" dropdown menu above the snapshot list and select "Copy Snapshot."
d. Choose the destination region where you want to copy the snapshot.
e. Click on the "Copy Snapshot" button to initiate the copy process. This may take some time depending on the size of the snapshot and the network speed.

Step 3. Monitor the Snapshot Copy Progress:
a. You can monitor the progress of the snapshot copy by navigating to the "Snapshots" section in the EC2 Dashboard of the source region.
b. Look for the snapshot you copied and check its status. It will change from "pending" to "completed" once the copy process is finished.

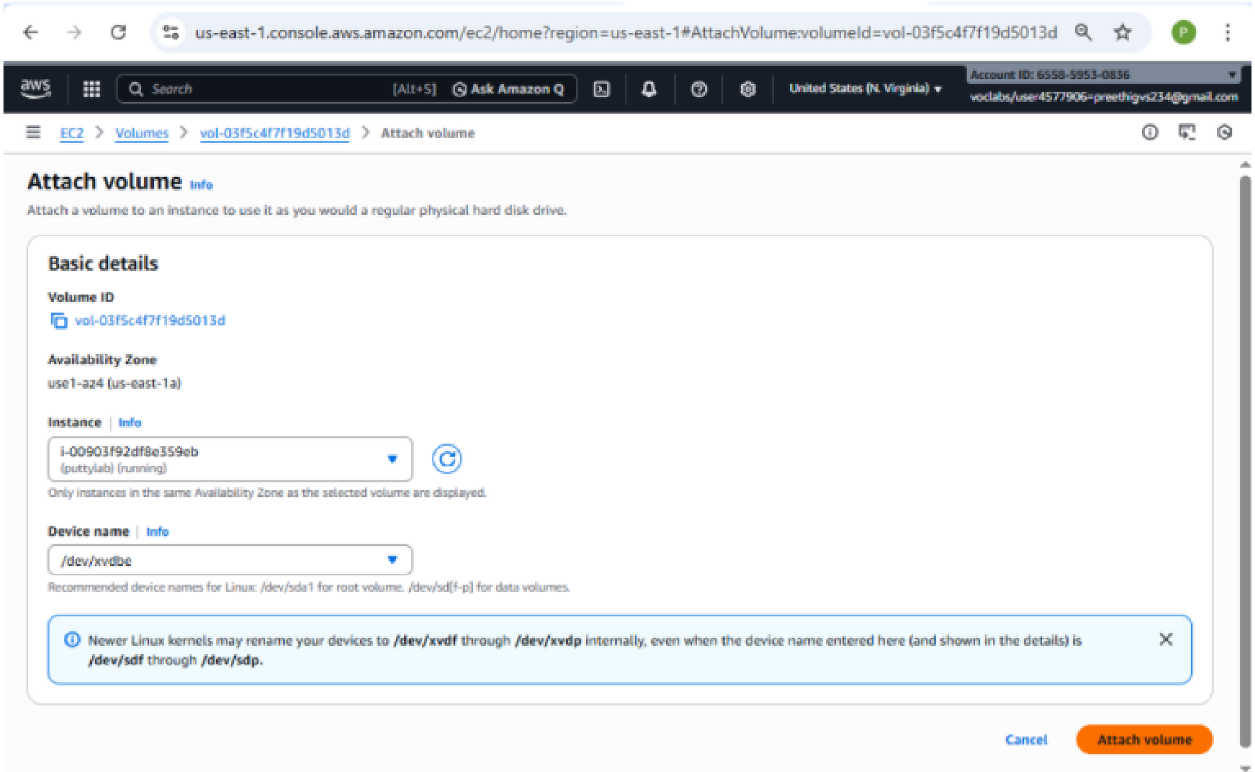
Step 4. Switch to the Destination Region:

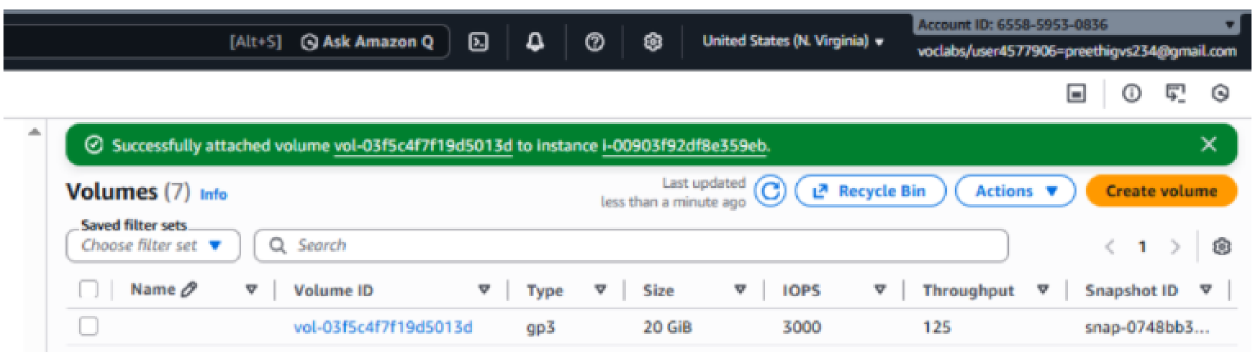
a. Use the region selector in the top-right corner of the AWS Management Console to switch to the destination region where you copied the snapshot.

Step 5. Create a Volume from the Snapshot:
a. In the EC2 Dashboard of the destination region, go to the "Snapshots" section.
b. Find the snapshot you copied from the source region.
c. Click on the snapshot, then click on the "Actions" dropdown menu and select "Create Volume."
d. Configure the volume settings, such as volume type, size, and availability zone.
e. Click on the "Create Volume" button to create the volume from the snapshot.

Step 6. Create the ec2 instance & Attach the Volume to an Instance:
a. Once the volume is created, navigate to the "Volumes" section in the EC2 Dashboard.
b. Find the newly created volume and select it.
c. Click on the "Actions" dropdown menu and select "Attach Volume."
D the EC2 instance to which you want to attach the volume and specify the device name.
e. Click on the "Attach" button to attach the volume to the instance.

We have created a snapshot of an EBS volume, copied it to another region, created a volume from the snapshot, and attached it to an instance in the destination region.


us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#AttachVolume:volumeId=vol-03f5c4f7f19d5013d
Search
[Alt+S]
Ask Amazon Q
United States (N. Virginia)
Account ID: 6558-5953-0836
voclabs/user4577906=preethigvs234@gmail.com
EC2 > Volumes > vol-03f5c4f7f19d5013d > Attach volume
Attach volume Info
Attach a volume to an instance to use it as you would a regular physical hard disk drive.
Basic details
Volume ID
vol-03f5c4f7f19d5013d
Availability Zone
use1-az4 (us-east-1a)
Instance Info
i-00903f92df8e359eb
(puttylab) (running)
Only instances in the same Availability Zone as the selected volume are displayed.
Device name Info
/dev/xvdbe
Recommended device names for Linux: /dev/sda1 for root volume. /dev/sd[f-p] for data volumes.
Newer Linux kernels may rename your devices to /dev/xvdf through /dev/xvdp internally, even when the device name entered here (and shown in the details) is /dev/sdf through /dev/sdp.
Cancel
Attach volume


[Alt+S]
Ask Amazon Q
United States (N. Virginia)
Account ID: 6558-5953-0836
voclabs/user4577906=preethigvs234@gmail.com
Successfully attached volume vol-03f5c4f7f19d5013d to instance i-00903f92df8e359eb.
Volumes (7) Info
Last updated
less than a minute ago
Recycle Bin
Actions
Create volume
Saved filter sets
Choose filter set
Search
1
Name
Volume ID
Type
Size
IOPS
Throughput
Snapshot ID
vol-03f5c4f7f19d5013d
gp3
20 GiB
3000
125
snap-0748bb3...

Week3

Elastic File Storage

Amazon Elastic File System (EFS) is a scalable and fully managed file storage service provided by Amazon Web Services (AWS). It allows you to create and configure file systems that can be accessed concurrently from multiple EC2 instances, providing a highly available and scalable storage solution for your applications and workloads.

Amazon EFS supports **for only Amazon Linux instance**:

Step 1: Launching EC2 Instances

1. Sign in to AWS Management Console.

2. Go to the EC2 dashboard.

3. Click on "Launch Instance".

4. Configure instance details:

 - Instance Name: **EFS-1**

5. Choose an Amazon Machine Image (AMI) **for Linux**.

6. Select instance type: t2.micro.

 - Key Pair: EFS

 - Network: Choose Subnet-1a

 - Security Group: **Create a new security group (SG) and add NFS and allow from anywhere.**

7. Launch the instance.

8. Repeat the above steps to launch another instance named **EFS-2 in Subnet-1b**with the same configuration.

Step 2: Creating an EFS File System

1. Go to the EFS service in the AWS Management Console.

2. Click on "Create file system".

3. Specify details:

 - Name: Optional

 - VPC: Default

 - Enable region button.

 Click on "Next".

4. In the Network settings:

 - Delete all security groups.

 - **Select the newly created security group (**NFS).---which is created while creating instance

 Click on "Next" and review the configuration.

 Click on "Create file system".

Step 3: Accessing the two EC2 instances named EFS-1& EFS -2 in **two different PowerShell** sessions and performing the specified tasks:

Accessing EFS-1 Instances in Two Different PowerShell Sessions:

1. Open Two PowerShell Sessions:

 - Open two separate PowerShell windows or tabs on your local machine.

For each Instance:

2. SSH into the Instance:

 - Use the SSH command to connect to the EFS-1 instance

 ssh -i [path-to-your-keypair.pem] ec2-user@[instance-public-ip]

3. Switch to Root User:

 - Gain root access by executing the following command:

sudosu

4. Create a Directory:

- Make a directory named "efs" using the following command:

mkdirefs

5. Install Amazon EFS Utilities:

- Use yum package manager to install the Amazon EFS utilities:

yum install -y amazon-efs-utils

6. List Files:

- Execute the following command to list files in the current directory:

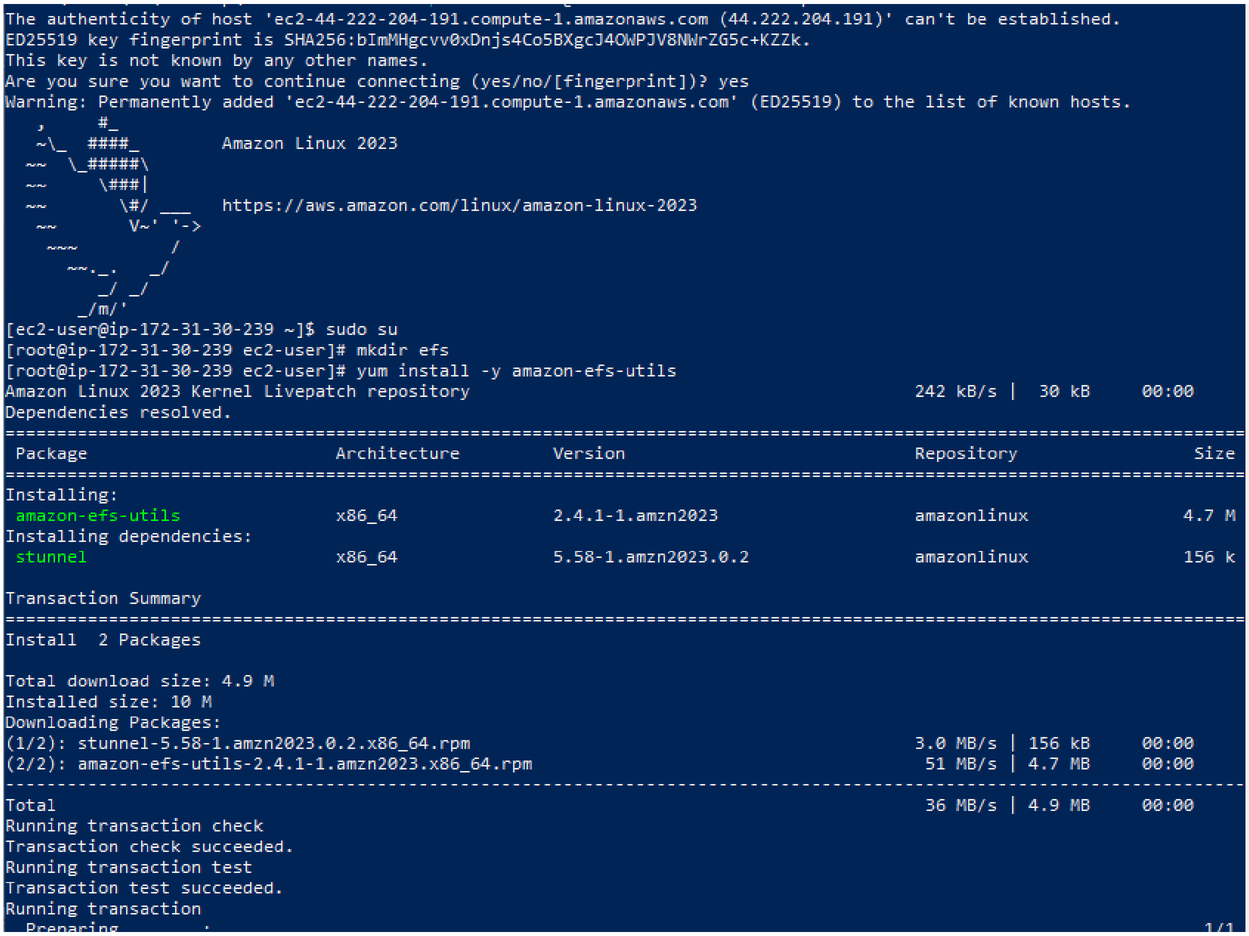
Ls

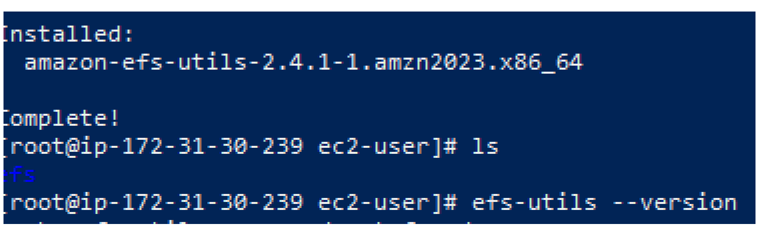
7. Verify Installation (Optional):

- Optionally, you can verify the installation of the Amazon EFS utilities by checking the version:

efs-utils --version

Repeat Steps 2-7 for ------------------------- EFS-2 Instance.





By following these steps, you'll have accessed each EFS-1 instance in separate PowerShell sessions, switched to the root user, created a directory named "efs," installed the Amazon EFS utilities, and listed files in the directory. This setup allows you to configure and manage each instance individually as needed.

Step 4: Attaching EFS to EC2 Instances

1. Go to the EFS service in the AWS Management Console.

2. Click on the target EFS file system.

3. Click on the "Attach" button.

4. Choose "**Mount via DNS**" option.

5. Copy the displayed command.,on both ec2 instances(powershell)

6. Paste and execute the copied command in the terminal to mount the EFS file system onto the instance.

Step 5: Verify and Test EFS

1. Change the directory to EFS on both instances.

2. **Create a file on one instance.**

3. Verify that the file automatically synchronizes and appears on the other instance.

By above steps, we can successfully create EC2 instances, configured them, created an EFS file system, and attached it to the instances in the same availability zone in the Mumbai region. Now, the instances can seamlessly access and share files stored in the EFS file system.

```
us-east-1a[root@ip-172-31-45-148 ec2-user]# sudo mount -t efs -o tls fs-047aaa8e0d54d6546:/ efs
Mount attempt 1/3 failed due to timeout after 15 sec, wait 0 sec before next attempt.
Mount attempt 2/3 failed due to timeout after 15 sec, wait 0 sec before next attempt.
^CTraceback (most recent call last):
  File "/sbin/mount.efs", line 4241, in <module>

[root@ip-172-31-45-148 ec2-user]# sudo mount -t efs -o tls fs-047aaa8e0d54d6546:/ efs
[root@ip-172-31-45-148 ec2-user]# df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs        4.0M     0  4.0M   0% /dev
tmpfs           481M     0  481M   0% /dev/shm
tmpfs           193M  492K  192M   1% /run
/dev/xvda1      8.0G  1.6G  6.4G  20% /
tmpfs           481M     0  481M   0% /tmp
/dev/xvda128     10M  1.3M  8.7M  13% /boot/efi
tmpfs            97M     0   97M   0% /run/user/1000
127.0.0.1:/     8.0E     0  8.0E   0% /home/ec2-user/efs
[root@ip-172-31-45-148 ec2-user]#
```

```
  File "/sbin/mount.efs", line 4241, in <module>

[root@ip-172-31-45-148 ec2-user]# sudo mount -t efs -o tls fs-047aaa8e0d54d6546:/ efs
[root@ip-172-31-45-148 ec2-user]# df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs        4.0M     0  4.0M   0% /dev
tmpfs           481M     0  481M   0% /dev/shm
tmpfs           193M  492K  192M   1% /run
/dev/xvda1      8.0G  1.6G  6.4G  20% /
tmpfs           481M     0  481M   0% /tmp
/dev/xvda128     10M  1.3M  8.7M  13% /boot/efi
tmpfs            97M     0   97M   0% /run/user/1000
127.0.0.1:/     8.0E     0  8.0E   0% /home/ec2-user/efs
[root@ip-172-31-45-148 ec2-user]# cd efs
[root@ip-172-31-45-148 efs]# echo "hello world">test.txt
[root@ip-172-31-45-148 efs]# ls
test.txt
[root@ip-172-31-45-148 efs]# _
```

```
[root@ip-172-31-15-228 ec2-user]# sudo mount -t nfs4 -o nfsvers=4.1 172.31.33.219:/ efs
^C
[root@ip-172-31-15-228 ec2-user]# sudo mount -t efs -o tls fs-047aaa8e0d54d6546:/ efs
[root@ip-172-31-15-228 ec2-user]# df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs        4.0M     0  4.0M   0% /dev
tmpfs           481M     0  481M   0% /dev/shm
tmpfs           193M  492K  192M   1% /run
/dev/xvda1      8.0G  1.6G  6.4G  20% /
tmpfs           481M     0  481M   0% /tmp
/dev/xvda128     10M  1.3M  8.7M  13% /boot/efi
tmpfs            97M     0   97M   0% /run/user/1000
127.0.0.1:/     8.0E     0  8.0E   0% /home/ec2-user/efs
[root@ip-172-31-15-228 ec2-user]# cd efs
[root@ip-172-31-15-228 efs]# ls -i
801337439624716203 test.txt
[root@ip-172-31-15-228 efs]# cat test.txt
hello world
[root@ip-172-31-15-228 efs]# _
```

Scenario-Based Questions on Amazon

EFS

1. Multiple EC2 instances across different Availability Zones must read and write the same files

at the

same time. Which AWS storage service fits this requirement and why?

Answer: Amazon EFS. EFS is a fully managed, scalable, and highly available file system that can

be

mounted concurrently by multiple EC2 instances across different AZs, allowing simultaneous

read and

write operations.

2. An application running on EC2 in three different AZs needs a shared file system with low operational

overhead. How would EFS support this setup?

Answer: EFS automatically provides a network file system accessible from EC2 instances in multiple AZs.

It handles scaling, replication across AZs, and maintenance, reducing operational overhead.

3. You mounted an EFS file system on one EC2 instance. Can another EC2 instance modify the same file

simultaneously? What happens in such a scenario?

Answer: Yes, multiple EC2 instances can access and modify the same file simultaneously. EFS supports

concurrent read/write, but applications should handle file locking or coordination to avoid data conflicts.

4. Your team tried to attach an EFS file system to an EC2 instance, but the mount command timed out.

What networking components would you check first?

Answer: Check the VPC subnet, security groups (ensure NFS TCP 2049 is allowed), route tables, and DNS

settings to ensure the EC2 can reach the EFS mount targets.

5. An EC2 instance in a private subnet is unable to mount EFS. What configuration might be missing at

the subnet or VPC level?

Answer: The subnet might lack a NAT gateway or internet access for DNS resolution, or DNS hostnames

and DNS resolution might be disabled in the VPC.

6. You created an EFS file system but forgot to create mount targets in all AZs. How will this affect EC2
instances in those AZs?
Answer: EC2 instances in AZs without mount targets will be unable to mount the EFS filesystem, causing
DNS resolution or connectivity failures.
7. Why does EFS require mount targets in each Availability Zone, and what happens if one AZ goes
down?
Answer: Mount targets provide network access to EC2 instances in that AZ. If an AZ goes down, EFS
remains available via mount targets in other AZs, ensuring high availability.
8. Two applications in different subnets need access to the same EFS file system. How should the EFS
networking and security groups be configured?
Answer: Create mount targets in each subnet's AZ. Use a security group for EFS that allows inbound NFS
(TCP 2049) from the EC2 instances' security groups.
9. Your security team wants only specific EC2 instances to access EFS. How can security groups be used
to enforce this?
Answer: Attach a security group to the EFS mount target that only allows inbound NFS (TCP 2049) from
the specific EC2 instances' security groups.
10. While mounting EFS, the command specifies -t nfs4. Why does EFS use NFS, and what

advantage

does it provide?

Answer: EFS uses NFSv4 because it is a standard, widely supported network file system protocol

that

allows multiple clients to read/write concurrently and supports file permissions and metadata.

11. An organization wants to migrate from EBS to EFS because multiple EC2 instances need

concurrent

access to data. What limitation of EBS is driving this decision?

Answer: EBS volumes can only be attached to a single EC2 instance at a time (except for EBS

Multi

Attach with certain restrictions). This prevents true concurrent access, which EFS supports.

12. An application team complains about increased latency when accessing EFS from EC2. What

factors

related to networking or AZ placement should be considered?

Answer: Latency may increase if EC2 instances and EFS mount targets are in different AZs, if the

network

is congested, or if security groups/route tables are misconfigured. Placement within the same

AZ

reduces latency.

13. Can an EFS file system be accessed from EC2 instances in different VPCs? If yes, what

networking

setup is required?

Answer: Yes, using **VPC peering**, **Transit Gateway**, or **AWS PrivateLink (EFS mount

targets

with interface endpoints)** to allow network traffic between VPCs while ensuring security

WEEK - 4
Simple Storage Service (S3)

1. **Versioning**: Versioning in Amazon S3 is a feature that allows you to keep multiple versions of an object in the same bucket. Whenever you upload a new version of an object with the same key (name) as an existing object, S3 doesn't overwrite the existing object but instead creates a new version of it. This means you can retain and access previous versions of objects, providing an additional layer of protection against accidental deletions or overwrites
   i. **Accidental Deletion Protection** :: With versioning enabled, if you accidentally delete an object, you can still retrieve previous versions of it, preventing data loss.
   ii. **Accidental Overwrite Protection**::Versioning prevents accidental overwrites of objects. Even if you upload a new version of an object with the same name, the previous versions are preserved.

1. **Cross-Region Replication**: Cross-region replication (CRR) in Amazon S3 is a feature that automatically replicates objects from one bucket in one AWS region to another bucket in a different AWS region. This provides redundancy and disaster recovery capabilities, ensuring that your data remains available even if an entire AWS region becomes unavailable.
   i. **Disaster Recovery**: Cross-region replication helps ensure business continuity by replicating critical data to a different geographic region, reducing the risk of data loss due to regional disasters or outages.
   ii. **Low-Latency Access**: Users in different geographic regions can access data from the nearest region, reducing latency and improving performance.
   iii. **Compliance**: Some regulatory requirements mandate data replication across multiple geographic regions for data sovereignty and compliance reasons.
2. **Event Notifications**: S3 can trigger events (e.g., object creation, deletion) that can be routed to AWS Lambda, SNS, or SQS for processing.
3. **Integration**: S3 seamlessly integrates with other AWS services such as AWS Lambda, Amazon CloudFront, AWS Glue, Amazon Athena, and Amazon EMR, enabling you to build powerful data processing and analytics workflows.

1. Creating an S3 bucket, upload an object, and enable public access

1️⃣ Create an S3 Bucket (with Public Access Enabled)

Step 1: Open S3 Service

- Login to AWS Management Console
- Go to Services → S3
- Click Create bucket

Step 2: Bucket Configuration

- Bucket name: my-public-bucket-123
- Region: Choose nearest region

Step 3: Disable Block Public Access (IMPORTANT)

Under Block Public Access settings for this bucket:

❌ Uncheck Block all public access

You must uncheck ALL options:

✔ Check I acknowledge that this bucket will become public

👉 Click Create bucket

2️⃣ Enable Public Access at Bucket Level (ACL)

Step 4: Open the Bucket

- Click on the bucket name
- Go to Permissions tab

Step 5: Edit Bucket ACL

- Scroll to Access Control List (ACL)
- Click Edit

Under Public access:

- Everyone (public access) → ✔ Read access

✓ **Save changes**

📌 **This allows public access at bucket level**

3️⃣ Upload Object to the Bucket

Step 6: Upload File

- Go to Objects tab
- Click Upload

- Click Add files
- Select file (e.g., image.jpg)

- Click Upload

4️⃣ Enable Public Access for Object (ACL)

Step 7: Object-Level ACL Permission

- Click on uploaded object
- Go to Permissions tab

- Scroll to Access Control List (ACL)
- Click Edit

Under Public access:

- Everyone (public access) → ✓ Read access

✓ Save changes

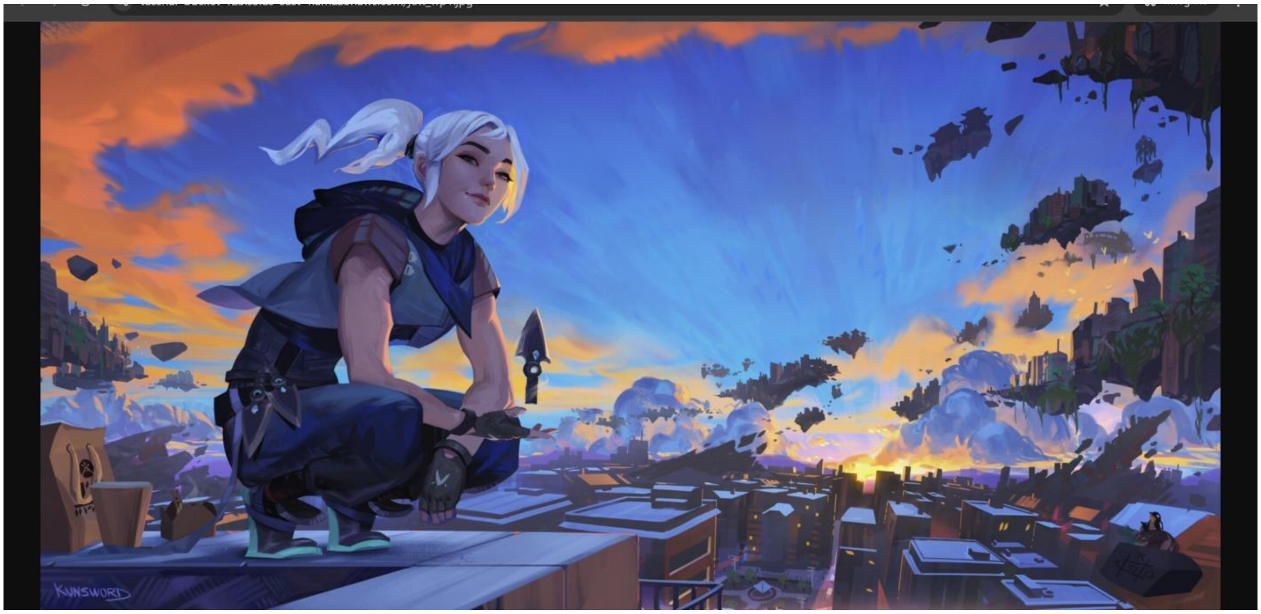
5️⃣ Verify Public Access

Step 8: Test Object URL

- Copy Object URL

- Open in browser (incognito)

✔ Object should be accessible without login



---

2. S3 Versioning & Cross-Region Replication

1️⃣Enabling Versioning in S3 (with Output Scenario)

🔷 Objective

To enable version control on an S3 bucket and observe object versions.

🔷 Steps to Enable Versioning

1. Open AWS Management Console
2. Navigate to S3
3. Select the bucket
4. Go to Properties tab
5. Scroll to Bucket Versioning
6. Click Edit
7. Select Enable
8. Click Save changes

✔ Versioning is now enabled

◈ **Output Scenario (What You Observe)**

Step A: Upload Object

- Upload file: report.pdf

📌 Output:

- Version ID is assigned automatically

Step B: Modify and Re-upload Same Object

- **Upload report.pdf again (same name)**

📌 Output:

- Old version is preserved
- New version becomes current

Step C: Delete Object

- Click Delete on report.pdf

📌 **Output:**

- Object appears deleted
- A Delete Marker is created
- Older versions still exist and can be restored

---

2 Cross-Region Replication (CRR)

◈ **Objective**

To replicate objects from one bucket to another in a different region using AWS-managed permissions.

◈ **Prerequisites (Very Important)**

✓ Versioning must be enabled on both buckets

✓ Source and destination buckets must be in different regions

✓ AWS Academy automatically handles replication permissions

Step 1️⃣ Create Destination Bucket

1. Go to S3 → Create bucket
2. Bucket name: dest-crr-bucket
3. Select different region

4. Complete bucket creation
5. Enable Versioning (same steps as above)

Step 2️⃣ Create Replication Rule (Without IAM Selection)

1. Open Source bucket
2. Go to Management tab

3. Click Replication rules
4. Click Create replication rule

Step 3️⃣ Configure Replication Rule

- Rule name: academy-crr-rule
- Status: Enabled
- Scope: Apply to all objects

Step 4️⃣ Choose Destination Bucket

- Select Another bucket
- Choose Destination bucket
- Confirm different region

📌 **AWS Academy Note:**

You will see an option like:

"AWS will create and manage the required permissions automatically"

✓ Select Use AWS-managed role / default role

Step 5️⃣ Replication Options

- **Replicate:**
  - ✓ New objects
  - ✕ Existing objects (optional, depends on Academy permissions)
- **Encryption: Leave default**

Click Save

✓ Replication rule is now active

3️⃣ CRR Output Scenario (What You Observe)

🔷 Step A: Upload New Object to Source Bucket

- Upload file: image.png

📌 Output:

- Object is uploaded normally

🔷 **Step B: Verify Destination Bucket**

- Open destination bucket
- Object image.png appears automatically
- Version ID is present

✓ Replication successful

---

3. Static Website Hosting in S3

(Using Bucket-Level & Object-Level Permissions – ACL Method)

◈ Objective

To host a static website in Amazon S3 using index.html and error.html, and allow public access using ACL-based bucket and object permissions.

◈ Prerequisites

- AWS / AWS Academy account
- Two files:
  - index.html
  - error.html

1□Create an S3 Bucket

1. Login to AWS Management Console
2. Go to Services → S3
3. Click Create bucket
4. **Enter:**
   - Bucket name: my-static-site-bucket-123
   - Region: Nearest region
5. **Under Block Public Access:**
   - ✕ Uncheck Block all public access
   - ✓ Acknowledge the warning
6. **Click Create bucket**

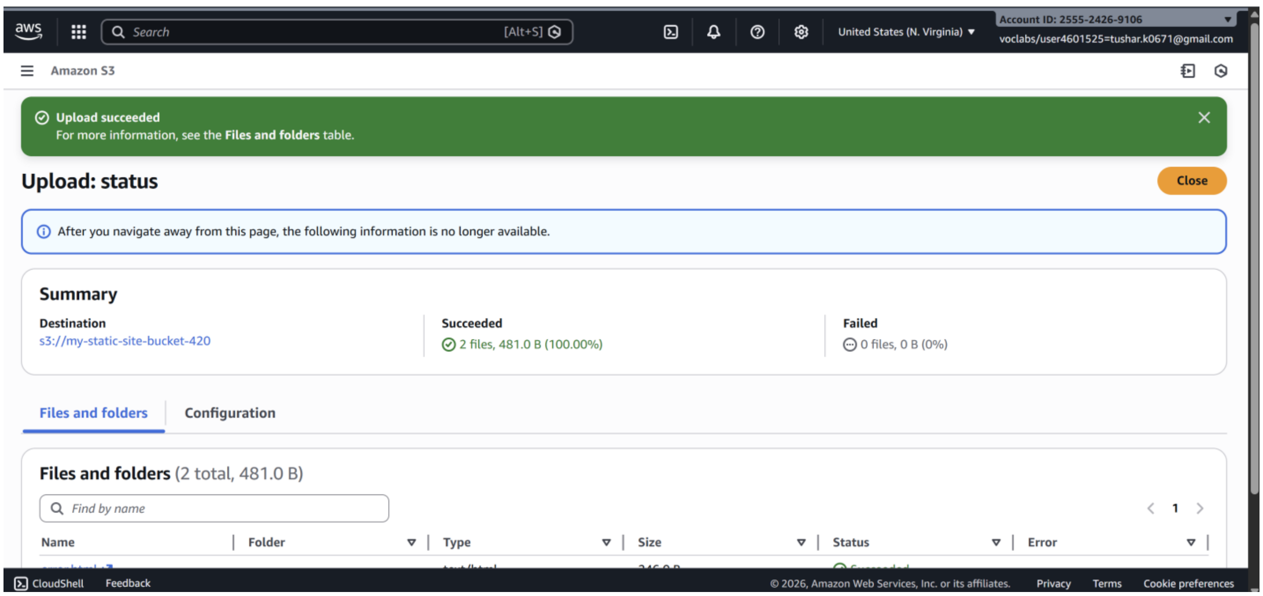
✓ Bucket created successfully

2□Upload Website Files

1. Open the bucket
2. Go to Objects tab

3. Click Upload
4. Click Add files
5. Select:
    - index.html
    - error.html
6. Click Upload

✓ Files uploaded



3️⃣ Enable Static Website Hosting

1. Go to Properties tab
2. Scroll to Static website hosting
3. Click Edit
4. Select Enable
5. Choose Host a static website
6. Enter:
    - Index document: index.html
    - Error document: error.html
7. **Click Save changes**

✓ Static website hosting enabled

4️⃣ Enable Bucket-Level Public Access (ACL)

🔷 **Purpose**

Allows public users to access objects inside the bucket.

---

1. Go to Permissions tab
2. Scroll to Access Control List (ACL)
3. Click Edit
4. Under Public access:
   - Everyone (public access) → ✓ Read
5. Click Save changes

✓ Bucket-level public read access enabled

5️⃣ Enable Object-Level Public Access (ACL)

Steps (For Each File)

1. Go to Objects tab
2. Click on index.html
3. Go to Permissions
4. Scroll to Access Control List (ACL)
5. Click Edit
6. Under Public access:
   - Everyone (public access) → ✓ Read
7. Save changes

🔁 **Repeat the same steps for error.html**

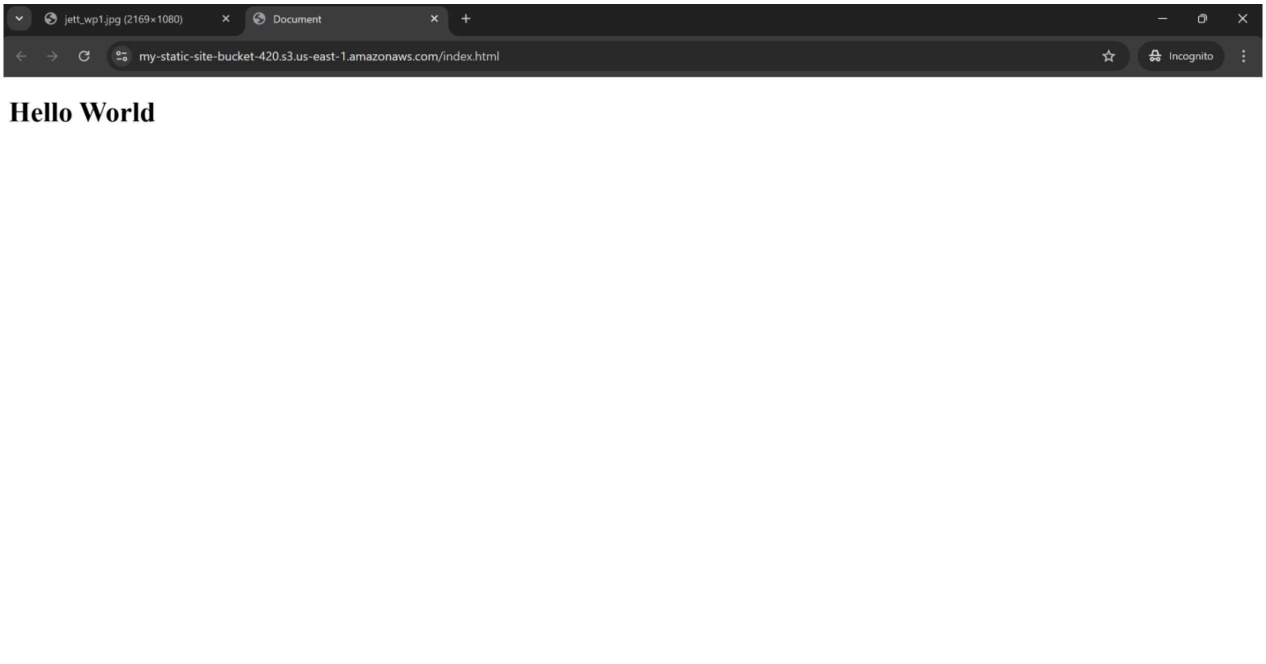
✓ **Object-level public access enabled**

6️⃣ Access the Static Website

1. Go to Properties
2. Scroll to Static website hosting
3. Copy the Bucket website endpoint

4. http://my-static-site-bucket-123.s3-website-region.amazonaws.com
5. Paste into browser

✓ index.html page loads successfully



7️⃣ Error Page Output Scenario

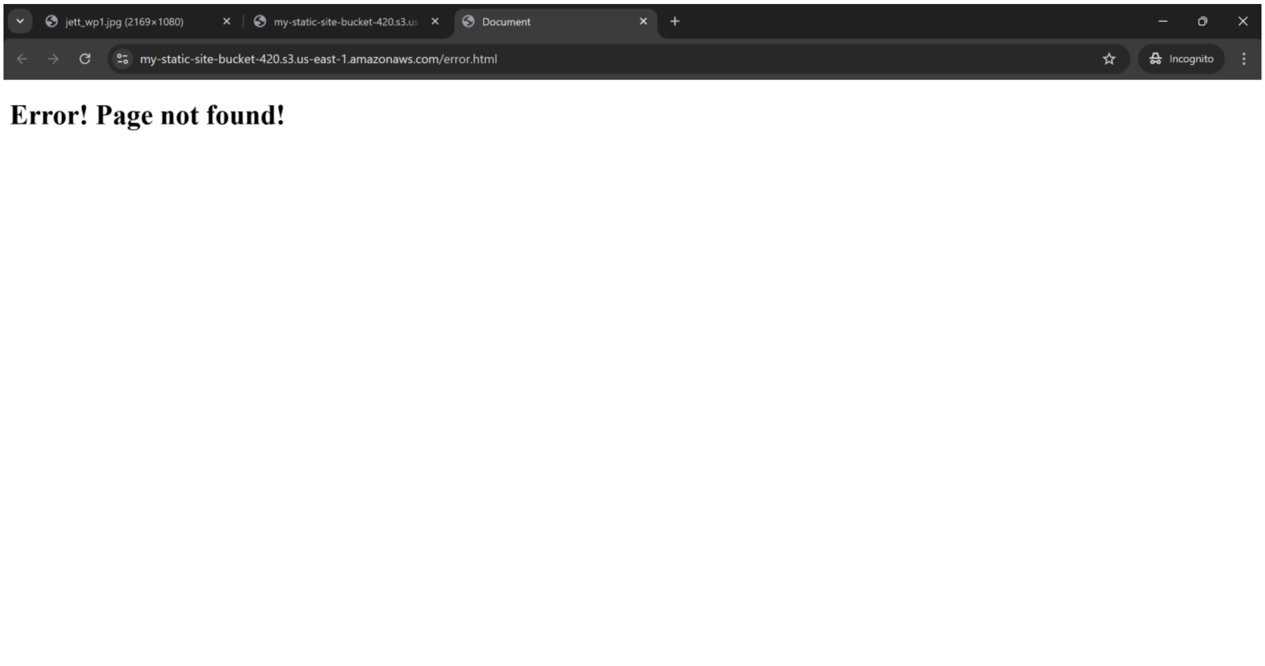
🔷 Test Error Page

- Open an invalid URL:
- http://bucket-name.s3-website-region.amazonaws.com/invalid.html

📌 Output:

- error.html page is displayed

✓ Error page working correctly



Scenario-Based Questions

1. A developer uploads an object to an S3 bucket but cannot access it from the browser, even though the bucket exists. What permission-related issue could be the cause?

Ans: The object may not have public read permission. Either the bucket policy, object ACL, or Block Public Access settings are preventing public access.

2. An S3 bucket policy allows public read access, but a specific object inside the bucket is still not accessible. Why might this happen?

Ans: The object's ACL might explicitly deny public read access, or Block Public Access is enabled at the bucket or account level, overriding the bucket policy.

3. A bucket has **no bucket policy**, but individual objects have public ACLs. Will users be able to access those objects? What determines the final access decision?

Ans: Yes, users *may* access the objects if object ACLs allow public read. Final access is determined by the combination of IAM policies, bucket policies, ACLs, and Block Public Access settings.

4. A company wants to ensure **no public access** to any object in an S3 bucket, even if someone mistakenly applies a public ACL. What S3 feature should be enabled?

Ans: Enable S3 Block Public Access (specifically “Block public ACLs” and “Ignore public ACLs”) to prevent any public exposure.

5. You attempt to create an S3 bucket named my-company-data but receive a naming error. What S3 bucket naming rules might have been violated?

Ans: The name may violate rules such as global uniqueness, using uppercase letters, underscores, or being too short/long, or resembling an IP address.

6. Two different AWS accounts try to create a bucket named project-backup-2026. One succeeds, the other fails. Why?

Ans: S3 bucket names are globally unique across all AWS accounts. Once one account creates it, no other account can use the same name.

7. You enabled **static website hosting** on an S3 bucket and uploaded files, but accessing the website URL returns **403 Forbidden**. What is the most likely cause?

Ans: The bucket or objects do not have public read permission. Static website hosting requires public access to objects via bucket policy or ACLs.

8. Static website hosting is enabled, but accessing the root URL shows an error saying **index.html not found**. What is the actual issue?

Ans: The specified index document (e.g., index.html) either does not exist in the bucket or the name/path is incorrect.

9. What HTTP error is returned when the **index document is missing** in S3 static website hosting, and why does this happen?

Ans: S3 returns 404 Not Found because it cannot locate the configured index document required to serve the root URL.

10. A bucket policy allows public access, but **static website hosting still doesn’t work**. What object-level setting must also be verified?

Ans: The objects themselves must have public read access. Bucket-level permissions alone are not sufficient for static website hosting.

11. You accidentally overwrite an important object in S3. Versioning was enabled earlier. How can the original file be recovered?

Ans: You can restore the original file by retrieving and re-copying a previous object version from the bucket's version history.

12. A user deletes an object from a versioned S3 bucket. Is the data permanently removed? Explain what actually happens.

Ans: No, the data is not permanently removed. S3 adds a delete marker, and previous versions remain recoverable unless explicitly deleted.

13. You enabled Cross-Region Replication (CRR) on a bucket, but objects are not replicating. What configuration might be missing?

Ans: An IAM role with replication permissions may be missing, or versioning may not be enabled on both source and destination buckets.

14. After enabling CRR, only newly uploaded objects are replicated, but old objects are not. Why?

Ans: CRR only replicates objects uploaded after replication is enabled. Existing objects must be replicated manually using S3 Batch Replication.

15. A company wants to replicate S3 data to another region for **disaster recovery and compliance**. What are the key prerequisites for enabling CRR?

Ans: Both buckets must have versioning enabled, proper IAM replication roles configured, and the destination bucket in a different AWS region.

Week 5: VPC Creation and Ec2 Instance Connection

��Step 1: Create the VPC

1. Go to **AWS Management Console → VPC**
2. Click **Create VPC**
3. Choose **VPC only**
4. Enter:
   - **Name**: `Custom-VPC`
   - **IPv4 CIDR block**: e.g. `10.0.0.0/16`
5. Click **Create VPC**

�� Step 2: Create Subnets (Public & Private)

Create subnets in same **Availability Zones** for high availability.

Example:

· **Public Subnet**: `10.0.1.0/24` (AZ-A)
· **Private Subnet**: `10.0.2.0/24` (AZ-A)

Steps:

1. Go to **Subnets → Create subnet**
2. Select your VPC
3. Choose AZ
4. Enter CIDR
5. Create subnet

Public Subnet

Private Subnet

�� Step 3: Create an Internet Gateway (IGW)

1. Go to **Internet Gateways**
2. Click **Create internet gateway**
3. Name it `Custom-IGW`
4. **Attach** it to your VPC
5.

�� Step 4: Create Route Tables

You need **separate route tables** for public and private subnets.

Public Route Table

1. Go to **Route Tables → Create route table**
2. Select VPC
3. Add route:
   - Destination: `0.0.0.0/0`
   - Target: **Internet Gateway**
4. Associate with **public subnet**

Private Route Table

· Keep default local route only (no IGW)
· Associate with **private subnet**

�� Step 5: Enable Auto-Assign Public IP (Public Subnet)

1. Select **Public Subnet**
2. Go to **Edit subnet settings**
3. Enable **Auto-assign public IPv4 address**

�� Step 6: Configure Security Groups

1. Create a **Security Group**
2. Add inbound rules:
   - HTTP (80) / HTTPS (443)
   - SSH (22) from trusted IP
3. Attach to EC2 instances

�� Step 7: Launch EC2 Instances

· Public EC2 → Public subnet
· Private EC2 → Private subnet
Attach correct security groups.

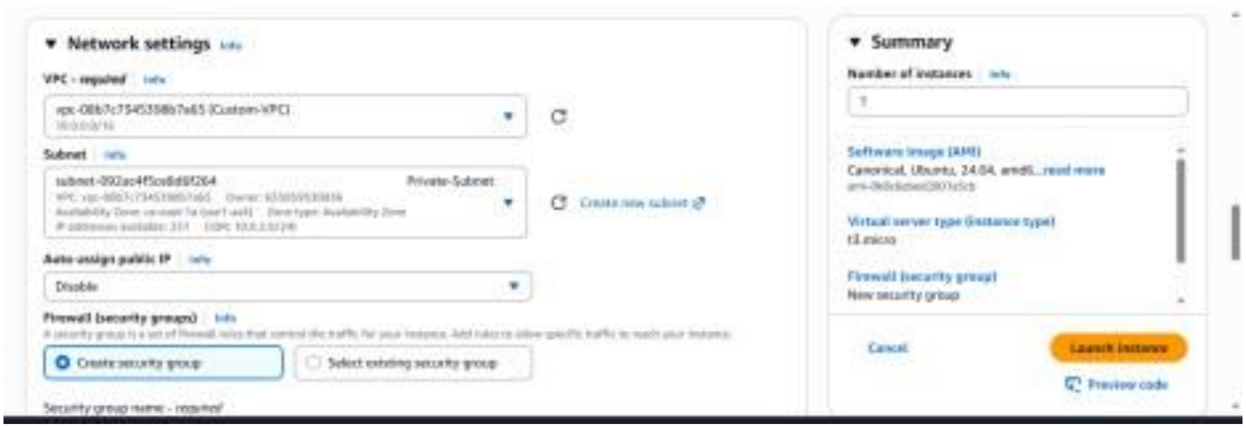
�� Step 8: Test the Setup

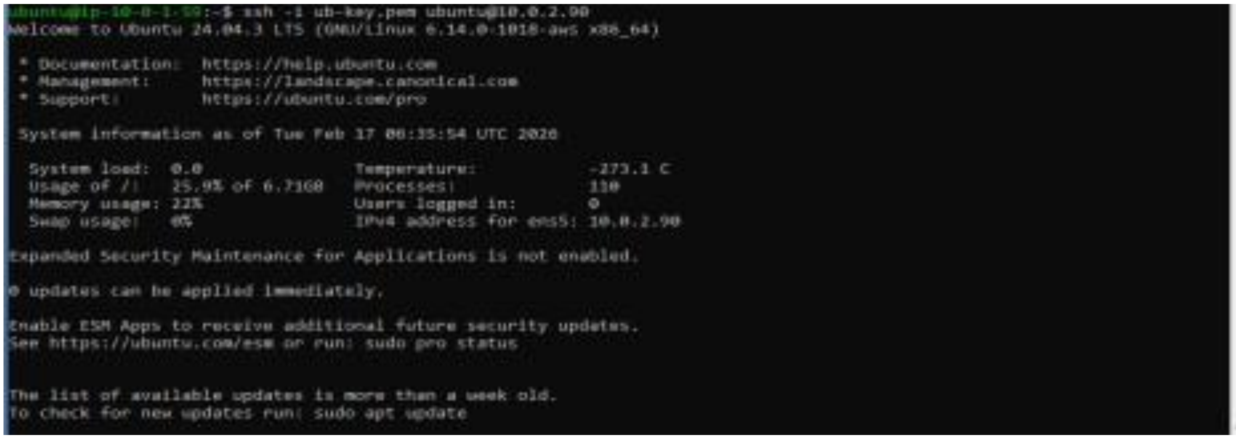
· Public EC2 → Internet access ✔

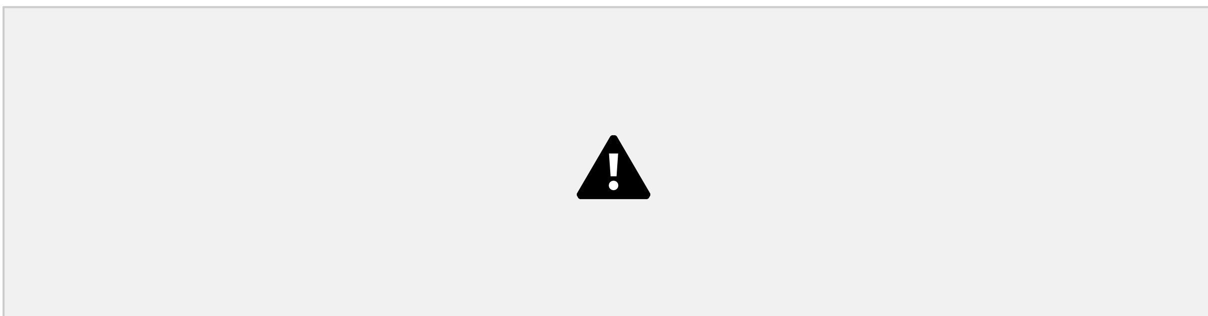
Private

· Private EC2 → Internet via NAT ✔

· Private EC2 → No direct inbound access ✔







Scenario Based Questions:

1. You have a web application where users access a website, but the database should not be exposed to the internet. How would you design the VPC? Which resources go into public and private subnets?
   Answer: Create public and private subnets in the VPC. Place Load Balancer and Web servers in public subnet, and App servers and RDS in private subnet. This ensures the database is secure and not directly accessible from the internet.
2. An EC2 instance in a public subnet cannot access the internet. What VPC components would you check and why?
   Answer: Check Internet Gateway is attached and route table has route 0.0.0.0/0 pointing to Internet Gateway. Verify instance has Public IP and security group allows outbound traffic. Also ensure Network ACL allows inbound and outbound traffic.
3. Your application server needs to connect to an RDS database securely. How would you configure security groups and subnets?
   Answer: Place both Application server and RDS in private subnets. Allow inbound database port like 3306 in RDS security group from App server security group only. Do not allow public access to RDS for better security.
4. You are asked to design a VPC for 500 servers today, but it should scale to 2,000 servers in the future. How would you choose the CIDR block?
   Answer: Choose CIDR block like /16 or /20 that supports more than 2000 IP addresses.

Example 10.0.0.0/16 provides 65,536 IP addresses. This ensures enough IP space for future expansion.

5. A company wants secure connectivity between its on-premises data center and AWS VPC. Which AWS services would you choose and why (VPN vs Direct Connect)?
   Answer: Use Site-to-Site VPN for quick setup and encrypted connection over internet. Use Direct Connect for high performance, low latency, and reliable private connection. Choose based on performance and cost requirements.
6. You need to connect two VPCs, but both use the same CIDR range. How would you solve this problem?
   Answer: VPC Peering cannot work with overlapping CIDR ranges. Use NAT Gateway, Private Link, or Transit Gateway with CIDR translation. Alternatively change CIDR block of one VPC.
7. Multiple micro services running in different subnets need to communicate securely. How would you design routing and security groups?
   Answer: Ensure route tables allow communication using local route within VPC. Use Security Groups to allow only required ports between services. Follow least privilege access principle.
8. Design a VPC for a 3-tier application (Web, App, DB) with high security and scalability. Explain subnets, route tables, gateways, and security groups.
   Answer: Place Web servers in public subnet and App and DB servers in private subnets. Use Internet Gateway for public subnet and NAT Gateway for private subnet internet access. Configure Security Groups to allow only required communication.
9. You suspect unusual traffic inside your VPC. How would you monitor and analyze network traffic?
   Answer: Enable VPC Flow Logs to capture network traffic. Use Cloud Watch and Cloud Trail to monitor activity. Analyze logs to identify suspicious traffic patterns. 10. Why can't a private subnet have an Internet Gateway directly attached? **Answer:** Internet Gateway attaches to VPC, not directly to subnet. Private subnet route table does not have route to Internet Gateway. This prevents direct internet access. 11. What happens if route tables are misconfigured in a VPC?
   Answer: Instances may lose internet or internal connectivity. Traffic may be dropped or routed incorrectly. Applications may stop working properly.
12. What happens if a route table has no local route?
   Answer: Local route enables communication within VPC. Without it, instances cannot communicate with each other. Internal networking will fail.
13. An EC2 instance allows traffic on port 80 in the security group, but traffic is still blocked. What could be the reason?
   Answer: Network ACL may be blocking traffic. Instance may not have Public IP. Route table or Internet Gateway may be misconfigured.
14. You need to create a VPC that will host 1,000+ EC2 instances across multiple AZs. How do you decide the CIDR block?
   Answer: Choose CIDR block like /16 or /20 to support enough IP addresses. Example 10.0.0.0/16 provides large IP range. Divide into subnets across availability zones.
15. Only a company's corporate IP should be able to SSH into EC2 instances. How would

you implement this securely in AWS VPC?
Answer: Configure Security Group inbound rule for SSH port 22. Set source as company corporate public IP only. This restricts unauthorized access.

16. An EC2 instance can send traffic out but cannot receive responses. Which VPC component might be misconfigured and why?
Answer: Network ACL may block inbound response traffic. Security Group inbound rules may be incorrect. Route table or NAT configuration may be wrong.

WEEK 6

VPC USING BASTION SERVER

1. VPC with Bastion Server

VPC & Subnet Design
VPC CIDR: 10.0.0.0/16
Public Subnet (Bastion Server): 10.0.1.0/24
Private Subnet (Database Server): 10.0.2.0/24

---

2. Security Group Configuration (VERY IMPORTANT)

Bastion Server Security Group (Public Subnet)
Allows SSH (22) from: My IP (Database Administrator)

Database Server Security Group (Private Subnet)

Allows MySQL / Aurora (3306) from:	custom	10.0.1.0/24 (Public subnet / Bastion subnet)
Allows SSH (22) from:	custom	Private IP of Bastion Server

Does NOT allow public access

---

3. Security Outcome

Database server is NOT publicly accessible
Only Bastion server can connect to DB server using SSH
External users cannot access DB directly

4. EC2 Instance Creation

Bastion Server
Subnet: Public Subnet (10.0.1.0/24)
Key pair: bastion.pem
Operating System: Ubuntu
Database Server
Subnet: Private Subnet (10.0.2.0/24)
Key pair: dbserver.pem
Operating System: Ubuntu

---

5. Copy DB Server Key to Bastion Server

Database server is in private subnet
Direct SSH from local system is NOT possible
DB server key must be copied to Bastion server
Command (PowerShell):
scp -i bastion.pem dbserver.pem ubuntu@<BASTION_PUBLIC_IP>:/home/ubuntu/
File is copied securely using SCP
dbserver.pem will now be available inside Bastion server

---

6. **Connect to Bastion Server**

SSH Command:
ssh -i bastion.pem ubuntu@<BASTION_PUBLIC_IP>
Verify key files:
ls
Output:
bastion.pem
dbserver.pem

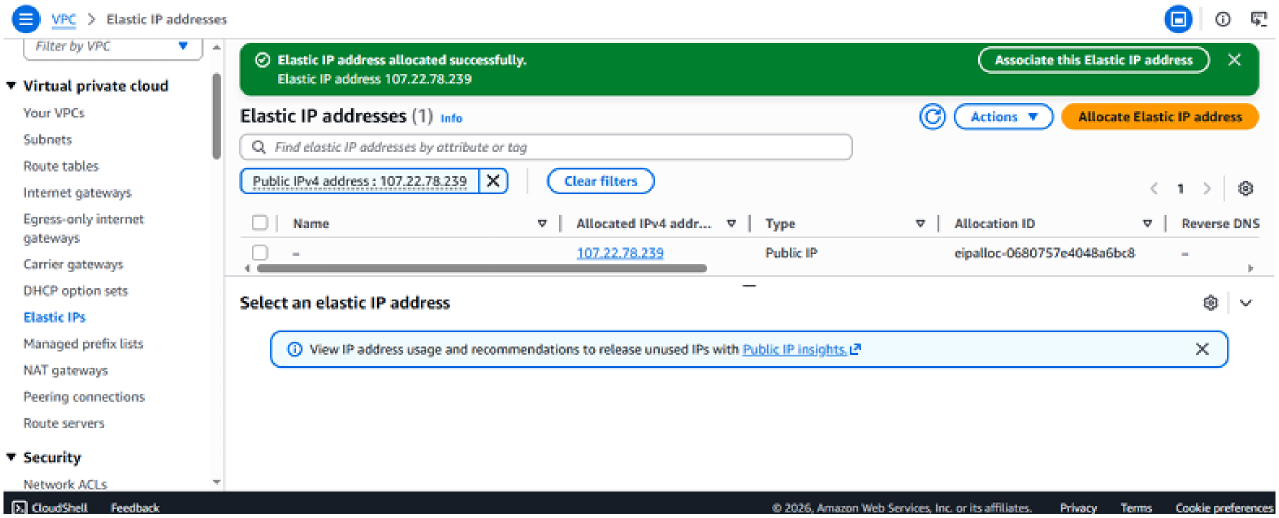
---

7. **Connect from Bastion to Database Server**

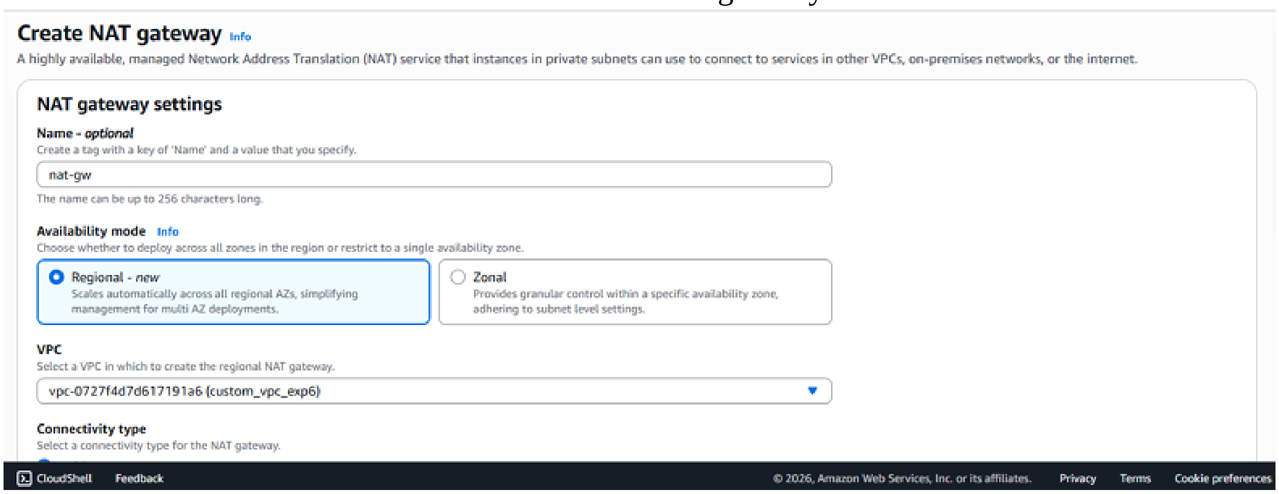
SSH from Bastion:
ssh -i dbserver.pem ubuntu@10.0.2.x
Connection to DB server is successful
Database administrator can now:
Update database configurations
Install required packages-------problem occurs
Restart MySQL service
?

VPC USING NAT GATEWAY

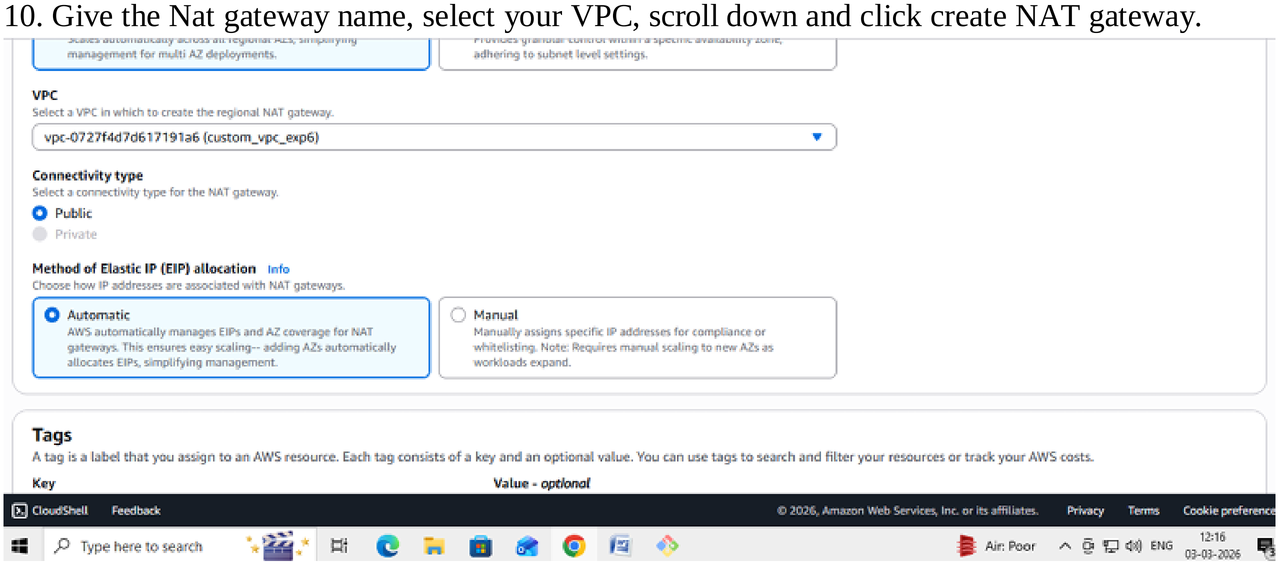
8. In the above steps it clear there is no internet access, if we want to install or update anything in the instances present in private subnet.



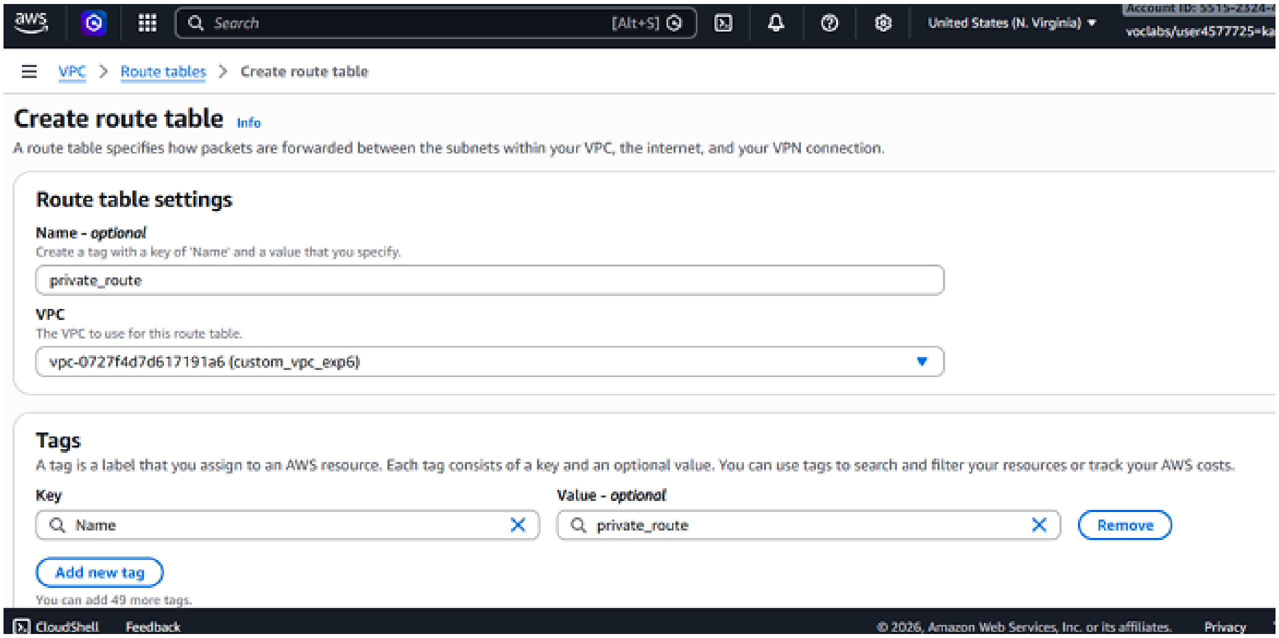
9. Network Address Translator (NAT Gateway) will solve this problem. Goto NAT Gateway in the left side menu bar of VPC and click on create NAT gateway.



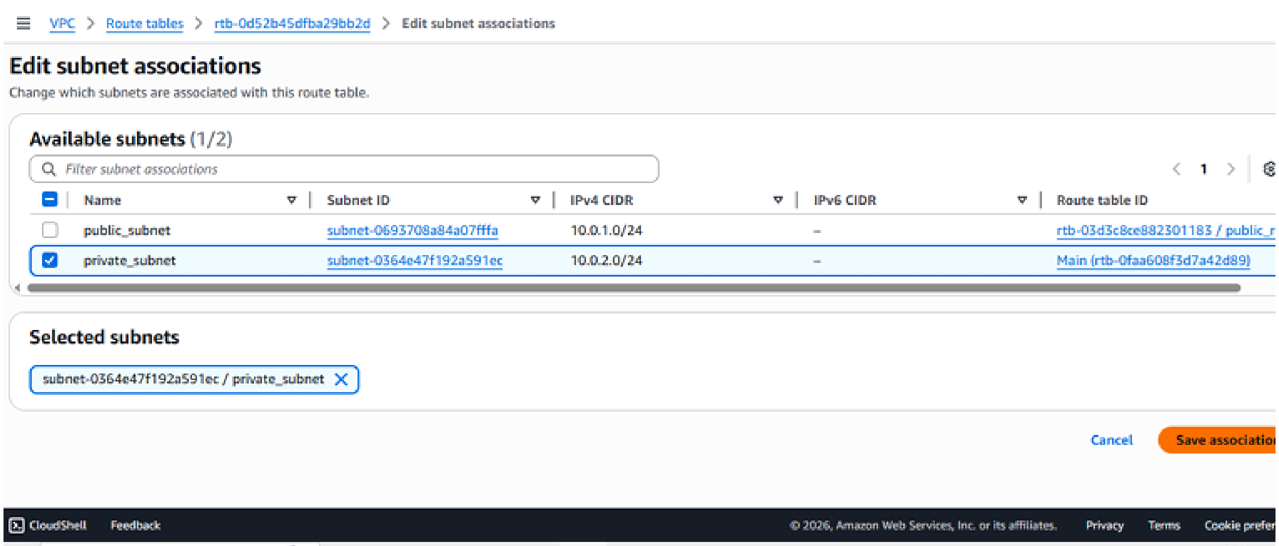
10. Give the Nat gateway name, select your VPC, scroll down and click create NAT gateway.



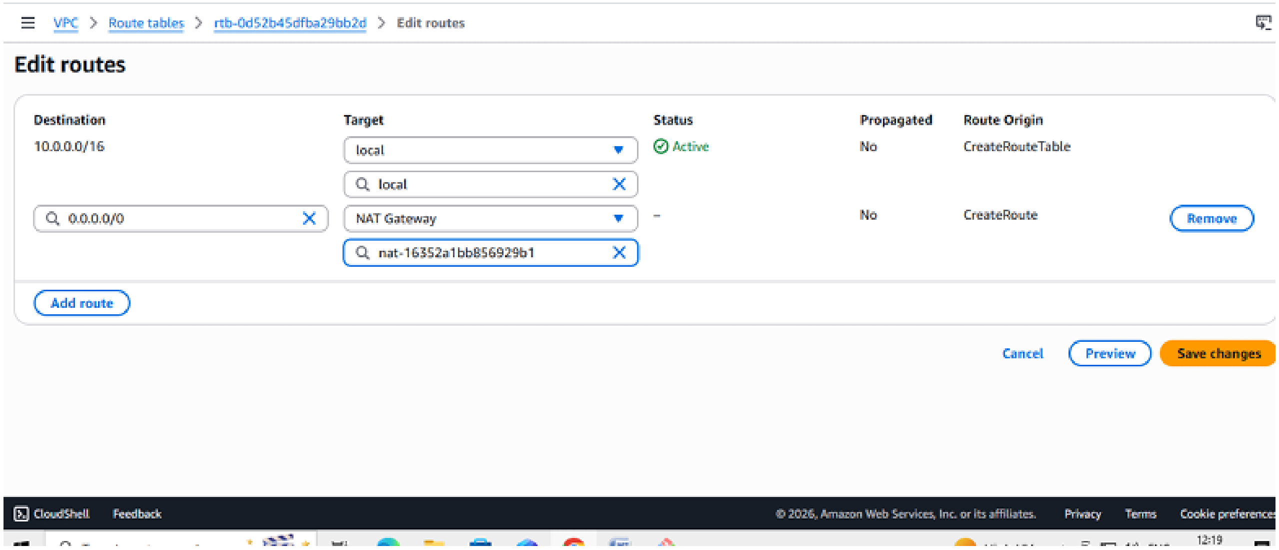
11. So Now create a route for accessing NATGateway. Goto Route table click on create route table and give the route table name as bastion route , select our customized vpc and click on create route.



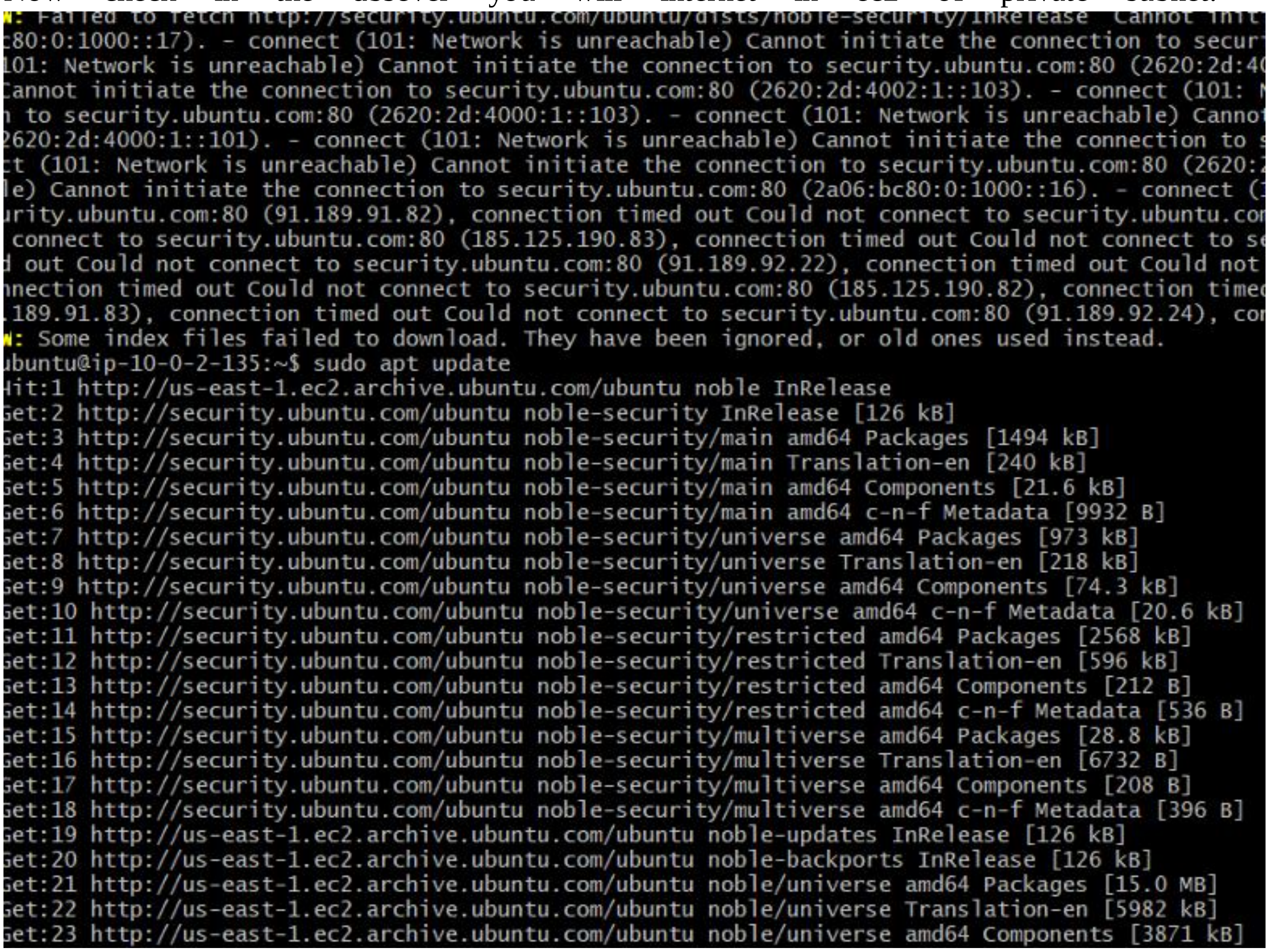
12. Select the route, click on actions, goto edit subnet associations and add private subnet to this bastion route.

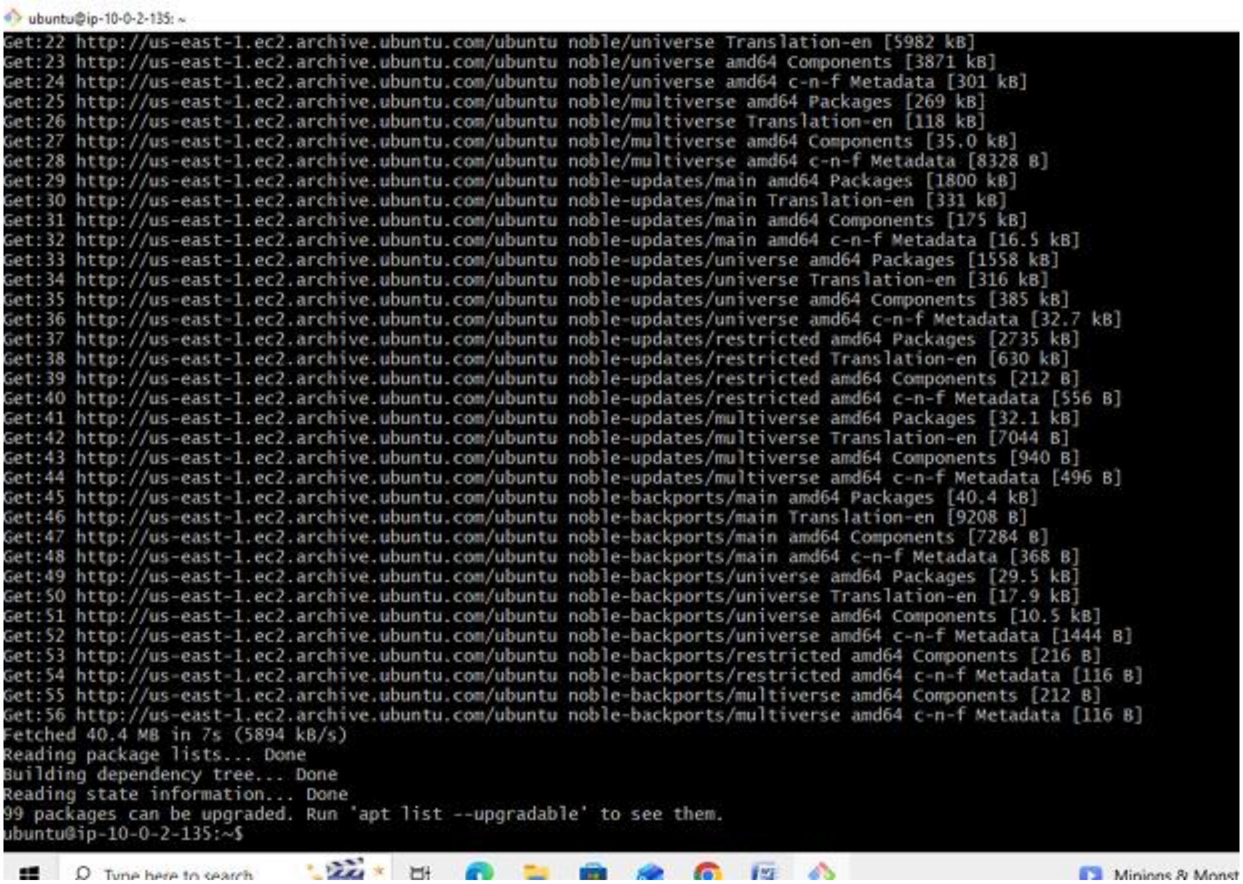


13. Again goto actions and click on edit routes, attach NAT Gateway with 0.0.0.0/0, select your NAT gateway name in the drop down menu and click on save changes.



14. Now check in the dbsever you will internet in ec2 of private subnet.





SCENARIO BASED QUESTIONS

1. Q: You have multiple private subnets in different AZs. Should you use one NAT Gateway or multiple NAT Gateways?
A: Use multiple NAT Gateways (one per AZ) for high availability and to avoid cross-AZ charges.

---

2. Q: Your private subnet instances can reach the internet, but the NAT Gateway shows no traffic. What could be the possible reasons?
A: Traffic may be routed via another path (IGW/VPC endpoint) or incorrect route table association.

---

3. Q: A NAT Gateway was accidentally created in a private subnet instead of a public subnet. What problem will occur, and how would you fix it?
A: NAT won't work due to no IGW access; fix by placing it in a public subnet with IGW route.

---

4. Q: An application in a private subnet needs outbound access only to specific services. How do you restrict traffic?
A: Use Security Groups, NACLs, or firewall/proxy rules to allow only specific destinations.

---

5. Q: Multiple teams share one NAT Gateway and traffic conflicts occur. How to redesign?
A: Use separate NAT Gateways and route tables per team for isolation.

---

6. Q: SSH via Bastion works, but yum/apt update fails in private instance. Why?
A: Missing or incorrect route to NAT Gateway or DNS configuration issue.

---

7. Q: Private instances access internet via NAT, but SSH from Bastion fails. Why?
A: Security group or NACL rules are blocking SSH from Bastion to private instance.

---

8. Q: Bastion server and NAT Gateway are both in a private subnet. What is the impact?
A: Neither internet access nor external SSH will work due to lack of public exposure.

---

9. Q: SSH to Bastion works, but SSH from Bastion to private EC2 fails. Why?
A: Private instance security group does not allow inbound SSH from Bastion.

---

10. Q: Why is a Bastion server not a replacement for a NAT Gateway?
A: Bastion provides SSH access only, while NAT Gateway enables outbound internet connectivity.

CCLAB-7

WEEK-7:

step-by-step procedure for setting up an AWS Lambda function to trigger whenever an object is uploaded to an S3 bucket and update a DynamoDB table:

STEP-1: Create the S3 Bucket

-Search for S3 bucket in amazon console

-Click on the create bucket in the S3 bucket console

-Select general purpose

-Name the bucket as you needed

-Enable Versioning

-Click on the create bucket

STEP-2: Create a AWS DynamoDB service table

-Search for dynamodb in the search bar of amazon

console -Click on the create table

-GIve any name to the table

Eg. newtable
-Name the row as unique with type string

STEP-3: Creating the lambda function

-Search for aws lambda

-Click on the create function

-Choose Author from scratch

-Type any name in the function name input

-Choose a runtime

Eg.Python

-In permissions choose the role type

If available labrole in the existing roles

Or

Create a new role with basic lambda permissions

-click on the create function

STEP-4: Adding trigger to connect the S3 bucket and DynamoDB

-Click on the Add trigger in the created function

-Search for S3 bucket in the trigger configuration

-Select the S3 bucket

-Choose the bucket which you want to apply the trigger

-Select All object create event in the event type, Acknoledge

-click on the add

STEP: 5 Creating the deplyoment code and monitoring

-Add the code for triggering when object is uploaded

Eg

```
import boto3

from uuid import uuid4

deflambda_handler(event, context):

    s3 = boto3.client("s3")

    dynamodb = boto3.resource('dynamodb')

    # Check if the 'Records' key is present in the event

    if'Records'in event:

        # Iterate over each record in the event

        for record in event['Records']:

            bucket_name = record['s3']['bucket']['name']

            object_key = record['s3']['object']['key']

            size = record['s3']['object'].get('size', -1)

            event_name = record.get('eventName', 'Unknown') # Use get()
method to handle missing keys
```

event_time = record.get('eventTime', 'Unknown') # Use get() method to handle missing keys

dynamoTable = dynamodb.Table('newtable')

dynamoTable.put_item(

Item={'unique': str(uuid4()), 'Bucket': bucket_name, 'Object': object_key,'Size': size, 'Event': event_name, 'EventTime': event_time})

else:

Log a message indicating that the 'Records' key is missing from the event

print("No 'Records' key found in the event.")

Create a policy if you are getting the Iam role problem Otherwise skip this

-Search for IAM console in AWS console

-Go to roles

-Select the Lamda function name like you gave in the labmda creation

-In permission policies click on add permissions

-Select create inline policy

-Click on the JSON in the policy editor

Write this policy code in that editor

{

"Version": "2012-10-17",

"Statement": [

{

"Effect": "Allow", "Action": "dynamodb:PutItem",

"Resource": "arn:aws:dynamodb:us-west 2:669420739378:table/newtable"

```
    }
  ]
}
```

-Click on the next

-Give any policy name

Eg.lambda-dynamo-policy

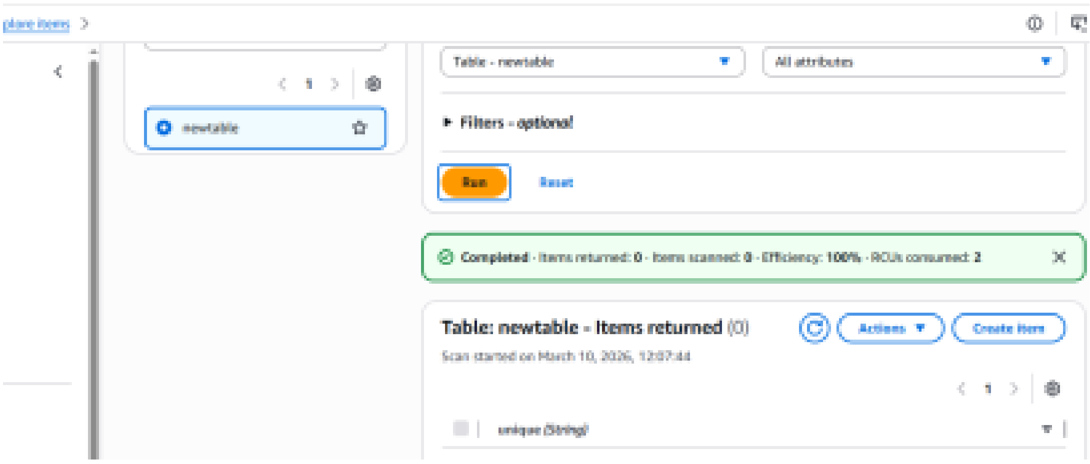
-Click on the create policy

STEP-6 : Upload any file(s) in the S3 bucket it will triggers and shows in both S3 bucket and the DynamoDB

-Click on the explore items in the DynamoDB

-click on the run

You'll able to see the objects in the dynamodb which is uploaded in the S3 bucket

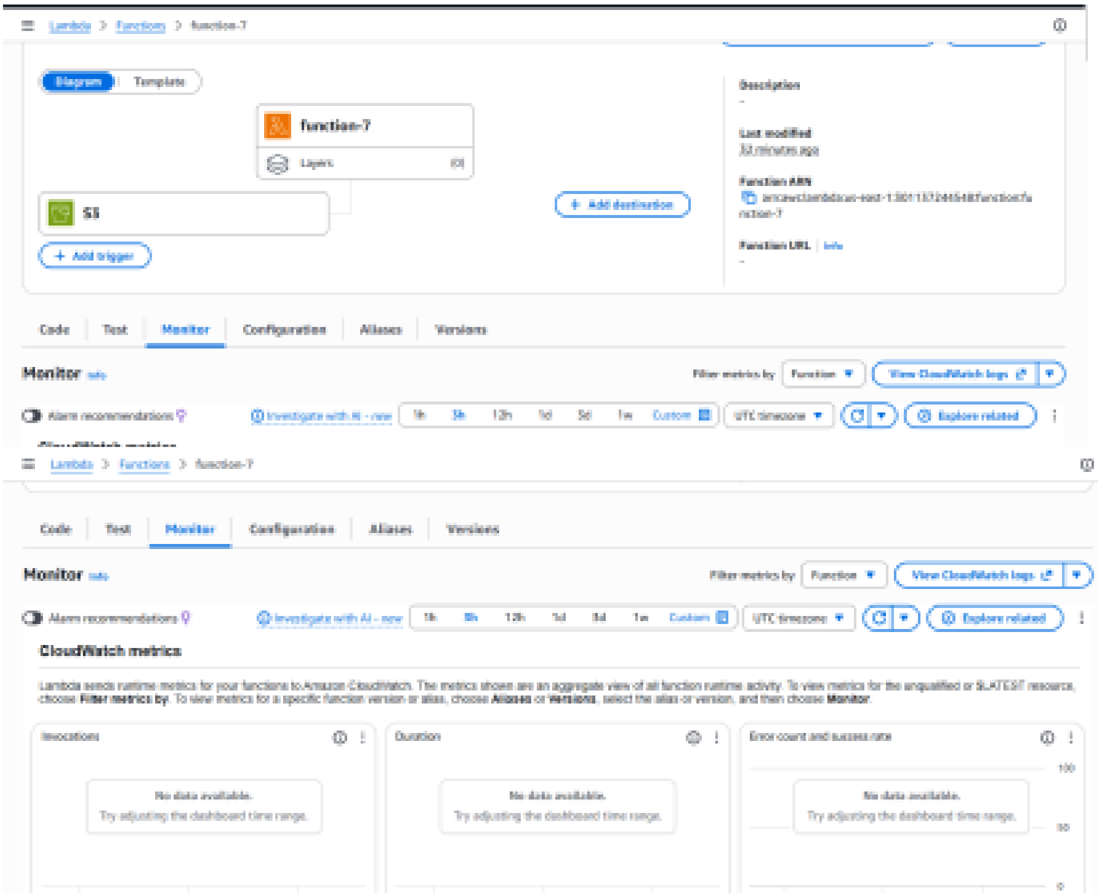


STEP-7: Monitoring using cloud watch

-Click on the Monitor tab in the lambda function

-Click on the view cloudwatch logs

-Click on the time lines and check the logs\



1.Q: A company wants to automatically process images whenever they are uploaded to an S3 bucket. The system should resize the images and store them in another bucket. Which AWS service can be used to automatically run the code when the image is uploaded?

A:Use AWS Lambda with an Amazon S3 Event Trigger. When an image is uploaded to the S3 bucket, the event automatically triggers the Lambda function which processes and resizes the image and stores it in another bucket.

2.Q: A developer created a Lambda function to store uploaded file details into a DynamoDB table, but the function fails with a permission denied error when writing to the database. What could be the reason and how can it be fixed?

A:The Lambda execution role does not have permission to write to DynamoDB.
This can be fixed by attaching an IAM policy (e.g., dynamodb:PutItem) to the Lambda execution role.
3.Q: An application needs to run a piece of code every time a new file is uploaded to an S3 bucket to extract metadata and store it in a database. How would you design this using AWS Lambda?

A:

1. **Configure Amazon S3 event notification for object upload.**
2. **Set the event trigger to AWS Lambda.**
3. **Lambda reads the uploaded file metadata.**
4. **Lambda stores the metadata into a database such as Amazon DynamoDB.**

4.Q: A Lambda function is configured with an S3 trigger, but it is not executing when files are uploaded. What possible configuration issues might cause this problem?

A: Possible issues include:

- **S3 event notification not configured properly.**
- **Wrong bucket or event type selected.**
- **Lambda permission not granted to S3.**
- **File upload not matching prefix/suffix filters.**

5.Q: A company wants to process only .jpg image files uploaded to an S3 bucket using Lambda. How can the trigger be configured so that the function runs only for those files?

A: Configure S3 event notification filters with a suffix filter .jpg so the Lambda function triggers only for .jpg files.

6Q: A Lambda function works correctly for a few uploads but fails when hundreds of files are uploaded simultaneously. What AWS feature helps Lambda handle large numbers of events?

A: AWS Lambda automatic scaling (concurrency) allows Lambda to handle large numbers of events by running multiple instances of the function simultaneously.

7.Q: A developer wants to store the table name and bucket name separately from the code so that they can easily change them later. Which Lambda feature can be used to achieve this?

A: Use Lambda Environment Variables to store configuration values like table names and bucket names.

8Q: A Lambda function executes but does not show any output in the console, making debugging difficult. Which AWS service can be used to view logs and troubleshoot the issue?

A: Use Amazon CloudWatch Logs to view Lambda execution logs and debug issues.

9.Q: An organization wants a function to run every day at 10 PM to clean temporary files stored in S3. How can Lambda be triggered automatically at a scheduled time?

A: Use Amazon EventBridge (CloudWatch Events) with a scheduled rule to trigger the Lambda function daily at 10 PM.

10.Q: A Lambda function processes uploaded files and stores metadata in DynamoDB, but duplicate records appear in the table when the same file is uploaded again. How can this issue be prevented?

A: Use a unique primary key in DynamoDB (such as file name or object key) and apply a conditional write (PutItem with condition expression) to prevent duplicate entries.

11.Q: A Lambda function processes files uploaded to S3, but the processing sometimes takes more than 10 minutes. What limitation of Lambda must be considered here?

A: AWS Lambda has a maximum execution time limit of 15 minutes. If processing exceeds this limit, another service like AWS Step Functions, ECS, or EC2 should be used.

12.Q: A developer wants to allow a Lambda function to read files from S3 and write records into DynamoDB. What type of AWS security configuration must be created?

A: An IAM Role with appropriate policies must be created for the Lambda function granting:

- s3:GetObject permission
- dynamodb:PutItem permission.

13.Q: A company wants to automatically send a notification email whenever a file is uploaded to S3 using Lambda. Which AWS services can be integrated to achieve this?

A: Use Amazon S3 → AWS Lambda → Amazon SNS (Simple Notification Service) to send email notifications.

14.Q: A Lambda function is triggered by S3 uploads, but the function keeps running multiple times for the same file. What could be the possible cause?

A:Possible causes include:

· Multiple S3 event notifications configured.

· Lambda writing back to the same bucket and retriggering itself. ·

Retry behavior due to failed executions.

15.Q: An organization wants to build a fully serverless application where uploaded files are processed automatically without managing servers. Which AWS services can be combined with Lambda to achieve this architecture?

A:
A serverless architecture can include:

· Amazon S3 – file storage

· AWS Lambda – processing logic

· Amazon DynamoDB – data storage

· Amazon API Gateway – API access

· Amazon SNS/SQS – notifications or messaging

WEEK-8---SNS,SQS & S3-SNS-SQS-LAMBDA SERVICES

1) Simple Notification Sevice (SNS)

A) Create SNS Topic and Send Message to Email

We will:

1. Create an SNS Topic
2. Create an Email Subscription
3. Confirm the subscription
4. Publish a test message

Step 1: Create SNS Topic

1. Login to **AWS Console**
2. In search bar → Type **SNS**
3. Open **Amazon SNS (Simple Notification Service)**
4. In left panel → Click **Topics**
5. Click **Create topic**

Configure:

- **Type** → Select **Standard**
- **Name** → MyEmailTopic
- Leave other settings default (unless required)

6. Click **Create topic**

Step 2: Create Email Subscription

1. Open your created topic
2. Go to **Subscriptions** tab
3. Click **Create subscription**

Configure:

- **Protocol** → Select **Email**
- **Endpoint** → Enter your email address

4. Click **Create subscription**

Step 3: Confirm Subscription

- Go to your email inbox
- You will receive a mail from **Amazon SNS**
- Click **Confirm subscription**

Now your email is successfully subscribed □

Step 4: Publish Test Message

1. Open your SNS topic
2. Click **Publish message**
3. Add:
    - **Subject** → Test Mail
    - **Message body** → Hello this is SNS test
4. Click **Publish message**

You will receive email notification.

B) Trigger Email When S3 Object is Uploaded (S3 → SNS → Email)

Now we extend this setup:

Flow:

S3 Bucket → Event Notification → SNS Topic → Email

Step 1: Create S3 Bucket

1. Search → **S3**
2. Click **Create bucket**

Configure:

- **Bucket name** → my-upload-bucket-2026
- Choose Region (Important: Same region as SNS topic)
- Keep default settings (unless specific requirement)

3. Click **Create bucket**

Step 2: Create SNS Topic (If not already created)

You can reuse the topic from Part A
OR create new topic like:

- Topic name → S3UploadNotification

Make sure email subscription is confirmed.

Step 3: Allow S3 to Publish to SNS (Access Policy)

1. Open SNS Topic
2. Go to **Access policy**
3. Click **Edit**

Add permission for S3 to publish.

If using Basic Editor:

Access Policy that needs to be configured on SNS:

a) replace SNS topic ARN

b) Replace s3 bucket ARN

c) Replace account

```
{

 "Version": "2012-10-17",

 "Id": "example-ID",

 "Statement": [

  {

   "Sid": "Example SNS topic policy",

   "Effect": "Allow",

   "Principal": {

    "Service": "s3.amazonaws.com"

   },

   "Action": [
```

```
        "SNS:Publish"
      ],
      "Resource": "SNS topic ARN",
      "Condition": {
        "ArnLike": {
          "aws:SourceArn": "s3 bucket ARN"
        },
        "StringEquals": {
          "aws:SourceAccount": "account id"
        }
      }
    }
  ]
}
```

(Modern AWS console often auto-generates this when configuring from S3)

Step 4: Configure S3 Event Notification

1. Open your S3 bucket
2. Go to **Properties**
3. Scroll to **Event notifications**
4. Click **Create event notification**

Configure:

- **Event name** → UploadNotification
- **Event types** → Select:
    - All object create events
- Destination:
    - Select **SNS topic**

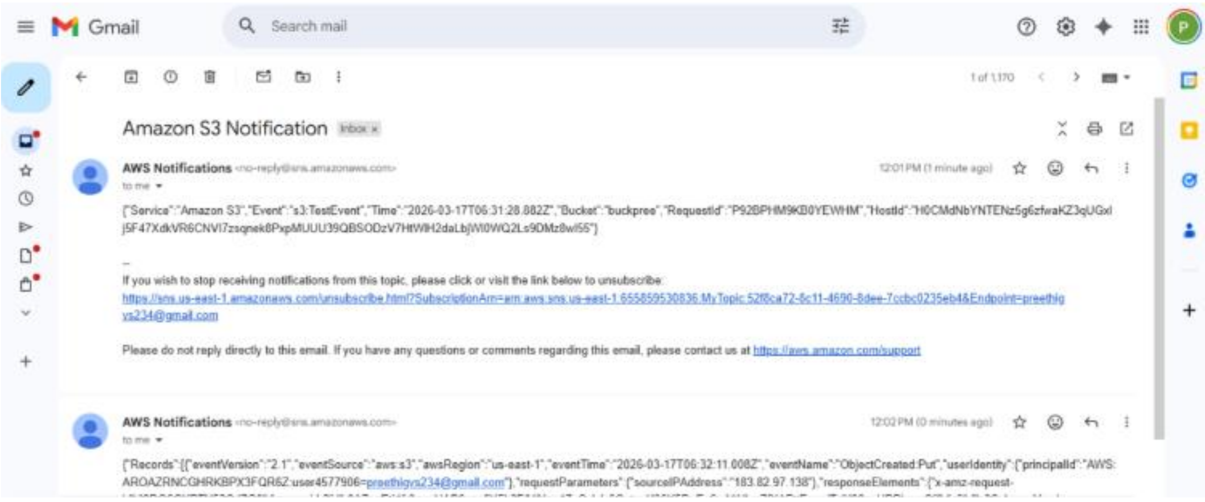
- Choose your topic (S3UploadNotification)

5. Click **Save changes**

Step 5: Test It

1. Upload any file into the S3 bucket
2. After upload completes:
    - S3 triggers event
    - SNS receives publish request
    - SNS sends email

You will receive mail about object upload.



Simple Queue Service (SQS)

2) Create SQS Queue and Send Message

We will:

1. Create an SQS Queue
2. Send a message to the Queue
3. Receive the message from the Queue

Step 1: Create SQS Queue

1. Login to **AWS Console**
2. In the **search bar → Type SQS**
3. Open **Amazon Simple Queue Service**
4. In the dashboard → Click **Create queue**

Configure:

• **Type → Select Standard**
• **Name → MyQueue**
• Leave other settings as **default**

5. Click **Create queue**

Your **SQS queue is now created successfully.**

Step 2: Send Message to Queue

1. Open the **MyQueue** queue
2. Click **Send and receive messages**
3. In the **Message body**, type:

Hello this is SQS test message

4. Click **Send message**

Your message is now stored in the queue.

Step 3: Receive Message from Queue

1. In the same page (**Send and receive messages**)
2. Click **Poll for messages**

You will see the message displayed.
Example output:
Hello this is SQS test message

3. After processing, click **Delete message** (optional)

Result:

- SQS queue created
- Message sent successfully
- Message received from queue

---

3) S3 Bucket Event → SNS → SQS → Lambda Processing

S3 bucket event (like file upload) triggers SNS, which sends the notification to SQS. Then **Lambda acts as the consumer and processes the message**.

Services used:

• Amazon S3
• Amazon Simple Notification Service
• Amazon Simple Queue Service
• AWS Lambda

S3 → SNS → SQS → Lambda Architecture

S3 Bucket (File Upload)

↓

SNS Topic

↓

SQS Queue

↓

Lambda Function (Consumer)

When a file is uploaded, S3 sends an event notification to SNS, which forwards the message to SQS. The Lambda function then reads and processes the message from SQS.

Step 1: Create S3 Bucket

1. Login to AWS Console
2. Search **S3**
3. Open **Amazon S3**
4. Click **Create bucket**

Configure:

• Bucket name → **mys3eventbucket123** (must be unique)
• Region → choose your region

Leave other settings default.

5. Click **Create bucket**

Step 2: Create SNS Topic

1. Search **SNS** in AWS Console
2. Open **Amazon SNS**
3. Click **Topics** → **Create topic**

Configure:

• Type → **Standard**
• Name → **MyS3SNSTopic**

4. Click **Create topic**

Copy the **SNS Topic ARN**.

Step 3: Create SQS Queue

1. Search **SQS** in AWS Console
2. Open **Amazon SQS**
3. Click **Create queue**

Configure:

• Type → **Standard**
• Name → **MyS3Queue**

4. Click **Create queue**

Copy the **Queue ARN**.

Step 4: Subscribe SQS to SNS

1. Open **MyS3SNSTopic**
2. Click **Create subscription**

Configure:

• Protocol → **Amazon SQS**
• Endpoint → Select **MyS3Queue**

3. Click **Create subscription**

Step 5: Add Access Policy to SQS

Open **MyS3Queue → Access Policy → Edit**

Replace:

• SQS ARN
• SNS ARN

```
{
  "Statement": [
    {
      "Effect": "Allow",
      "Principal": {
        "Service": "sns.amazonaws.com"
```

```
      },
      "Action": "sqs:SendMessage",
      "Resource": "SQS ARN",
      "Condition": {
        "ArnEquals": {
          "aws:SourceArn": "SNS ARN"
        }
      }
    }
  ]
}
```

Click **Save**.

Step 6: Add Access Policy to SNS

Open **SNS Topic → Edit access policy**

Replace:

- SNS topic ARN
- S3 bucket ARN
- Account ID

```
{
  "Version": "2012-10-17",
  "Id": "example-ID",
  "Statement": [
    {
      "Sid": "Example SNS topic policy",
      "Effect": "Allow",
```

```
      "Principal": {
        "Service": "s3.amazonaws.com"
      },
      "Action": [
        "SNS:Publish"
      ],
      "Resource": "SNS topic ARN",
      "Condition": {
        "ArnLike": {
          "aws:SourceArn": "S3 bucket ARN"
        },
        "StringEquals": {
          "aws:SourceAccount": "Account ID"
        }
      }
    }
  ]
}
```

Save the policy.

Step 7: Configure S3 Event Notification

1. Open **S3 Bucket (mys3eventbucket123)**
2. Go to **Properties**
3. Scroll to **Event notifications**

Click **Create event notification**

Configure:

• Event name → **S3UploadEvent**
• Event type → **All object create events**

Destination:

• **SNS Topic → MyS3SNSTopic**

Click **Save changes**.

Step 8: Test the Architecture

1. Open the **S3 bucket**
2. Click **Upload file**
3. Upload any file (example: **test.txt**)

Now:

File Upload → S3

↓

Notification → SNS

↓

Message → SQS

Step 9: Verify Message in SQS

1. Open **MyS3Queue**
2. Click **Send and receive messages**
3. Click **Poll for messages**

You will see the **S3 event notification message**.

Step 10: Configure Lambda as Consumer

Now configure Lambda to **automatically process messages from SQS**.

Create Lambda Function

1. Search **Lambda** in AWS Console
2. Open **AWS Lambda**
3. Click **Create Function**

Configure:

• Author from scratch
• Function name → **SQSConsumerFunction**
• Runtime → **Python 3.x**

Click **Create function**

Add SQS Trigger

1. Open **SQSConsumerFunction**
2. Click **Add Trigger**
3. Select **SQS**
4. Choose **MyS3Queue**

Click **Add**

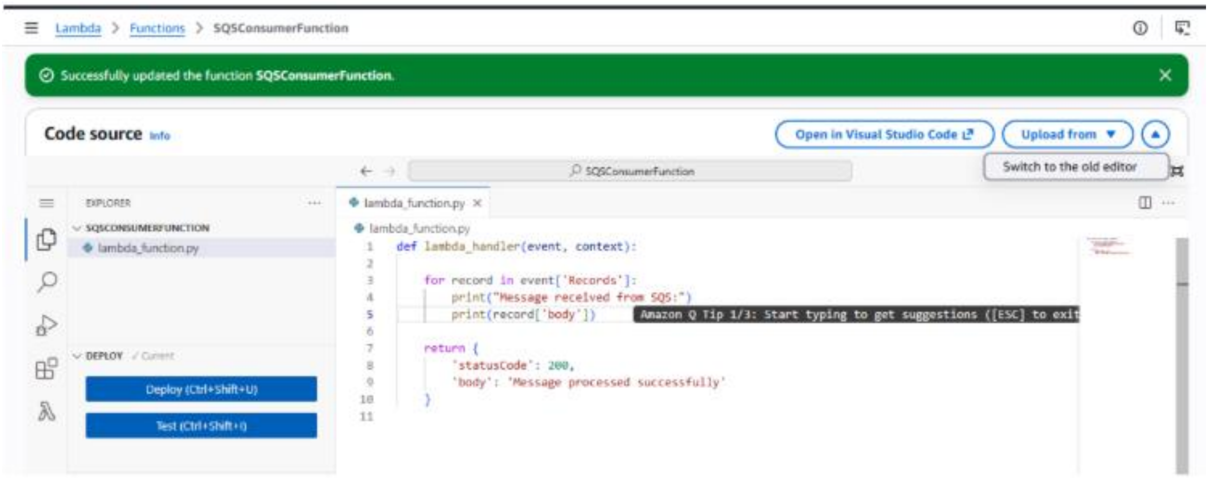
Now Lambda will automatically read messages from SQS.

Lambda Code

Replace code with:

```
def lambda_handler(event, context):


    for record in event['Records']:

        print("Message received from SQS:")

        print(record['body'])


    return {

        'statusCode': 200,

        'body': 'Message processed successfully'

    }
```

Click **Deploy**.



Final Architecture

User uploads file

↓

Amazon S3

↓

SNS Topic

↓

SQS Queue

↓

Lambda Function (Consumer)

↓

Processing / Logs / Database

Result

When a file is uploaded to S3:

1. S3 sends a notification to SNS
2. SNS forwards the message to SQS

3. SQS stores the message
4. Lambda automatically reads and processes the message

This creates a **serverless event-driven architecture commonly used in real cloud applications**.

Scenario Based Questions (SNS, SQS, S3-SNS-SQS-Lambda)

Q1. A company wants to send notifications to multiple users whenever an important system event occurs. The same message must be delivered to email, SMS, and application

subscribers simultaneously. Which AWS service can be used to implement this publish-subscribe messaging system?

Ans. Amazon SNS

Q2. A developer created an SNS topic and added an email subscription, but the subscriber is not receiving any notifications. What could be the possible reason for this issue?

Ans. The email subscription is not confirmed by the recipient.

Q3. An organization wants to automatically notify administrators via email whenever a new file is uploaded to an S3 bucket. How can this be implemented using AWS services?

Ans. Configure S3 event notifications to trigger Amazon SNS and subscribe administrators via email.

Q4. A developer configured an S3 bucket to trigger an SNS notification on file upload, but the message is not reaching the SNS topic. What configuration or permission issue might cause this problem?

Ans. Missing permissions in the SNS topic policy to allow S3 to publish messages.

Q5. A company wants to store incoming messages temporarily so that applications can process them later without losing data even if the system is busy. Which AWS messaging service should be used for this requirement?

Ans. Amazon SQS

Q6. A developer sends messages to an SQS queue, but when checking the queue no messages appear. What possible reasons could cause this issue?

Ans. Messages may have already been consumed, sent to the wrong queue, or are delayed due to delay settings.

Q7. A company wants to process thousands of messages coming from different applications without overloading the processing system. How does Amazon SQS help in handling this situation?

Ans. Amazon SQS buffers messages and enables asynchronous processing, allowing systems to scale and avoid overload.

Q8. A developer wants to ensure that messages are processed automatically from an SQS queue without manually polling the queue. Which AWS service can be configured to automatically consume messages from SQS?

Ans. AWS Lambda

Q9. An application uploads files to S3, and the company wants the upload event to trigger a workflow that sends notifications and processes the message later. How can this be designed using S3, SNS, SQS, and Lambda?

Ans. S3 triggers Amazon SNS, SNS sends messages to Amazon SQS, and AWS Lambda processes messages from SQS.

Q10. A developer configured an SNS topic to send messages to an SQS queue, but the queue is not receiving any messages. What policy configuration might be missing?

Ans. Missing SQS queue policy allowing SNS to send messages to the queue.

Q11. A company wants to decouple its application components so that failures in one component do not affect others. Which AWS service combination can be used to implement loosely coupled communication?

Ans. Amazon SNS and Amazon SQS

Q12. A Lambda function is configured to process messages from an SQS queue, but the function is not being triggered. What possible configuration issue might cause this problem?

Ans. Missing event source mapping between SQS and AWS Lambda or insufficient permissions.

Q13. An organization wants to ensure that uploaded files in S3 trigger notifications, store messages reliably, and process them asynchronously. Which serverless architecture using AWS services can achieve this?

Ans. S3 → Amazon SNS → Amazon SQS → AWS Lambda

Q14. A developer wants to monitor the processing of messages handled by a Lambda function triggered by SQS. Which AWS service can be used to view logs and troubleshoot the function execution?

Ans. Amazon CloudWatch

Q15. A company wants to build a fully serverless event-driven system where file uploads automatically trigger notifications and background processing without managing servers. Which AWS services can be combined to build this architecture?

Ans. Amazon S3, Amazon SNS, Amazon SQS, and AWS Lambda

Q16. A company wants to send a notification to multiple applications whenever a new order is placed in their system. Each application should receive the same message and process it independently. Which AWS service can be used to implement this publish-subscribe communication model?

Ans. Amazon SNS

Q17. A developer configured an SNS topic and connected it to an SQS queue, but the messages published to SNS are not appearing in the queue. What configuration or permission might be missing in this setup?

Ans. Missing SQS access policy allowing SNS topic to send messages.

Q18. An application generates a large number of events, and the development team wants to ensure that these events are stored safely until they are processed by another service. Which AWS service can be used to reliably store and deliver these messages?

Ans. Amazon SQS

Q19. A company uploads files to an S3 bucket and wants to process these files using Lambda, but only after the messages is safely stored to avoid data loss during high traffic. Which AWS architecture using SNS and SQS can help achieve this requirement?

Ans. S3 → Amazon SNS → Amazon SQS → AWS Lambda

Q20. A developer notices that messages remain in an SQS queue even after being processed by a Lambda function. What configuration or step might be missing to ensure messages are removed after processing?

Ans. The Lambda function is not successfully completing execution or not deleting messages (visibility timeout or error handling issue).

WEEK-9- Implementation of Elastic Load Balancer with Auto Scaling using AWS EC2 Instances.

Elastic Load Balancer (ALB)

Step 1: Launch Two EC2 Instances

1.1 Launch EC2 Instance #1

1. Go to **AWS EC2 Console → Launch Instance**.
2. **Name**: webserver-1
3. Choose an **AMI** (Amazon Linux 2).
4. Select **Instance Type** (e.g., t2.micro).
5. Add **Storage** (default is fine).
6. Configure **Security Group**:
    - Allow **SSH (22)** from your IP
    - Allow **HTTP (80)** from anywhere (0.0.0.0/0)
7. **Launch** → Select/Create Key Pair → Launch.

    Run these commands after connecting VM

```
yum update -y
yum install httpd -y
systemctl start httpd
systemctl enable httpd

echo  "This is Server 1" > /var/www/html/index.html
```

✅ Now you create second **EC2 instances** running Apache, with message **This is Servier 2** each serving a simple web page.

Repeat it for second instance

```
yum update -y
yum install httpd -y
systemctl start httpd
systemctl enable httpd
```

```
echo  "This is Server 2" > /var/www/html/index.html
```

Step 2: Create a Load Balancer

AWS provides **Elastic Load Balancer (ELB)**. We will create an **Application Load Balancer (ALB)**.

2.1 Configure Security Group for Load Balancer

- Go to **EC2 → Security Groups → Create Security Group**
    - Name: lb-sg
    - Allow **HTTP (80)** from anywhere
    - Optional: HTTPS (443) if using SSL

2.2 Create the Load Balancer

1. Go to **EC2 → Load Balancers → Create Load Balancer → Application Load Balancer**
2. **Basic Configuration**:
    - Name: my-alb
    - Scheme: Internet-facing
    - IP address type: IPv4
3. **Listeners**:
    - Default: HTTP (80)

2.3 Configure Availability Zones

- Select the deafault **VPC** where your EC2 instances exist
- Select **two /all subnets** (one for each instance if possible)

2.4 Configure Security Groups

- Choose the **lb-sg** security group created earlier

2.5 Configure Target Groups

1. **Create a new target group**:
    - Name: web-servers-tg
    - Target type: Instance
    - Protocol: HTTP
    - Port: 80
    - Health checks: / (default)
2. **Register Targets**:
    - Select EC2 INSTANCES HERE --- **webserver-1** and **webserver-2**
    - Click **Include as pending** → **Register**

2.6 Review and Create

- Review all settings → Click **Create Load Balancer**
- AWS will take a few minutes to provision the ALB

Step 3: Verify Load Balancer

1. Go to **Load Balancers → Select ALB → Description → DNS name**
2. Copy the **DNS name** (something like my-alb-123456.us-east-1.elb.amazonaws.com)
3. Paste it in a browser → You should see either **WebServer-1** or **WebServer-2** page.
    - Refresh multiple times → Load balancer distributes traffic between the two instances

---

Step 4: Optional – Test Health Checks

- Go to **Target Groups → web-servers-tg → Targets**
- Ensure **Status** of both instances is healthy

If a server is unhealthy, check the bootstrapping script or security group rules.

Step-by-Step: Auto Scaling Group with Load Balancer

Step 1: Create an AMI from Existing EC2 Instance

1. Go to **EC2 Console → Instances**
2. Select the EC2 instance you already configured with bootstrapping
3. Click **Actions → Image and Templates → Create Image**
4. **Provide Image Name**: e.g., my-webserver-AMI
5. Optional: Add description
6. Click **Create Image**
7. Wait for the AMI to become **available** (status: available in **AMIs** section)

✅ This AMI will be used for launching new instances automatically in ASG.

Step 2: Create a Launch Template

1. Go to **EC2 → instances->Launch Templates → Create Launch Template**
2. **Launch Template Name**: my-launch-template
3. **AMI**: Select the AMI created in Step 1 (my-webserver-AMI)
4. **Instance Type**: t2.micro (or your choice)
5. **Key Pair**: Select your existing key pair
6. **Security Groups**: Select the **Load Balancer Security Group** (allow HTTP 80)
7. **User Data**: Optional (already baked into AMI)
8. Click **Create Launch Template**

✅ Launch Template is ready for Auto Scaling.

Step 3: Create Auto Scaling Group (ASG)

1. Go to **EC2 → Auto Scaling Groups → Create Auto Scaling Group**
2. **Name**: webserver-asg
3. **Launch Template**: Select my-launch-template
4. **Template Version**: Select **Version 1**
5. Click **Next**

Step 3.1: Network Settings

1. **VPC**: Default VPC (or your existing VPC)
2. **Availability Zones (AZs)**: Select **all** subnets (for high availability)
3. **Load Balancer**: Enable **Attach to an existing load balancer**
   - Select your **existing Load Balancer**
4. **Health Check**: HTTP
   - Set **Health Check Grace Period**: 300 seconds
5. Click **Next**

Step 3.2: Configure Group Size

1. **Desired Capacity**: 2 (initial instances)
2. **Minimum Capacity**: 1
3. **Maximum Capacity**: 4
4. Click **Next**

Step 3.3: Review and Create ASG

1. Review all settings
2. Click **Create Auto Scaling Group**

✅ Auto Scaling Group is now ready and attached to your Load Balancer.

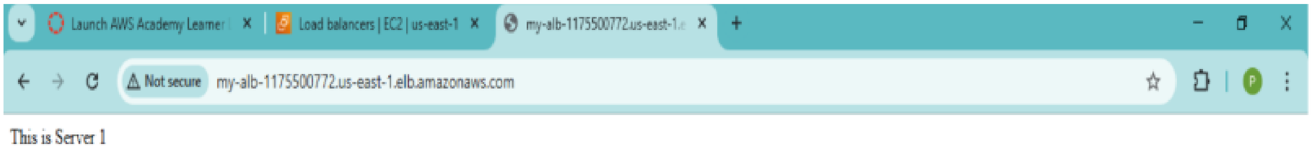
Step 4: Verify Load Balancer and Auto Scaling

1. Go to **Load Balancers → Select your Load Balancer**
2. Click **Instances → Target Group → Registered Targets**
   - You will see **2 healthy instances** initially
3. Check **Auto Scaling Group → Instances**
   - You may see up to **4 instances** if scaling triggers
4. Test **auto recovery**:
   - **Terminate one of the instances**
   - After a few seconds, **ASG will launch a new instance automatically**
   - Traffic will automatically be **distributed by the Load Balancer**

Step 5: Test the Load Balancer

1. Copy the **Load Balancer DNS Name** from ALB
2. Open in a browser → You should see your web page from one of the instances
3. Refresh multiple times → traffic alternates between instances

✅ This confirms **Auto Scaling + Load Balancer** is working correctly.



Summary of Flow

1. **Create AMI** → snapshot of working EC2
2. **Create Launch Template** → using AMI, security group
3. **Create Auto Scaling Group** → attach Launch Template, set min/desired/max, attach Load Balancer
4. **Verify instances** → ASG launches/replaces instances automatically
5. **Test Load Balancer** → distributes traffic between all active instances

---

Scenario Based Questions (ELB & Auto Scaling)

1. A company hosts a web application on two EC2 instances, but sometimes one server becomes overloaded while the other remains idle.

Which AWS service can distribute incoming traffic evenly across both servers?

A) Elastic Load Balancer

2. A developer created multiple EC2 instances for a web application but wants users to access the application using a single entry point instead of multiple server IP addresses.

Which AWS service can solve this problem?

A) Elastic Load Balancer

3. An organization notices that during peak hours their application receives heavy traffic and servers become slow. During non-peak hours, many servers remain unused.

Which AWS feature can automatically increase or decrease the number of servers based on demand?

A) Auto Scaling

4. A company deployed a load balancer but users are still getting errors because traffic is being sent to an unhealthy server.

Which load balancer feature helps detect and stop sending traffic to failed servers?

A) Health Checks

5. A developer wants to automatically launch additional EC2 instances when CPU utilization exceeds 70%.

Which AWS service configuration should be used to implement this requirement?

A) Auto Scaling Policy (Target Tracking / Scaling Policy)

6. An application receives traffic from users worldwide, and the company wants to ensure that requests are distributed across multiple EC2 instances for high availability.

How can Elastic Load Balancer help in this architecture?

A) By distributing traffic across multiple EC2 instances in multiple Availability Zones

7. A developer configured an Auto Scaling Group, but new instances are not launching even when traffic increases.

What configuration might be missing or incorrect?

A) Scaling Policy or CloudWatch Alarm not configured

8. A company wants to maintain a minimum of two running servers at all times, but automatically add more servers when traffic increases.

Which Auto Scaling configuration parameters should be defined?

A) Minimum Capacity, Desired Capacity, Maximum Capacity

9. A web application running on EC2 instances behind a load balancer suddenly crashes when traffic spikes.

How can combining Elastic Load Balancer with Auto Scaling improve system reliability?

A) ELB distributes traffic and Auto Scaling adds/removes instances automatically

10. A developer created an Auto Scaling Group but forgot to attach it to a load balancer.

What potential problem might occur in handling user requests?

A) Traffic will not be distributed and users may not reach new instances

11. A company wants to replace unhealthy EC2 instances automatically to maintain application availability.

Which AWS feature can automatically detect and replace failed instances?

A) Auto Scaling (Health Check Replacement)

12. A developer wants to monitor the performance of EC2 instances and trigger scaling actions based on metrics such as CPU usage or network traffic.

Which AWS service can provide these monitoring metrics?

A) Amazon CloudWatch

13. A startup expects sudden spikes in website traffic during a promotional event and wants the infrastructure to automatically handle the load.

How can Auto Scaling help manage this situation?

A) Automatically scales out during high traffic and scales in when traffic decreases

14. A developer wants to ensure that incoming user requests are routed to servers based on HTTP or HTTPS protocols.

Which type of AWS load balancer should be used?

A) Application Load Balancer

WEEK-10--Elasticbeanstalk

Prerequisites

- A web application ready (e.g., ROOT.war, .zip, or code)----**YOU CAN YOURS MAVEN PROJECT HERE**

Steps to Create Elastic Beanstalk Application

Step 1: Open Elastic Beanstalk

1. Login to AWS Console
2. Under Compute section ->Search for **Elastic Beanstalk**
3. Click **Create Application**

Step 2: Create Application

1. Enter **Application Name** (e.g., MyApp)
2. Add description (optional)
3. Click **Create**

Step 3: Create Environment

1. Click **Create Environment**
2. Choose:
    - **Web Server Environment** (for web apps)
3. Click **Select**

Step 4: Configure Environment

1. Enter **Environment Name**

2. Choose **Platform**:
   - Java (for .war)
   - Node.js / Python / PHP / etc.
3. Under **Application Code**:
   - Select **Upload your code**
   - Upload your .war or .zip file

Step 5: Configure Service Access

1. Service Role → Select existing -→**LabRole**
2. EC2 Key Pair → Optional (for SSH)
3. EC2 Instance Profile → select (e.g., LabInstanceProfile)

Step 6: Configure Network

1. Select **VPC** (default is fine)
2. Select **Subnets**
3. Enable **Public IP**

Step 7: Instance & Scaling Setup

1. Choose **Instance type** (e.g., t2.micro – free tier)
2. Set:
   - Min instances: 1
   - Max instances: 2 (or more)
3. Load Balancer → Enabled (optional but recommended)

Step 8: Review & Create

1. Click **Review**
2. Click **Create**

Deployment Process

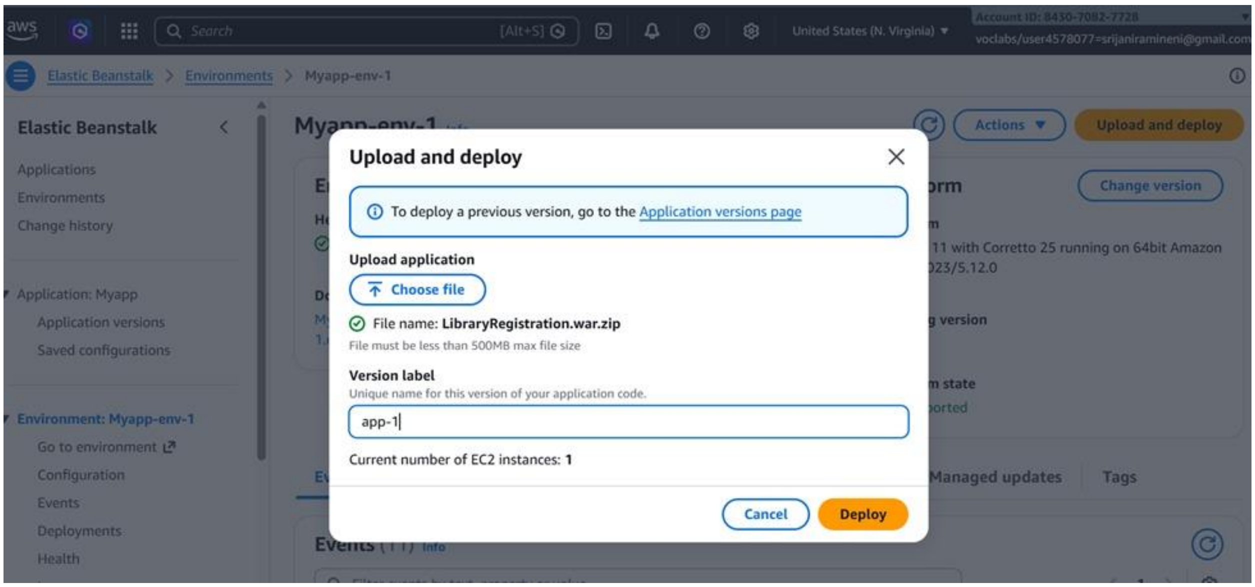
- AWS automatically creates:
  - EC2 instances
  - Load Balancer
  - Auto Scaling Group
  - Security Groups

Access Your Application

- After deployment:
  - You will get a **URL (domain link)**
  - Open it in browser → Your app will run

-

Updating Application

1. Go to your environment
2. Click **Upload and Deploy**
3. Upload new version
4. 

SCENARIO BASED QUESTIONS

Scenario-Based Questions

Scenario 1: Application Not Loading

Q1. What could be the possible reasons?

- Application crashed or not started
- Wrong port or missing entry file

Q2. Could deployment or configuration be incorrect?
Yes. Wrong platform or incorrect packaging can prevent the app from running.

Q3. How would you troubleshoot this problem?
Check **Events** for failure messages and logs like eb-engine.log to identify startup errors.

Scenario 2: Incorrect Application Structure

Q1. How does application structure affect deployment?
Elastic Beanstalk requires files at the **root level**; incorrect structure prevents detection of the app.

Q2. What happens if files are inside an extra folder?
EB cannot find the entry file, causing deployment failure or blank output.

Q3. How would you fix the structure issue?
Move required files (e.g., .war, app.js) to the root and repackage the ZIP correctly.

Scenario 3: Wrong Platform Selection

Q1. Why is platform selection important?
Platform defines runtime (Java, Node, Python) and how the app is executed.

Q2. What errors might occur?
App won't start or shows runtime errors due to mismatch.

Q3. How can you correct the platform?
Recreate the environment with the correct platform (e.g., Tomcat for .war).

Scenario 4: Environment Health Warning

Q1. Possible reasons for warning?

- Failed health checks
- High response time or errors

Q2. Which sections to check first?
Events, Logs, and Metrics.

Q3. How to resolve?
Fix health check URL, reduce errors, and optimize performance.

Scenario 5: Application Errors in Logs

Q1. What do such errors indicate?
The application is failing during execution.

Q2. Could misconfiguration cause this?
Yes, missing environment variables or resources can break the app.

Q3. How to analyze logs?
Identify the first error, read stack trace, and trace the failing component.

Scenario 6: Deployment Failure

Q1. What could be wrong with the package?
Missing entry file or incorrect ZIP structure.

Q2. Which logs to check?
eb-engine.log and Events tab.

Q3. How do packaging errors affect deployment?
EB cannot start the app, leading to complete failure.

Scenario 7: Works Locally but Not on Cloud

Q1. What environment differences cause this?
OS differences, ports, or file paths.

Q2. Could dependencies/config be responsible?
Yes, missing dependencies or incorrect configs.

Q3. How to debug?
Compare local vs cloud setup and analyze logs.

Scenario 8: Blank Output

Q1. What causes no output?
Routes not defined or frontend not loading.

Q2. Could runtime/config issues be responsible?
Yes, incorrect responses or missing files.

Q3. How do logs help?
Logs show request handling and response errors.

Scenario 9: Application URL Not Accessible

Q1. Could environment still be launching?
Yes, during initialization phase.

Q2. What status indicators to check?
Environment status and health (Pending, Green).

Q3. How can Events help?
They show deployment progress and failures.

Scenario 10: Resource Management Issue

Q1. Cost implications?
Charges continue for running resources.

Q2. Which services run in background?
EC2, Load Balancer, Auto Scaling.

Q3. How to terminate resources?
Delete environment and verify all services are stopped.

Scenario 11: High Traffic Handling

Q1. How does EB handle scaling?
Using Auto Scaling Groups.

Q2. What configurations are needed?
Scaling policies and load balancer.

Q3. What if scaling is not configured?
App slows down or crashes under load.

Scenario 12: Application Update Not Reflected

Q1. Could deployment issues be the reason?
Yes, old version may still be active.

Q2. How can caching affect output?
Browser or CDN may serve old content.

Q3. How to ensure latest version?
Redeploy properly and clear cache.

WEEK 11—Amazon LEX

Step 1: Open Amazon Lex

1. Login to AWS Console
2. Search **Amazon Lex**
3. Click **Create bot**

Step 2: Create Bot

1. Choose:
   - **Create a blank bot**
2. Enter:
   - Bot name: HotelBookingBot
3. IAM role → Create new role
4. Remaining default→next
5. Language: English
6. Voice (optional)
7. Click **Done**

Step 3: Create Intent

1. Go to **Intents**
2. Name: BookHotel
3. Click **Create**

Step 4: Add Sample Utterances

Add user sentences (how user talks):

I want to book a hotel

Book a room

Reserve hotel

I need a room

These are **Utterances**

Step 5: Create Slots (User Inputs)

Slots collect user information.

Add Slot

1. Click **Add slot**
2. Fill:
   - Slot name: age

- Slot type: AMAZON.Number
- Prompt: *Which city do you want?*

Mark as **Required**

Add IF Conditions (Conditional Logic)

Used to control flow based on slot values

Example Condition:

If {age} < 18

Steps:

1. Go to **Intent → Slots -> under Advance options**
2. Scroll to **Slot capture: success response ->Conditional branching**
3. Click **Add condition**

Example:

Condition:

If {age} < 18

Response:

Your not eligible for hoot booking?

Step 6: Create Slots (User Inputs)

Slots collect user information.

Add Slot

3. Click **Add slot**
4. Fill:
    - Slot name: location
    - Slot type: AMAZON.City
    - Prompt: *Which city do you want?*
5. Mark as **Required**

Repeat for other slots.

1. Click **Add slot**
2. Fill details:
    - **Slot name:** checkin
    - **Slot type:** AMAZON.Date
    - **Prompt:**
      "What is your check-in date?"
3. Mark as **Required**

   (Optional but recommended)
    - Add **Retry prompt**:
      "Please provide a valid date (e.g., 2026-03-25)"

4. Click **Save**

Slot 7: Number of Nights (nights)

Steps:

1. Click **Add slot** again
2. Fill details:
    - **Slot name:** nights
    - **Slot type:** AMAZON.Number
    - **Prompt:**
      "How many nights will you stay?"
3. Mark as **Required**
4. Click **Save**

Step 7: Create Custom Slot Type

Save intent

Click on Back to intents

If you want custom values:

1. Go to **Slot Types**
2. Click **Add slot type ->add blank slot type**
3. Name: RoomType

Add values:

Single

Double

Suite

Save slot type

Now use this in slot: go back to intent

- Slot name: RoomType
- Slot type: RoomType

Step 9: Add Response Cards (Card Groups)

Used for buttons (quick replies)

Steps:

1. Go to **slot prompts -> more prompt options**
2. Click **Add -> add card groups**

Example:

Card Title: Select Room Type

click on add Buttons:

- Single → "Single"
- Double → "Double"
- Suite → "Suite"

Step 10: Configure Responses

Initial Response:

Welcome to Hotel Booking! What is your name?

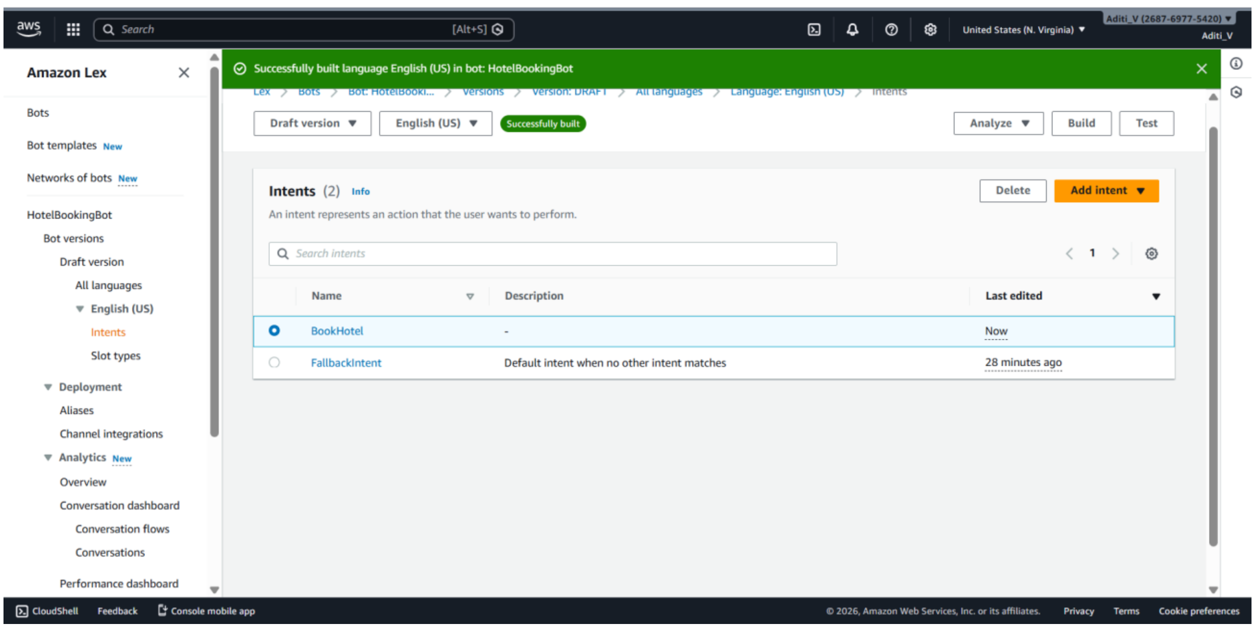
Confirmation:

Do you want to confirm booking in {location} for {nights} nights?

Step 11: Build & Test

1. Click **Build**
2. Use **Test chatbot** panel
3. Try:

Book a hotel



Sample output

User: Book a hotel

Bot: What is your age?

User: 25

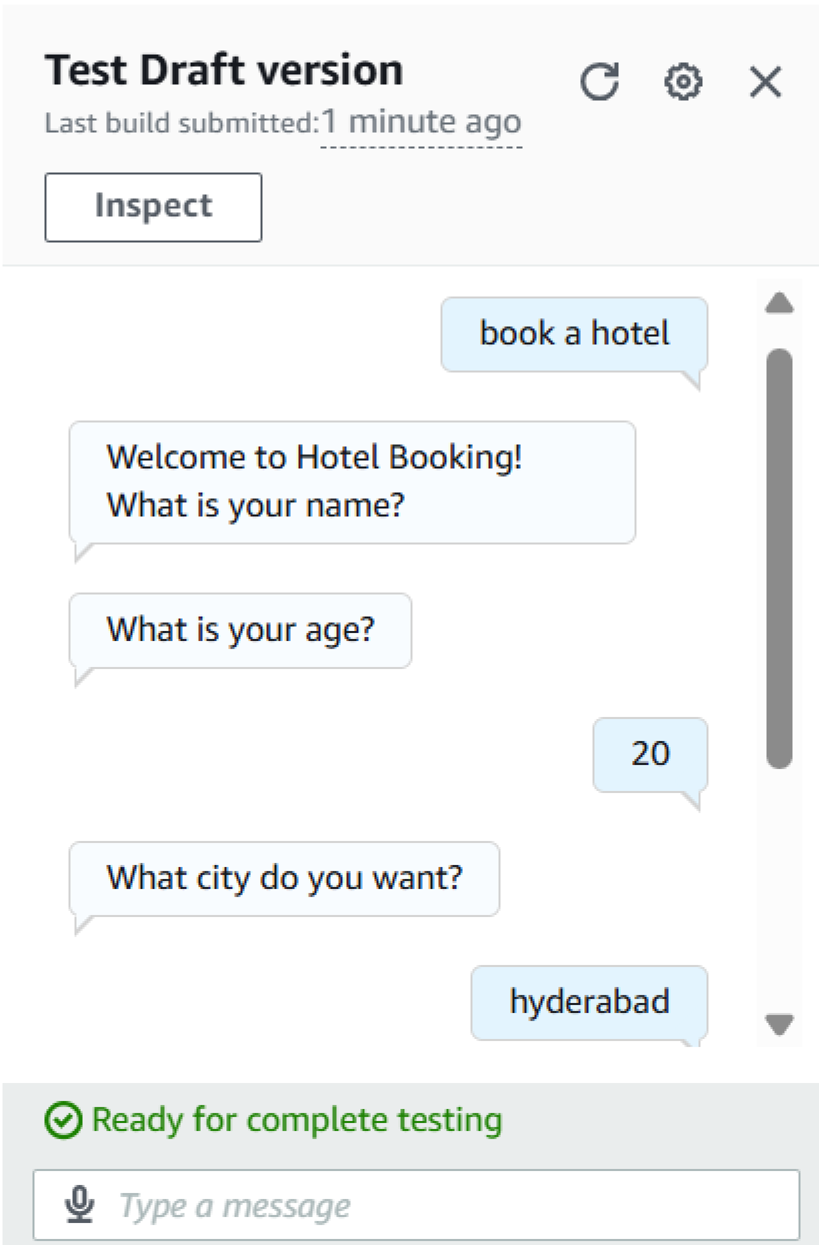
Bot: Which city do you want?

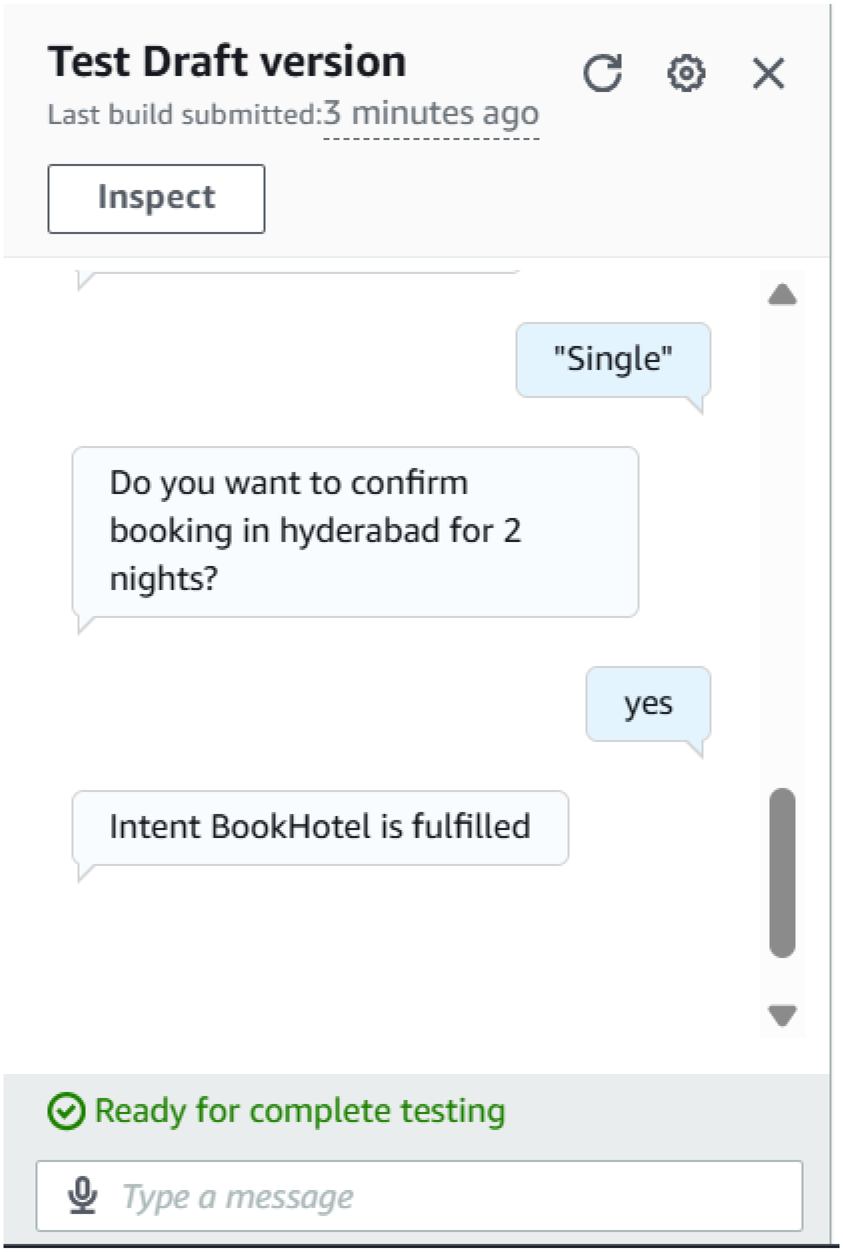
User: Hyderabad

Bot: Check-in date?

User: Tomorrow

Bot: Number of nights?
User: 2
Bot: Select room type
User: Double
Bot: Do you want to confirm booking in Hyderabad for 2 nights?
User: Yes
Bot: Booking confirmed.





SCENARIO BASED QUESTIONS

Scenario 1: Bot Not Responding

You created a chatbot, but when you test it, the bot does not respond to user input.

- What could be the possible reason for no response?
- Did you build and publish the bot correctly?

- How would you troubleshoot this issue?

Bot may not respond if it is not **built/published**, has no matching intent, or configuration errors—troubleshoot by building, checking intents, and logs in Amazon CloudWatch.

Scenario 2: Incorrect Intent Matching

A user types a valid query, but the chatbot responds with the wrong intent.

- What could cause incorrect intent recognition?
- How do utterances affect intent matching?
- How can you improve accuracy?

Wrong intent occurs due to poor or overlapping utterances, so improve accuracy by adding diverse and clear training phrases in Amazon Lex.

Scenario 3: Missing Slot Values

The chatbot asks for user input, but fails to capture required slot values.

- What is a slot in Amazon Lex?
- Why might slot values not be captured properly?
- How would you fix slot configuration issues?

A slot is a user input variable, and values may fail due to incorrect configuration, so fix by marking slots required and using proper slot types.

Scenario 4: Fulfillment Not Working

The bot collects all user inputs but does not perform the expected action.

- What is fulfillment in Amazon Lex?
- Could Lambda integration be misconfigured?
- How would you verify the backend processing?

Fulfillment is the backend action after data collection, and it may fail if AWS Lambda is misconfigured, so verify integration and logs.

Scenario 5: Bot Not Accessible

You created the bot successfully, but users are unable to access it.

- Did you deploy or publish the bot?
- Could permissions or channel integration be missing?
- How would you make the bot accessible to users?

The bot may not be accessible if not published or lacks permissions, so deploy it and configure alias and access settings.

Scenario 6: Wrong Language Understanding

The chatbot fails to understand user queries written in slightly different formats.

- How does Amazon Lex process natural language?
- Why is training with multiple utterances important?
- How can you improve NLP performance?

Lex uses NLP to map input to intent, and accuracy improves by training with multiple varied utterances.

Scenario 7: Versioning Issue

You updated the bot, but users still experience the old behavior.

- Did you create a new version of the bot?
- Why is alias configuration important?
- How do you ensure users access the latest version?

Users see old behavior if alias points to an old version, so update alias to the latest bot version.

Scenario 8: Lambda Permission Error

The chatbot is connected to a Lambda function, but it fails during execution.

- What type of permission error might occur?
- How are IAM roles used in this integration?
- How would you fix the permission issue?

Permission errors occur when IAM roles lack access, so grant required permissions using AWS Identity and Access Management.

Scenario 9: Bot Gives Default Response

The chatbot always gives a fallback/default response instead of proper answers.

- What is a fallback intent?
- Why does the bot fail to match user input?
- How can you improve intent recognition?

Fallback intent triggers when no intent matches, so improve recognition by adding better utterances.

Scenario 10: Multi-turn Conversation Issue

The chatbot fails to continue conversation properly after asking follow-up questions.

- What is dialog management in Amazon Lex?
- How are slots used in multi-turn conversations?
- How can conversation flow be improved?

Dialog management controls conversation flow, and issues occur if slots are not properly configured or sequenced.

Scenario 11: Integration with Web Application

A company wants to integrate the chatbot into their website.

- How can Amazon Lex be integrated with web applications?
- What services or APIs are required?
- What challenges might occur during integration?

- Amazon Lex can be integrated using APIs and SDKs, but challenges include authentication and UI handling.

Scenario 12: Performance Issue

The chatbot responds slowly when many users interact simultaneously.

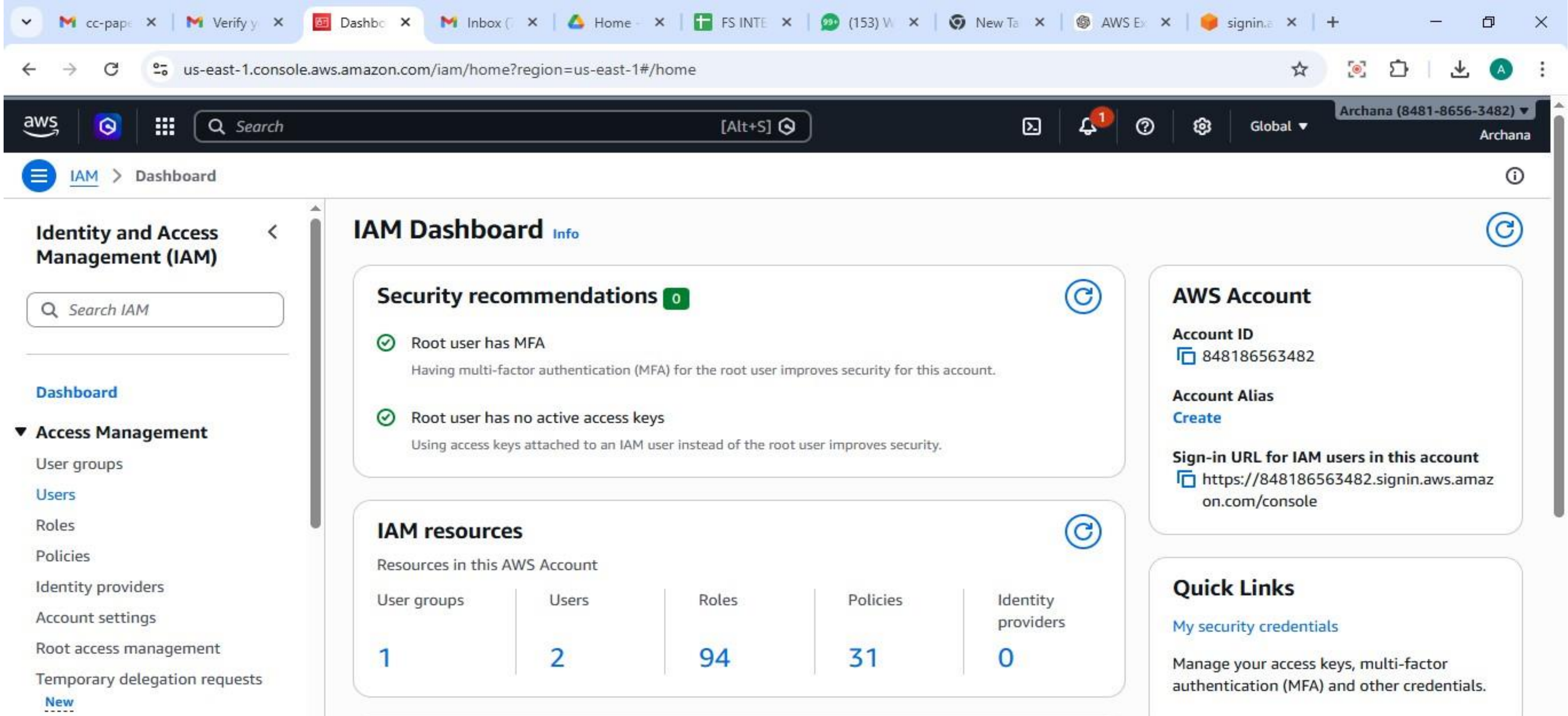
- What could cause performance delays?
- How can scalability be handled in Amazon Lex?
- What AWS services can support high traffic handling?

Delays occur due to high load or slow backend, and scalability is handled using services like AWS Lambda and API Gateway.

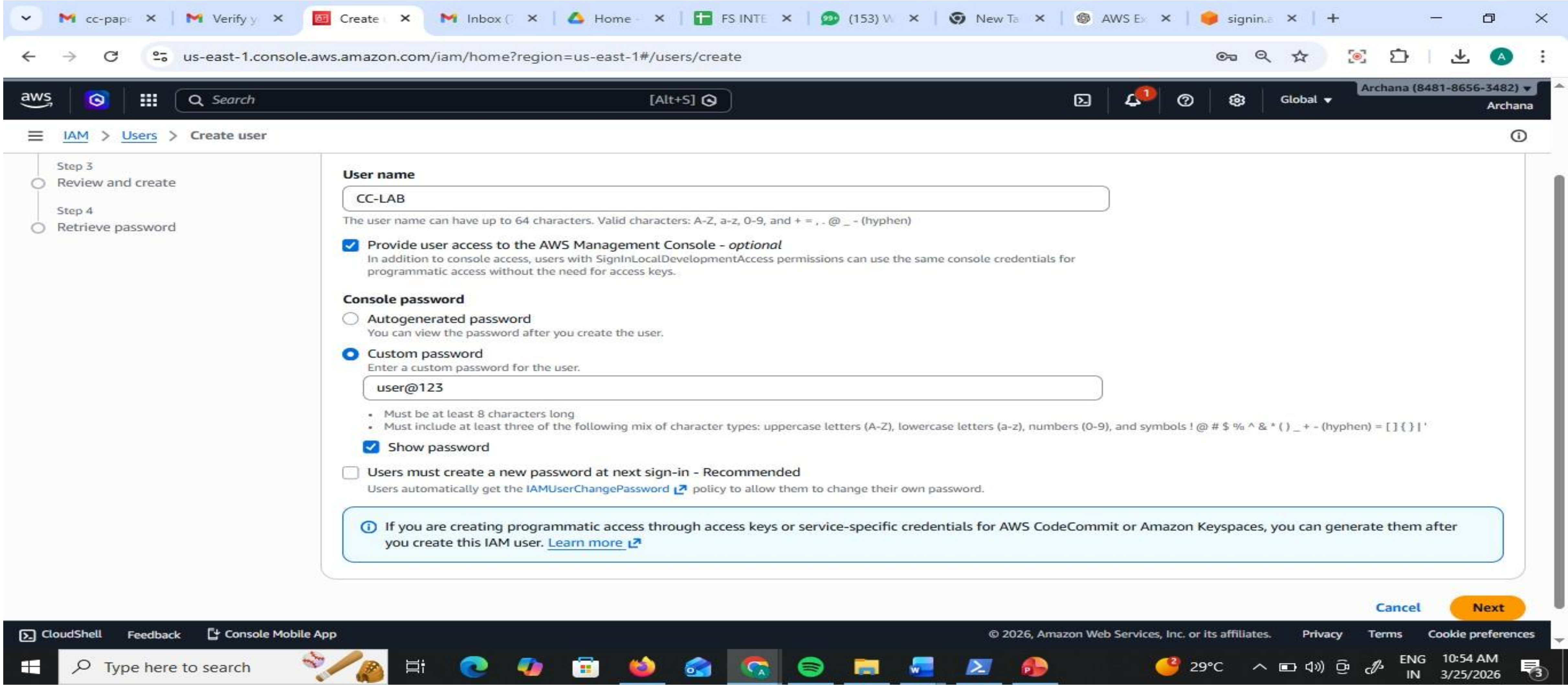
WEEK-12
IAM

1. Administrative Setup (Root User)

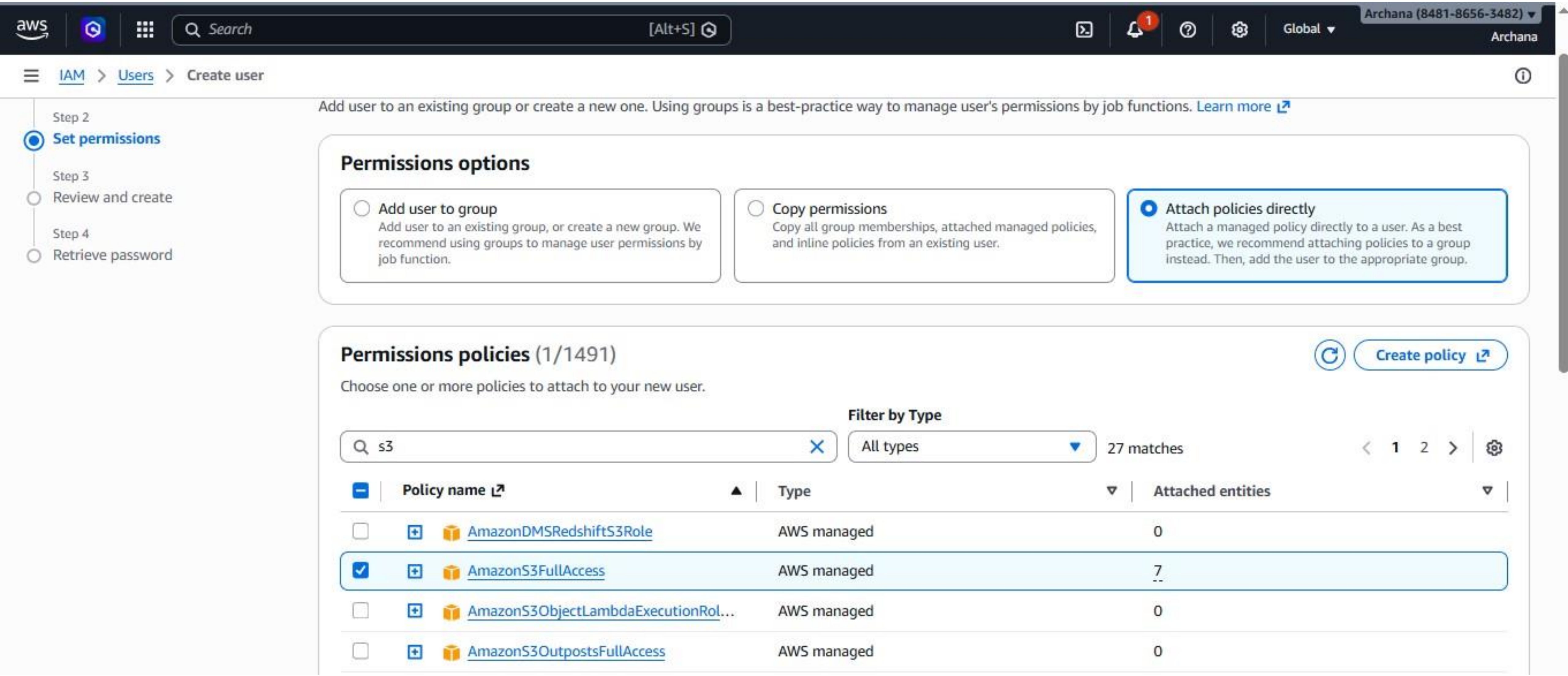
Step 1: Log in to the AWS Management Console as the Root User.
Step 2: Search for IAM under the "Security, Identity, & Compliance" category.
Step 3: Navigate to Users and click Create user.



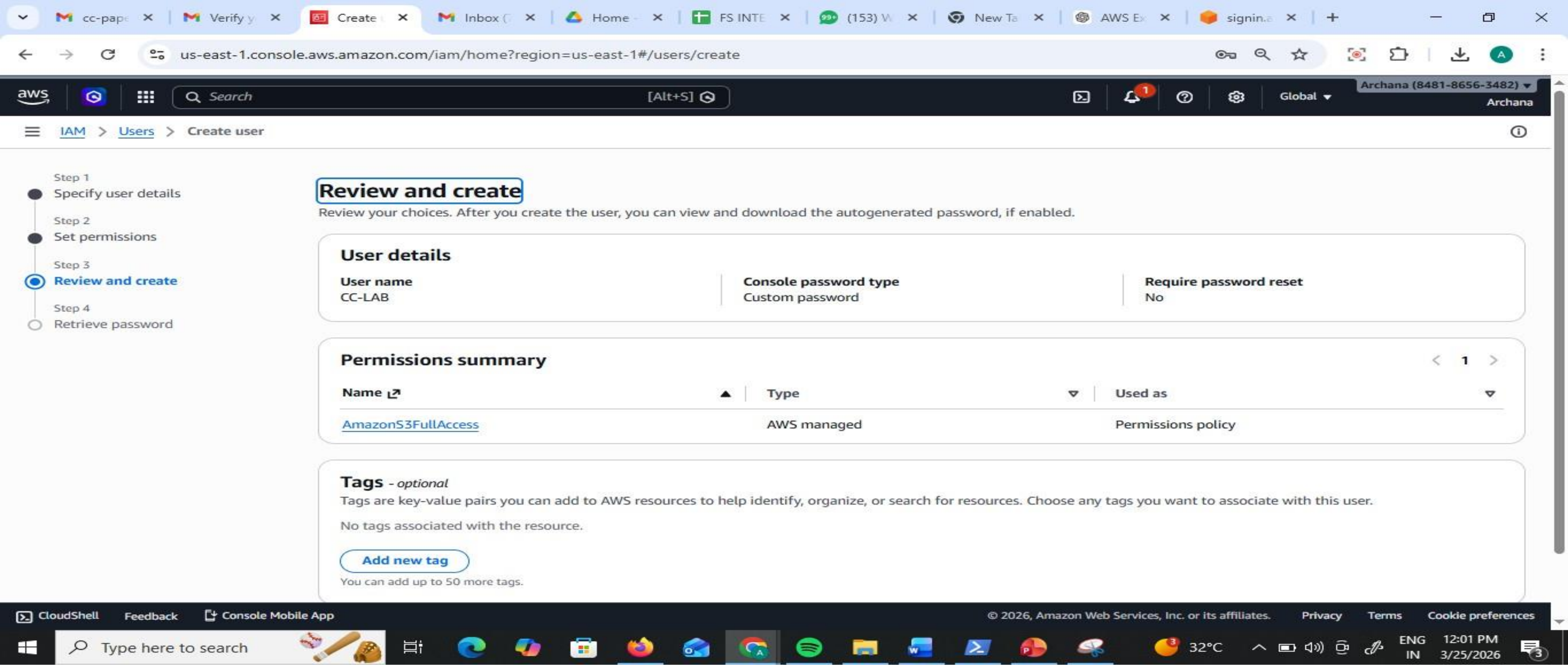
Step 4: User Details: Enter a name (e.g., S3_Specialist). Check the box for "Provide user access to the AWS Management Console." – custom password.



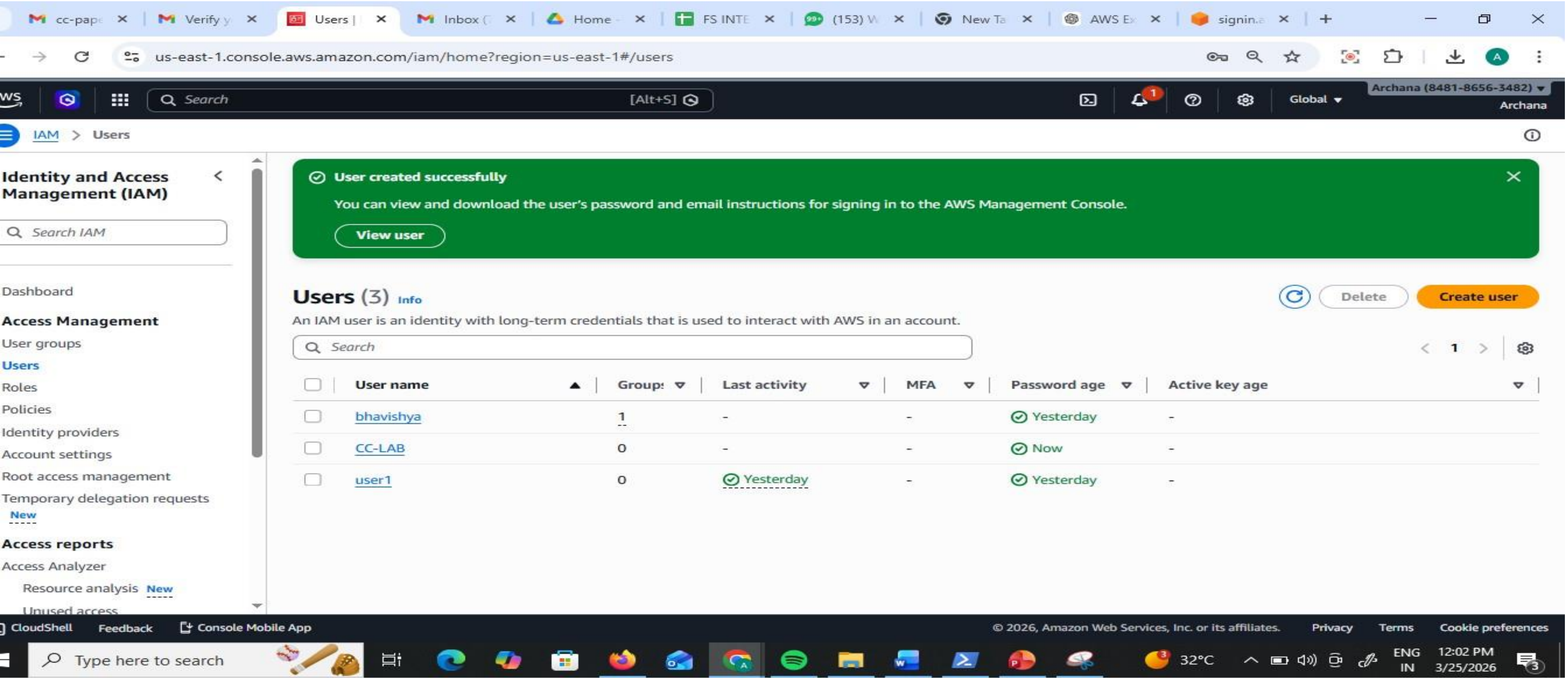
Step 5: Set Permissions: Select "Attach policies directly." Search for and select AmazonS3FullAccess.



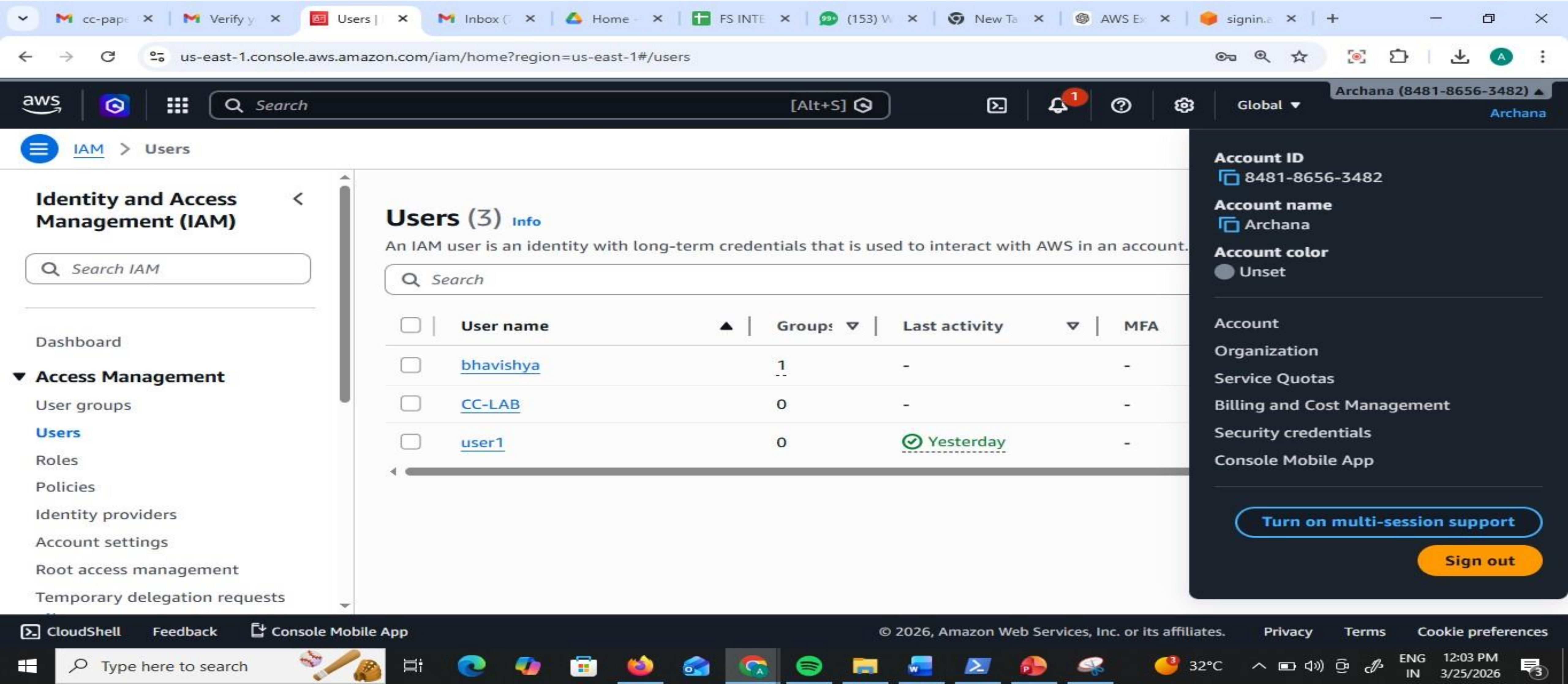
Step 6: Review & Create: Complete the wizard. On the final page, click Download .csv file.



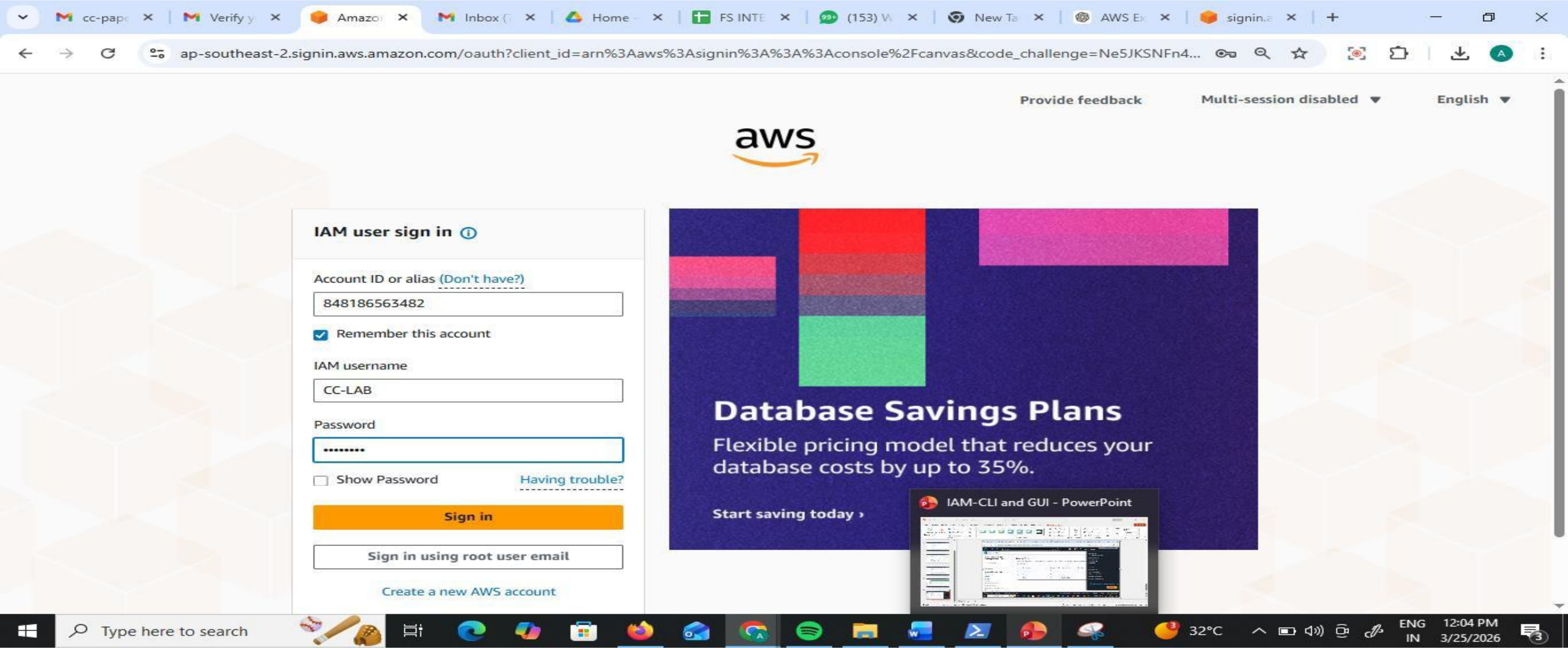
Now you see CC-LAB user created



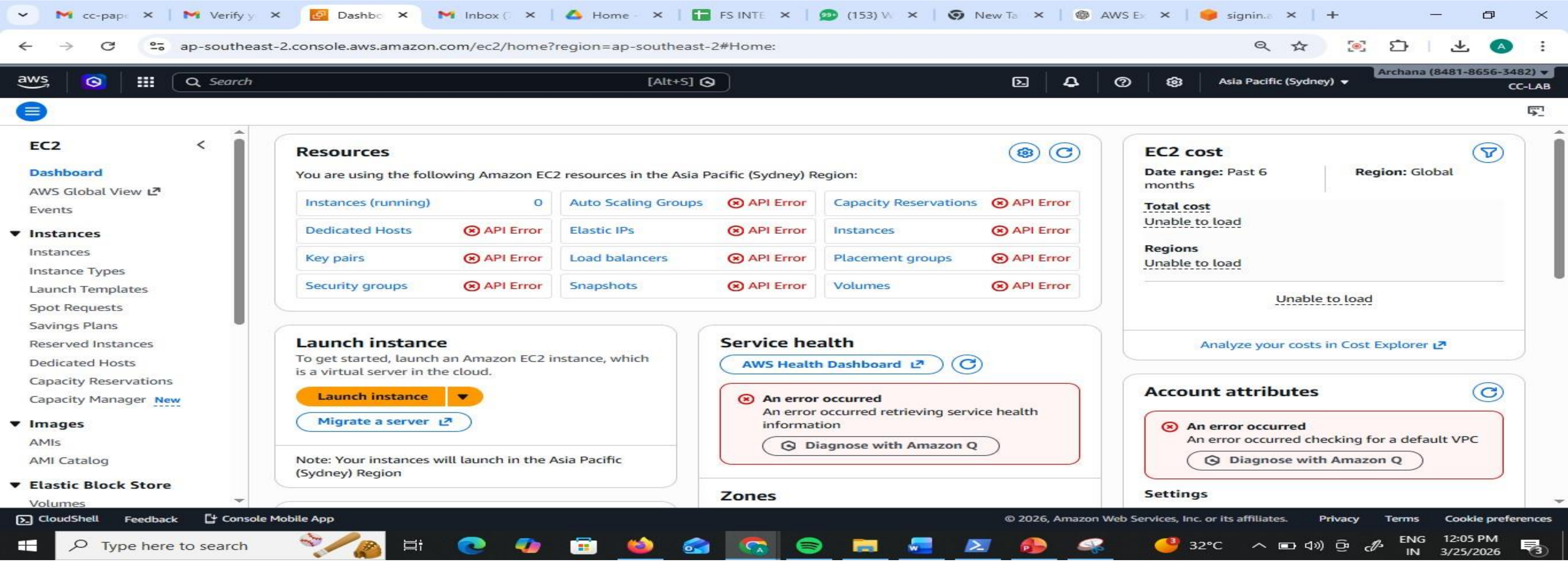
Step 7: Copy your 12-digit Account ID (found in the top-right dropdown or the .csv) and Sign Out.



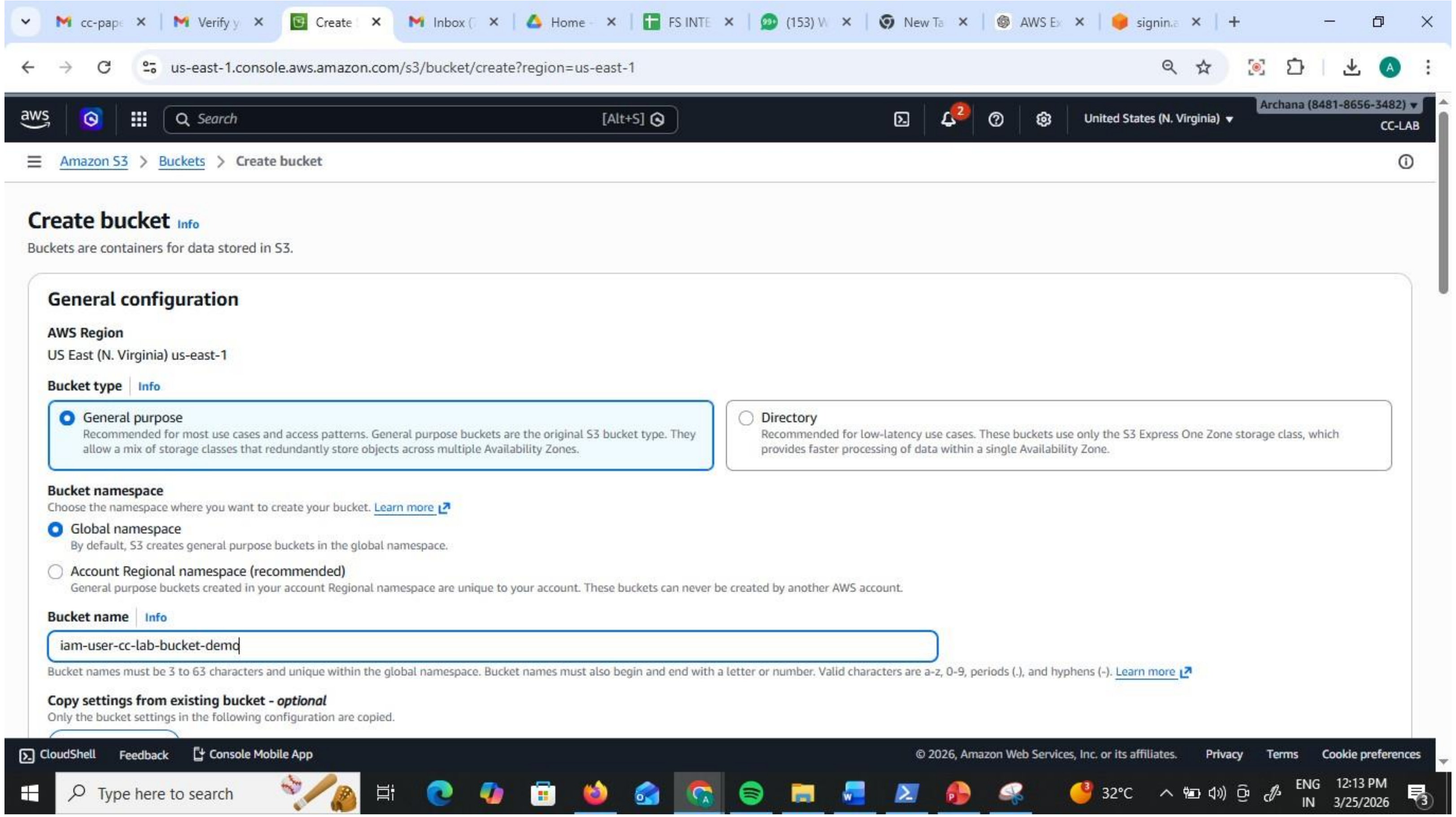
Step 8: Use the Sign-in URL from the .csv (or go to the AWS sign-in page and select IAM User).
Step 9: Enter the Account ID, then the IAM Username and Password from the .csv.



Test 1 (Failure): Navigate to **EC2**. Try to view instances. You will see "API Error" or "Access Denied." This proves the user is restricted.



Test 2 (Success): Navigate to **S3**. Create a new bucket. It will work perfectly because of the attached policy.



us-east-1.console.aws.amazon.com/s3/buckets/iam-user-cc-lab-bucket-demo?region=us-east-1&tab=objects

Amazon S3 > Buckets > iam-user-cc-lab-bucket-demo

Successfully created bucket "iam-user-cc-lab-bucket-demo"

iam-user-cc-lab-bucket-demo Info

Objects | Metadata | Properties | Permissions | Metrics | Management | Access Points

Objects (0)

Copy S3 URI | Copy URL | Download | Open | Delete | Actions | Create folder | Upload

Objects are the fundamental entities stored in Amazon S3. You can use Amazon S3 inventory to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. Learn more

Find objects by prefix

Name	Type	Last modified	Size	Storage class

No objects

You don't have any objects in this bucket.

Upload

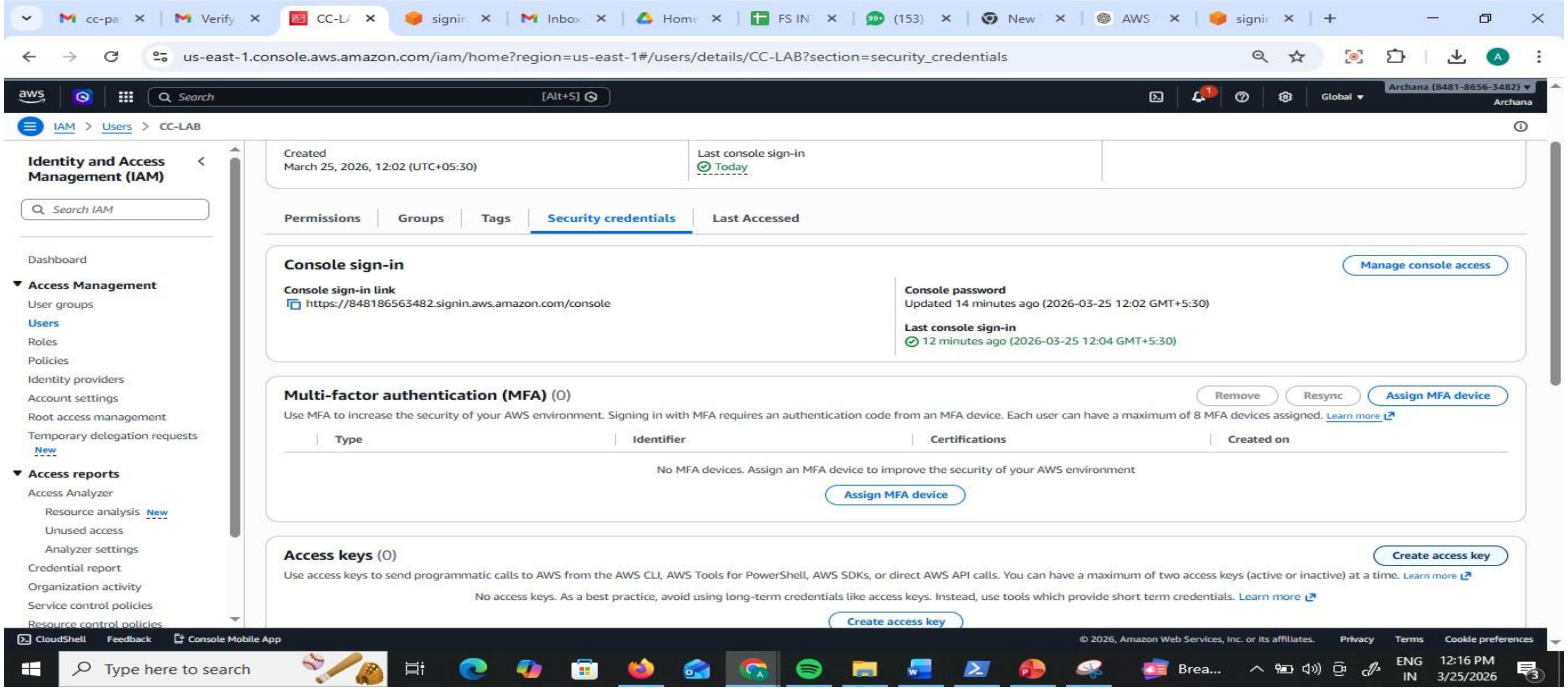
Part B: CLI Access (Programmatic Access)

Objective: Access AWS resources from a standalone system (CMD/Terminal) using security keys.

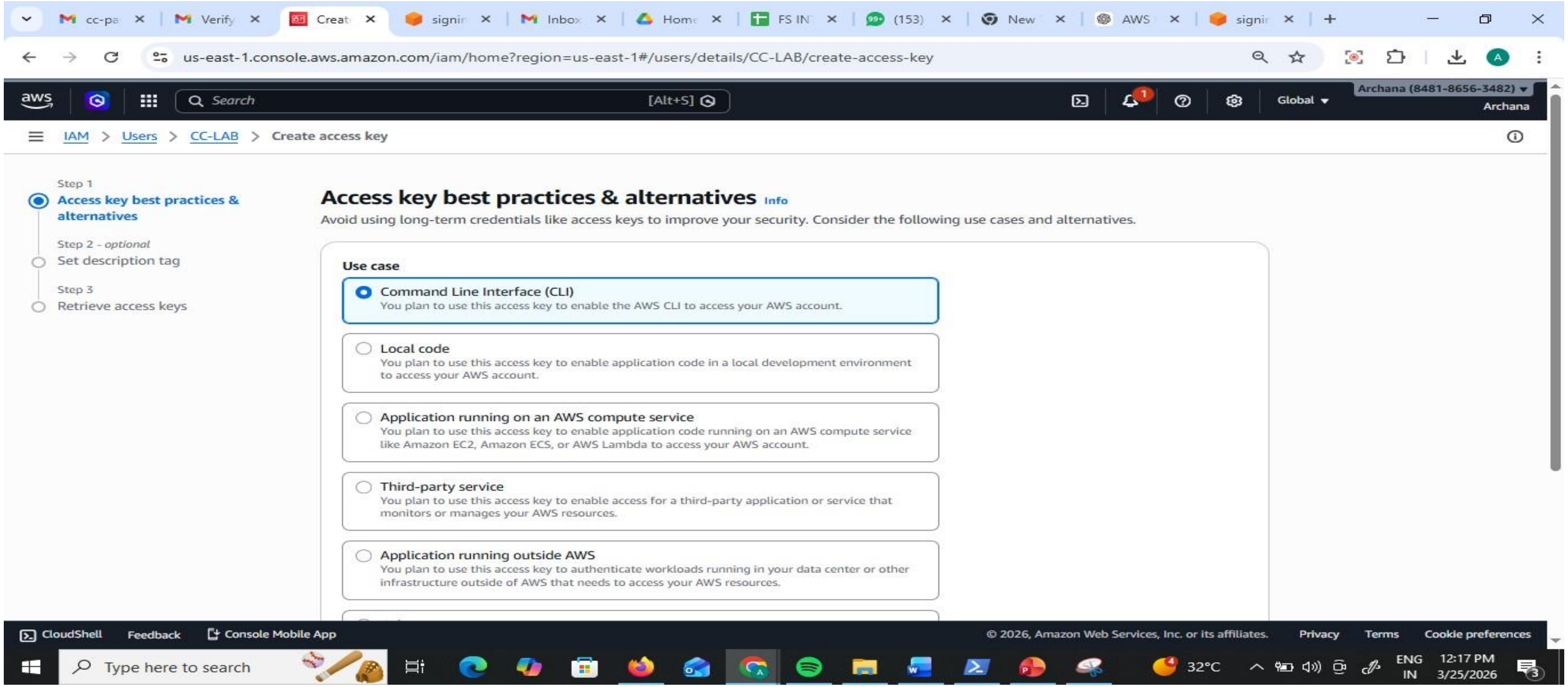
1. Credential Generation

Step 1: While logged in as an Admin/Root, go to IAM > Users and click on your user.

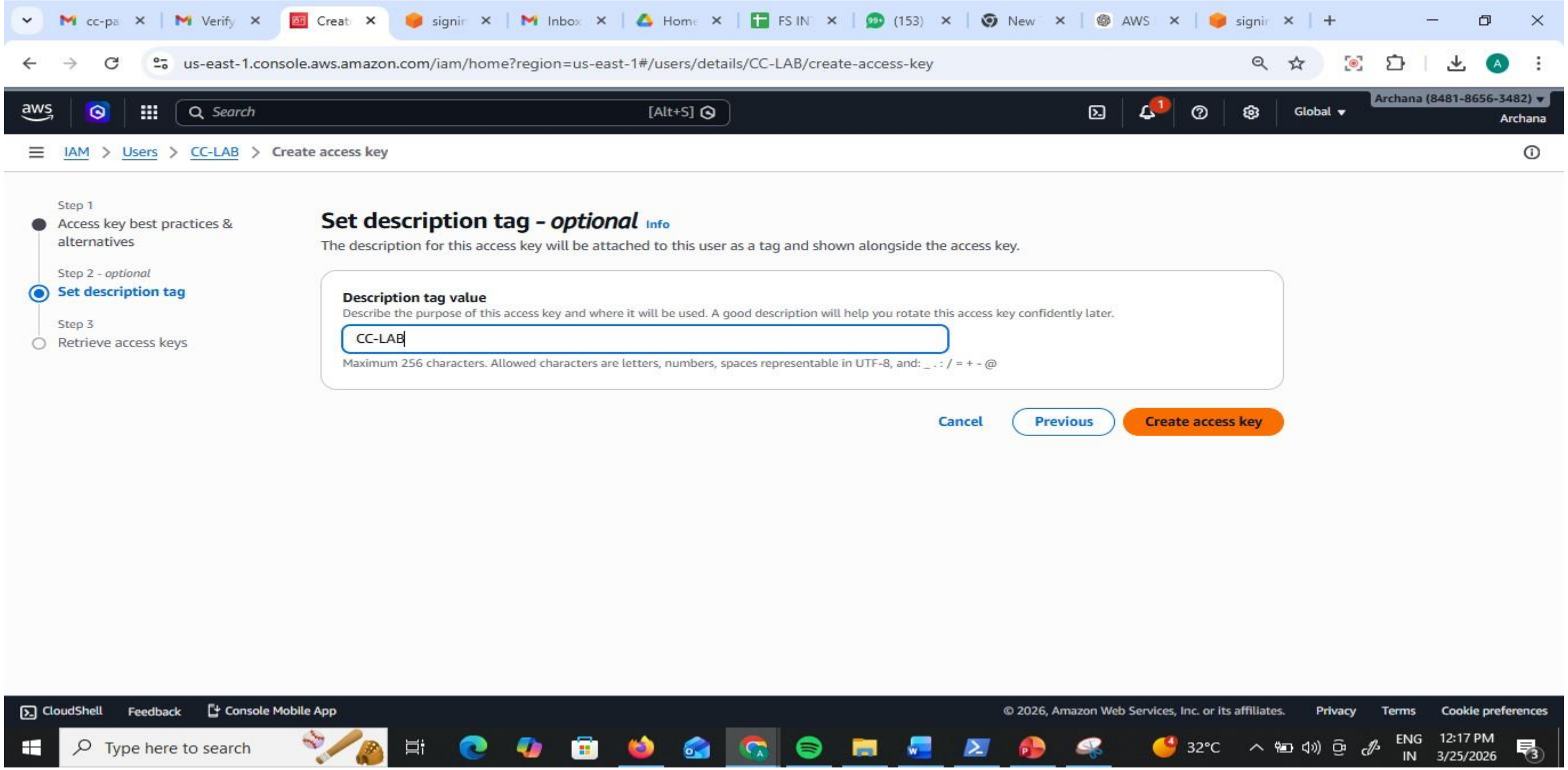
Step 2: Go to the Security credentials tab. Scroll to Access keys and click Create access key.

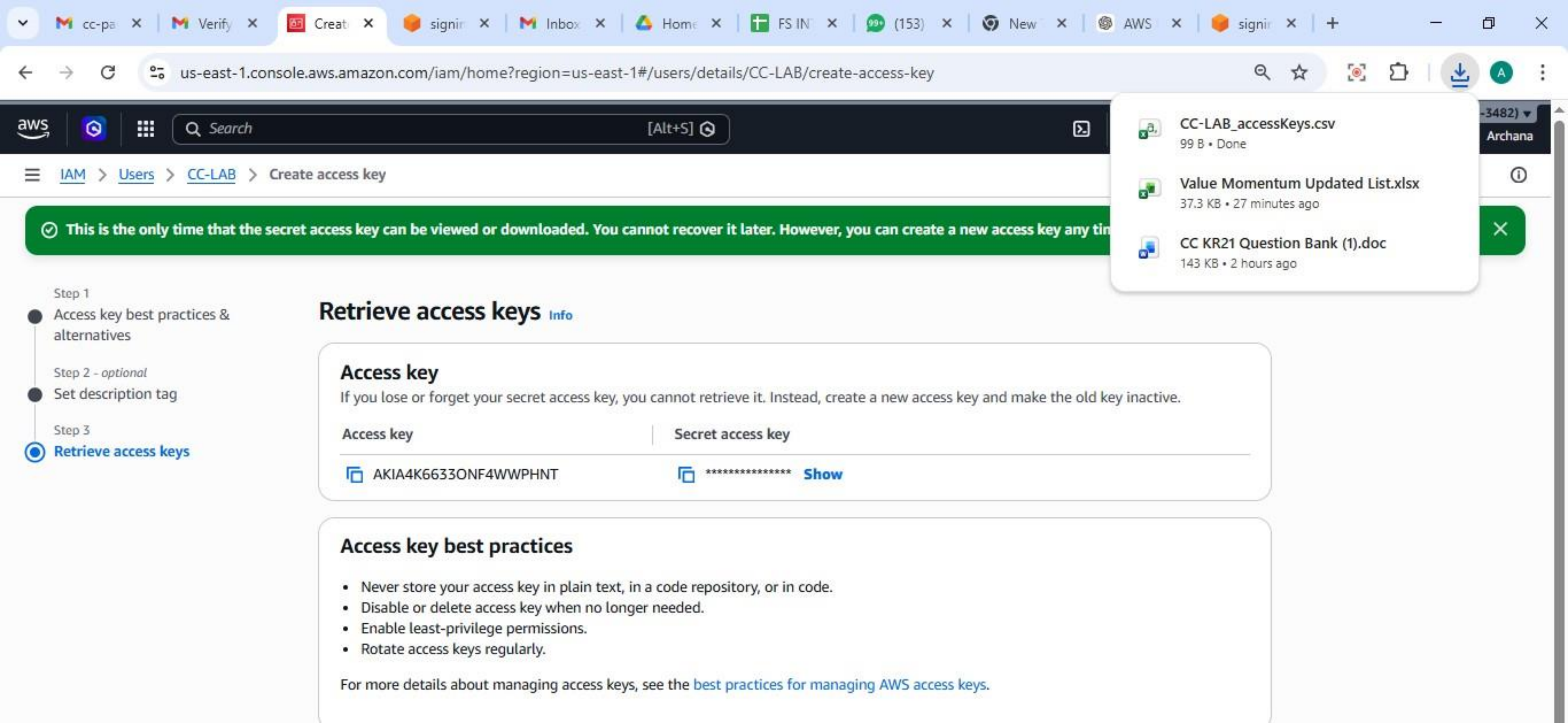


Step 3: Select Command Line Interface (CLI). Acknowledge the warning and click Next.



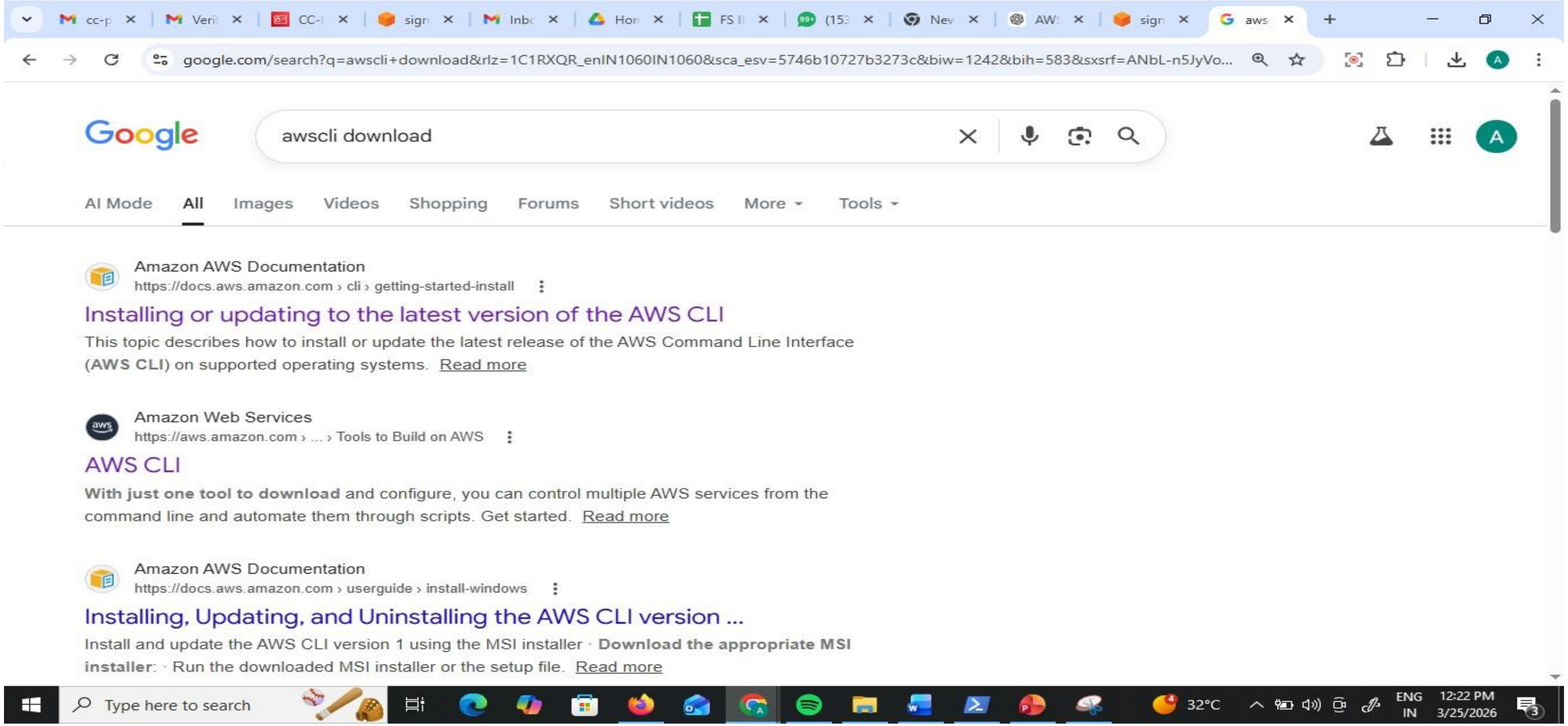
Step 4: Download the .csv file. > Critical: This contains the Access Key ID and Secret Access Key. If lost, you must delete and recreate them.




cc-pa
Verify
Creat
signir
Inbox
Home
FS IN
(153)
New
AWS
signir
us-east-1.console.aws.amazon.com/iam/home?region=us-east-1#/users/details/CC-LAB/create-access-key
aws
Search
[Alt+S]
-3482)
Archana
CC-LAB_accessKeys.csv
99 B • Done
Value Momentum Updated List.xlsx
37.3 KB • 27 minutes ago
CC KR21 Question Bank (1).doc
143 KB • 2 hours ago
IAM > Users > CC-LAB > Create access key
This is the only time that the secret access key can be viewed or downloaded. You cannot recover it later. However, you can create a new access key any tin
Step 1
Access key best practices & alternatives
Step 2 - optional
Set description tag
Step 3
Retrieve access keys
Retrieve access keys Info
Access key
If you lose or forget your secret access key, you cannot retrieve it. Instead, create a new access key and make the old key inactive.
Access key
Secret access key
AKIA4K6633ONF4WWPHNT
*************** Show
Access key best practices
Never store your access key in plain text, in a code repository, or in code.
Disable or delete access key when no longer needed.
Enable least-privilege permissions.
Rotate access keys regularly.
For more details about managing access keys, see the best practices for managing AWS access keys.

2. Standalone System Configuration

Step 5: Install the AWS CLI Tool from google on your computer



For installation instructions, expand the section for your operating system.

▶ **Linux**

▶ **macOS**

▶ **Windows**

Troubleshooting AWS CLI install and uninstall errors

If you come across issues after installing or uninstalling the AWS CLI, see Troubleshooting errors for the AWS CLI for troubleshooting steps. For the most relevant troubleshooting steps, see Command not found errors, The "aws --version" command returns a different version than you installed, and The "aws --version" command returns a version after uninstalling the AWS CLI.

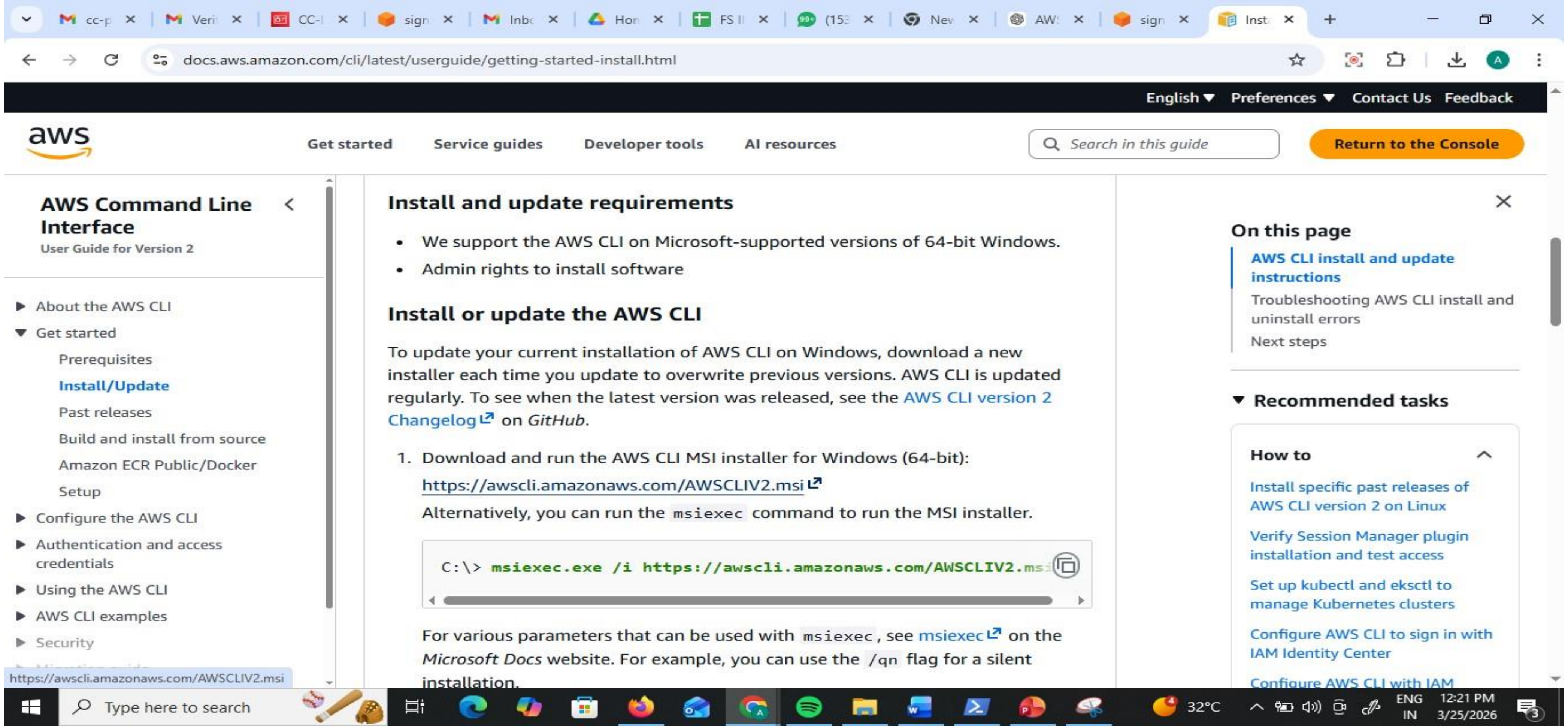
On this page
- AWS CLI install and update instructions
- Troubleshooting AWS CLI install and uninstall errors
- Next steps

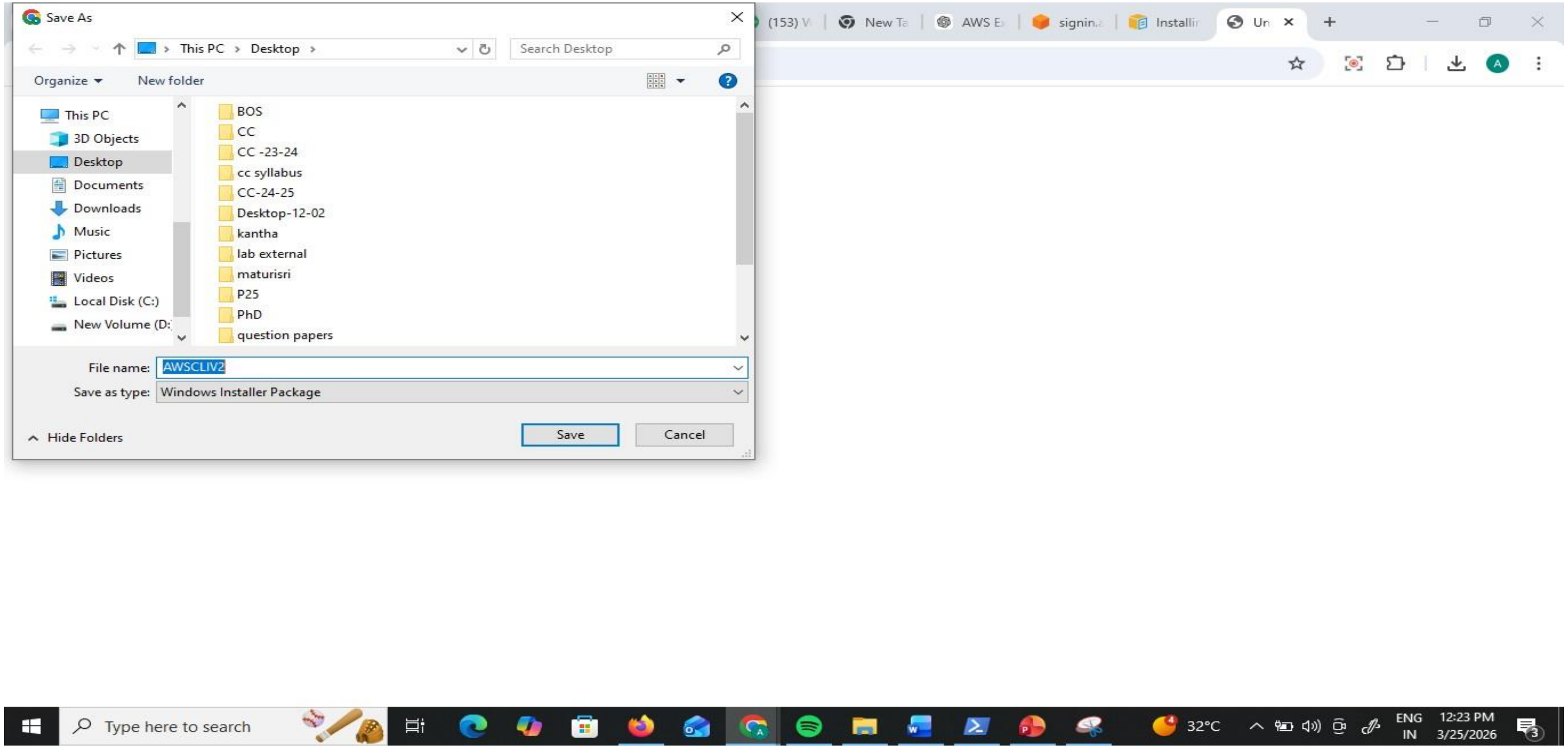
▼ Recommended tasks

How to
- Install specific past releases of AWS CLI version 2 on Linux
- Verify Session Manager plugin installation and test access
- Set up kubectl and eksctl to manage Kubernetes clusters
- Configure AWS CLI to sign in with IAM Identity Center
- Configure AWS CLI with IAM

Download

https://awscli.amazonaws.com/AWSCLIV2.msi




Save As
This PC
Desktop
Search Desktop
Organize
New folder
This PC
3D Objects
Desktop
Documents
Downloads
Music
Pictures
Videos
Local Disk (C:)
New Volume (D:
BOS
CC
CC -23-24
cc syllabus
CC-24-25
Desktop-12-02
kantha
lab external
maturisri
P25
PhD
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File name:
AWSCLIV2
Save as type:
Windows Installer Package
Hide Folders
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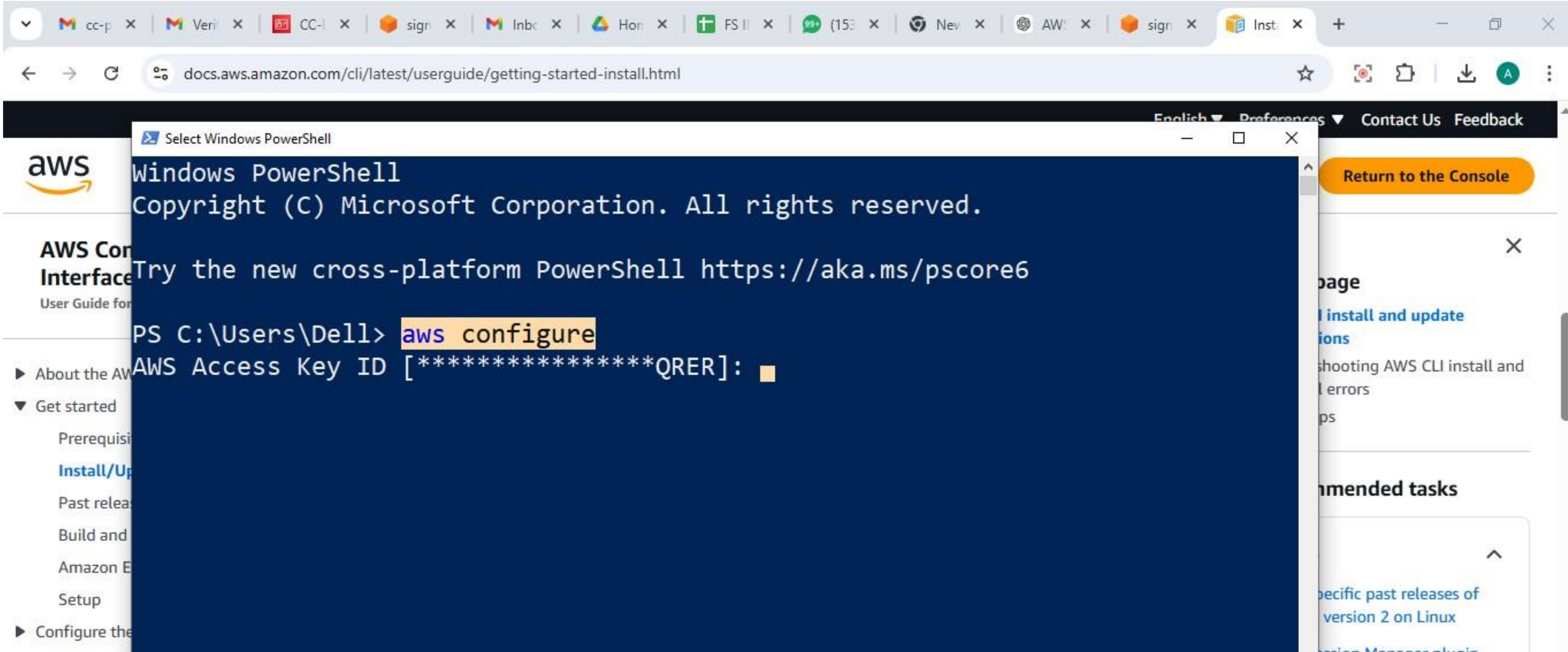
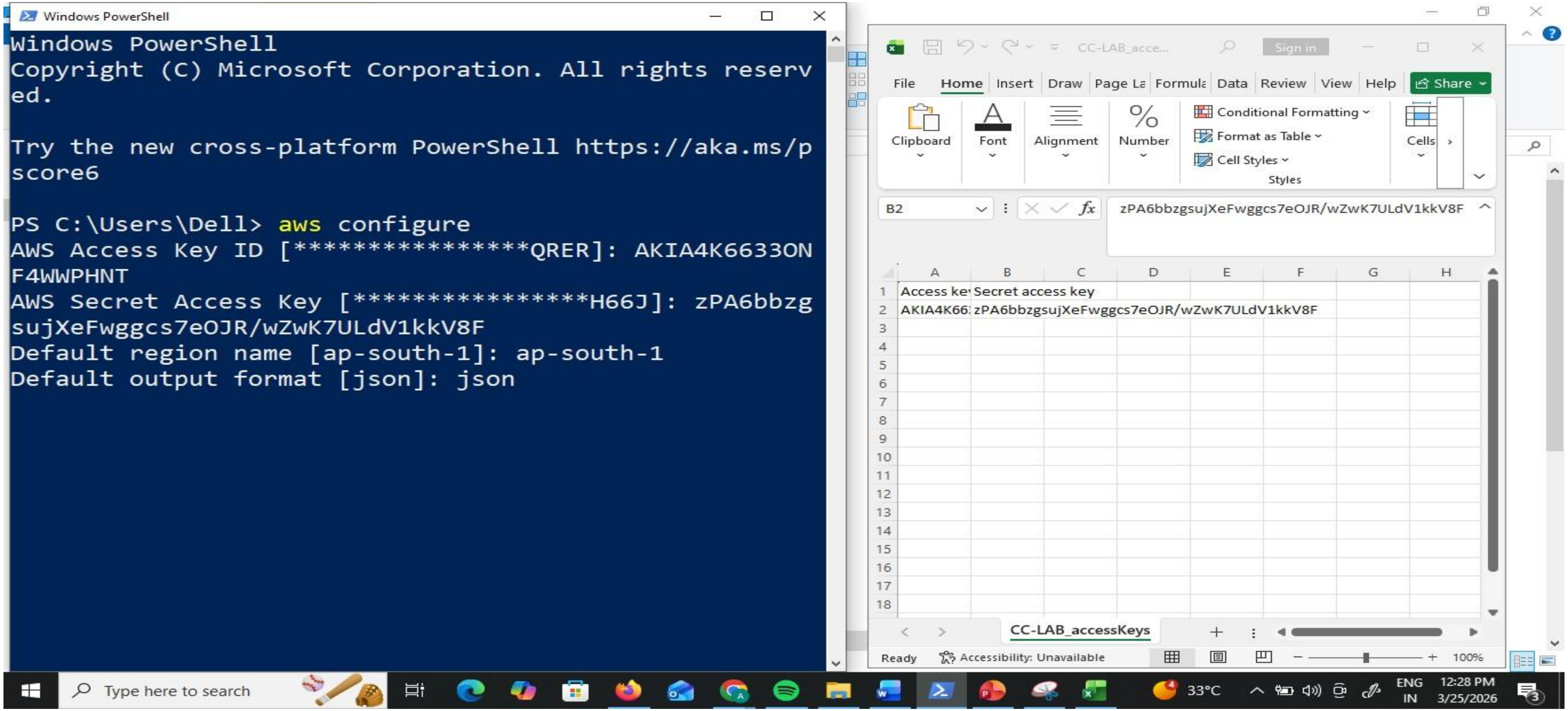
3/25/2026

Step 6: Open your Command Prompt (CMD) or Terminal.
Step 7: Type aws configure and press Enter.

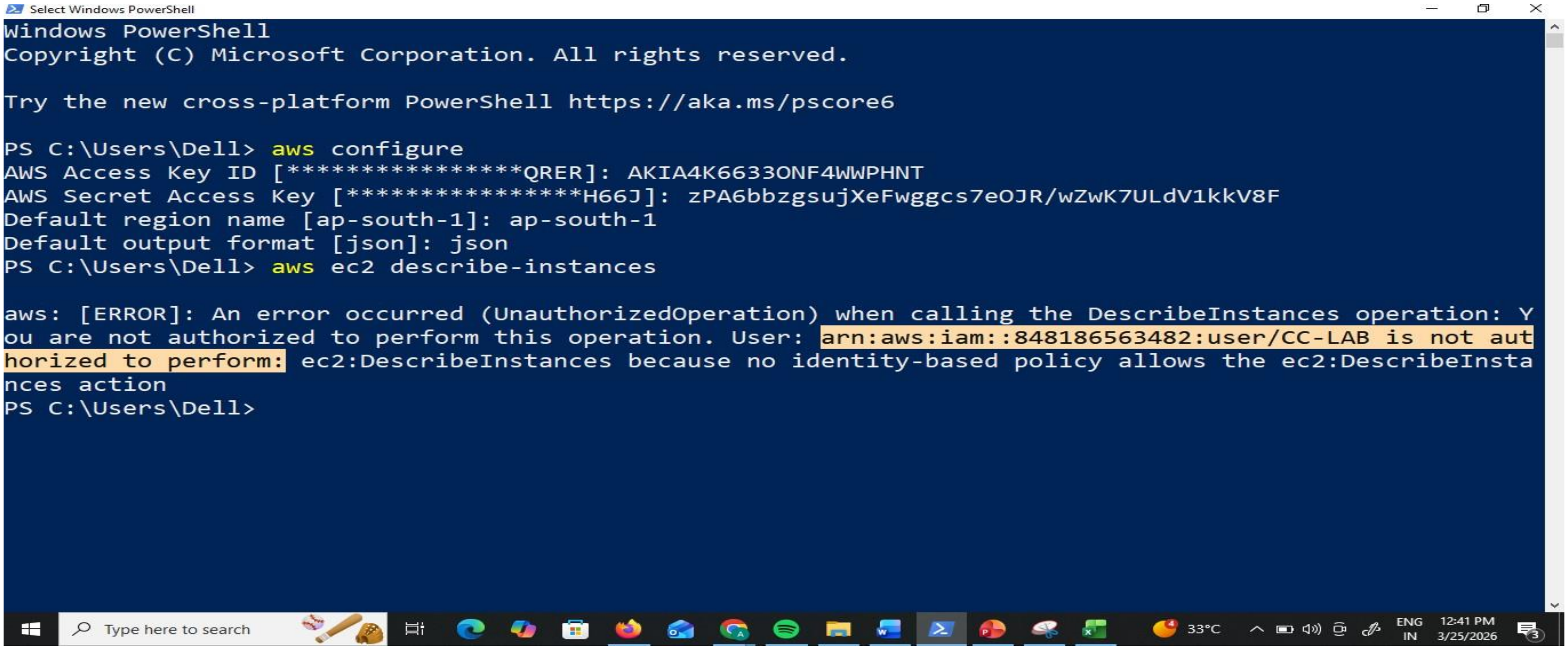


Step 8: Provide the following from your .csv:
Access Key ID
Secret Access Key
Default region: (e.g., ap-south-1)
Default output: json



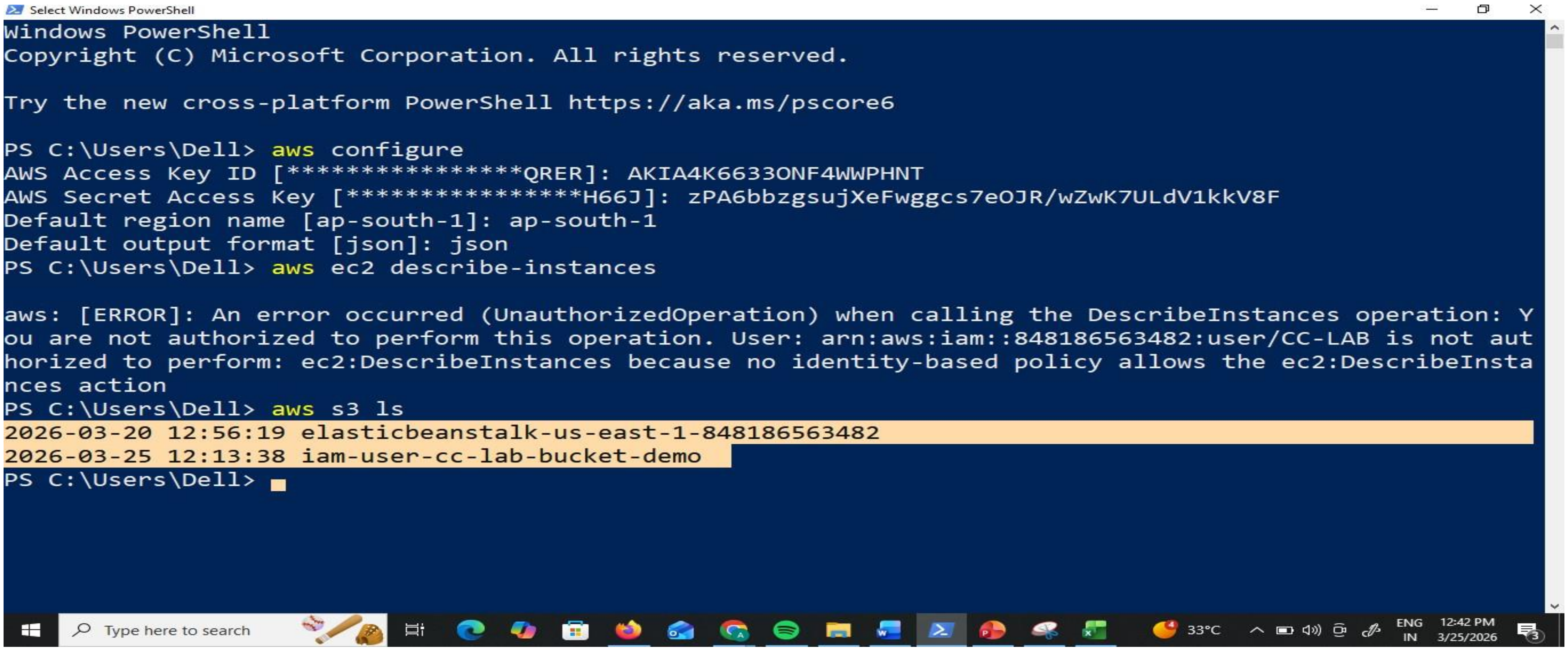
Run aws ec2 describe-nstances.
Result: You will get an (AccessDenied) error.

This confirms that the CLI is respecting the same IAM policy as the GUI.



3. Verification via Commands

Step 9: Run aws s3 ls. This should list your buckets



Step 10: Run aws s3 mb s3://your-unique-bucket-name. (Success)

Select Windows PowerShell

```
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Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Users\Dell> aws configure
AWS Access Key ID [****************QRER]: AKIA4K6633ONF4WWPHNT
AWS Secret Access Key [****************H66J]: 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
Default region name [ap-south-1]: ap-south-1
Default output format [json]: json
PS C:\Users\Dell> aws ec2 describe-instances

aws: [ERROR]: An error occurred (UnauthorizedOperation) when calling the DescribeInstances operation: Y
ou are not authorized to perform this operation. User: arn:aws:iam::848186563482:user/CC-LAB is not aut
horized to perform: ec2:DescribeInstances because no identity-based policy allows the ec2:DescribeInsta
nces action
PS C:\Users\Dell> aws s3 ls
2026-03-20 12:56:19 elasticbeanstalk-us-east-1-848186563482
2026-03-25 12:13:38 iam-user-cc-lab-bucket-demo
PS C:\Users\Dell> aws s3 mb iam-user-demo-cli

aws: [ERROR]: An error occurred (ParamValidation): <S3Uri>
Error: Invalid argument type
PS C:\Users\Dell> aws s3 mb s3://iam-user-demo-cli
make_bucket: iam-user-demo-cli
PS C:\Users\Dell>
```

Type here to search

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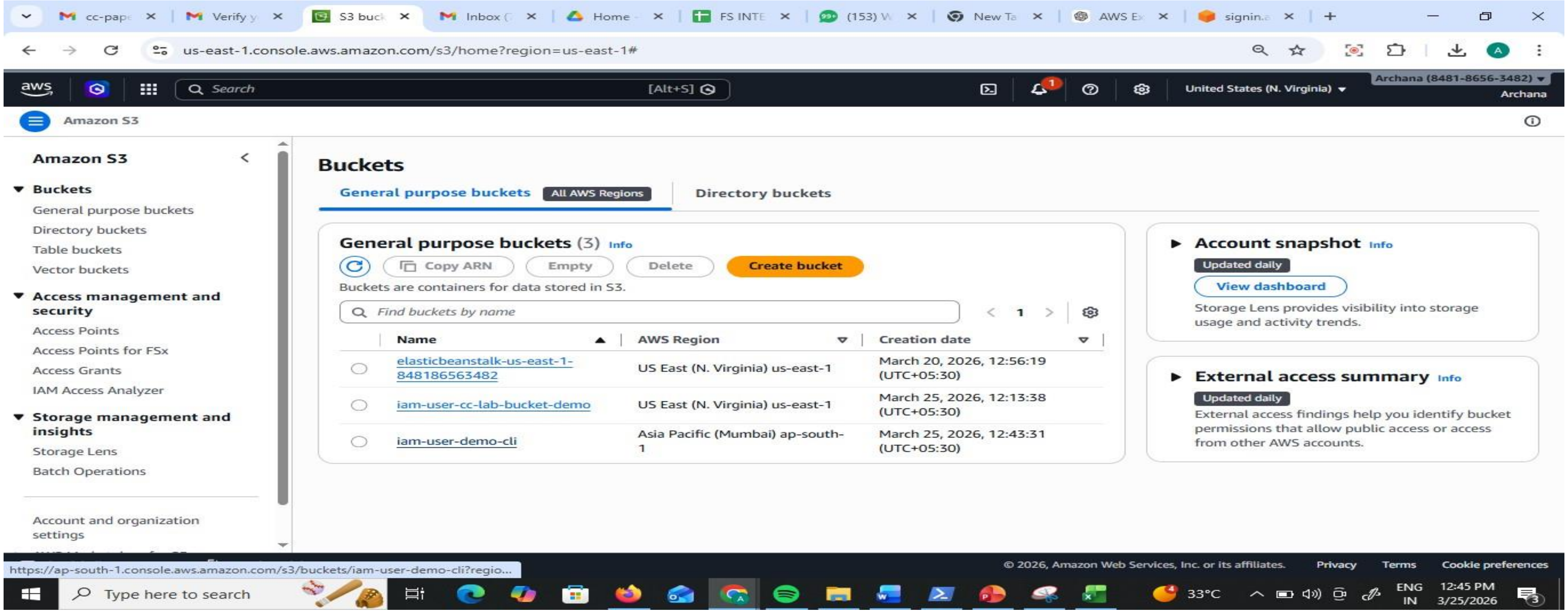
Check in console you see the bucket created from CLI



Scenario 1: Security Risk

Q: You logged into AWS using the root user for daily work. What's wrong here?

✓Scenario 2: Account Compromise

Q: Your AWS bill suddenly increases and you suspect root user compromise. What steps will you take?

✓Scenario 3: Root Credentials Lost

Q: What if you forgot root user password?

✓Scenario 4: Root Access Keys

Q: Why should you avoid creating access keys for root user?

✓Scenario 5: New Developer Joining

Q: A new developer joins your team. How do you give access?

✅Scenario 6: Least Privilege

Q: A user needs only S3 read access. What will you do?

✅Scenario 7: Temporary Access

Q: A contractor needs access for 2 weeks.

✅Scenario 8: Access Denied Error

Q: User gets “Access Denied” while accessing EC2. Why?

✅Scenario 9: Multiple Users Same Permissions

Q: 50 users need same access. What is best practice?

✅Scenario 10: Configure CLI

Q: You installed AWS CLI. How do you configure it?

✅Scenario 11: Create IAM User via CLI

Q: How to create IAM user using CLI?

✅Scenario 12: Attach Policy via CLI

Q: Give S3 read access to user using CLI

✅Scenario 13: Create Access Keys via CLI

Q: How to generate access keys for IAM user?

✅Scenario 14: List IAM Users

Q: How to check all users using CLI?

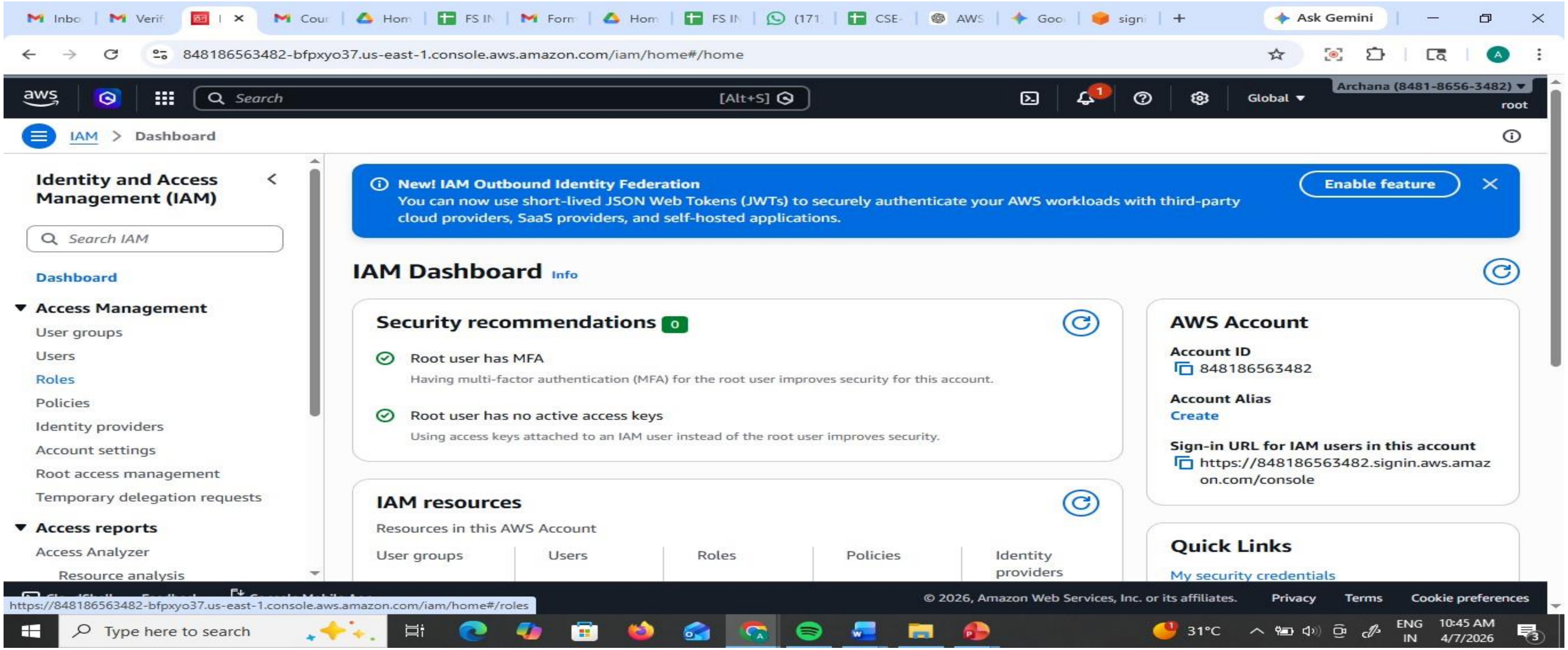
✅Scenario 15: Delete IAM User via CLI

Q: How to safely delete a user?

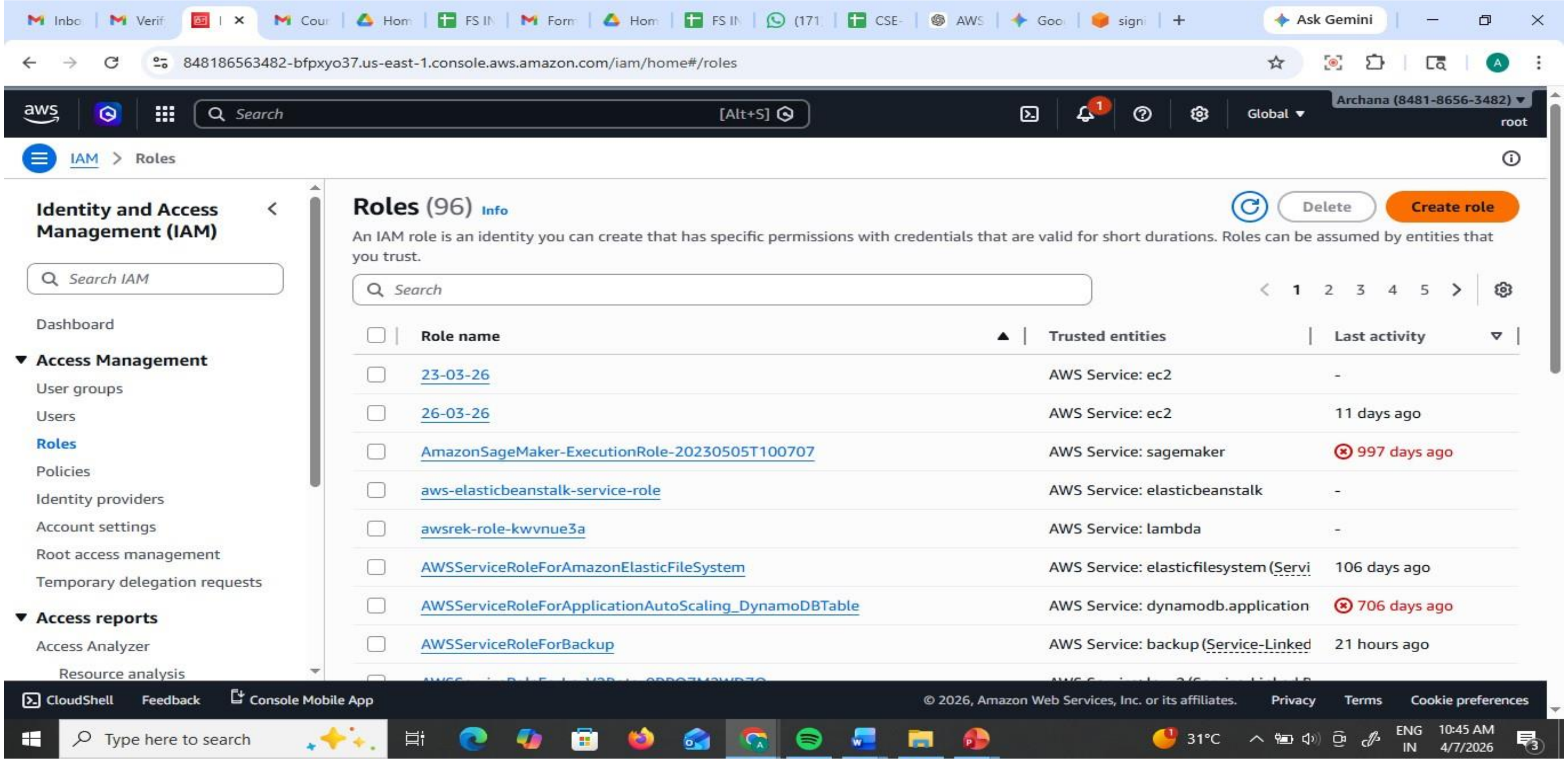
1Q: CLI command fails with AccessDenied. What to check?

Week-13 IAM Roles

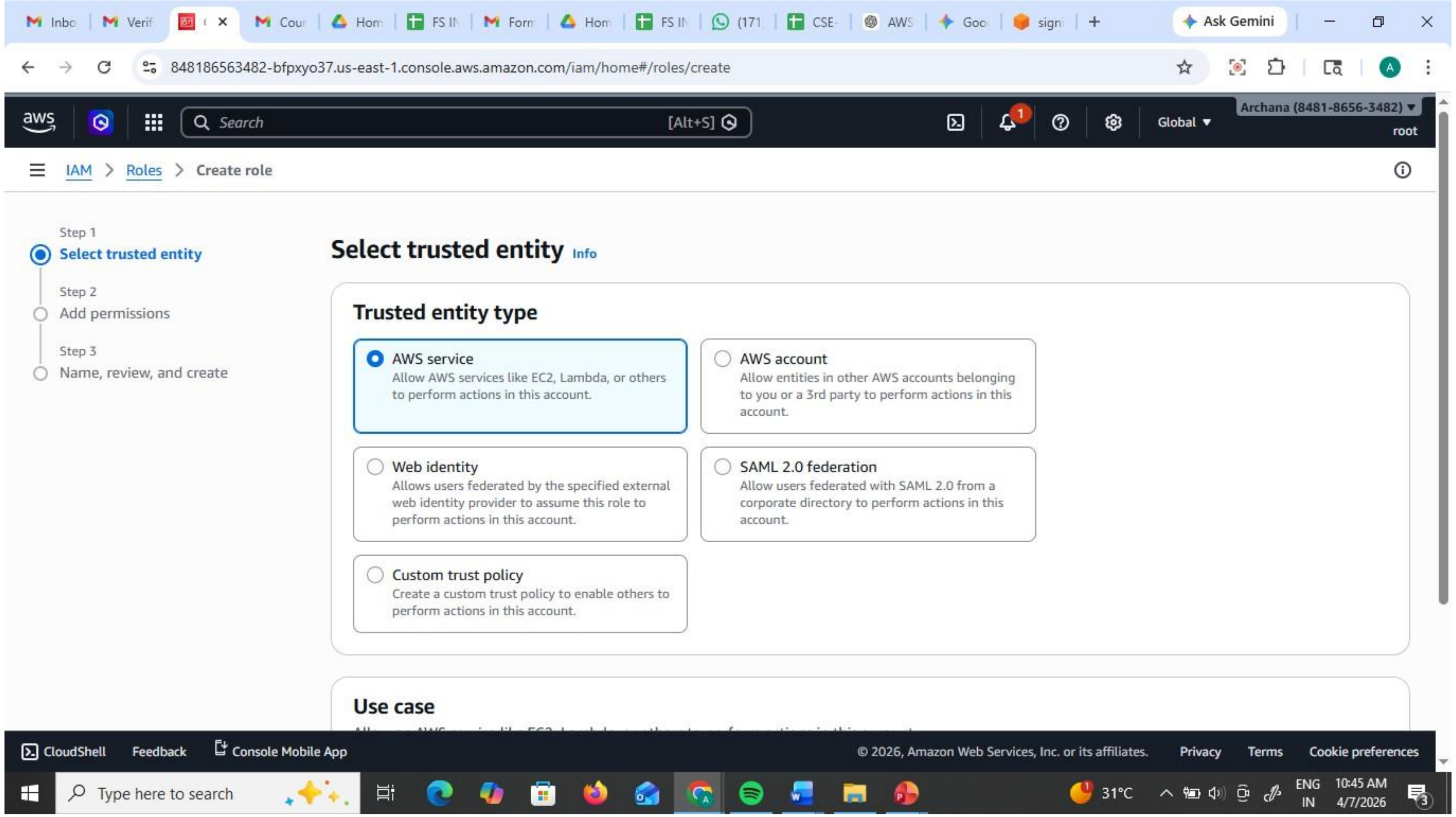
Procedure: Attaching a Role to an EC2 Instance



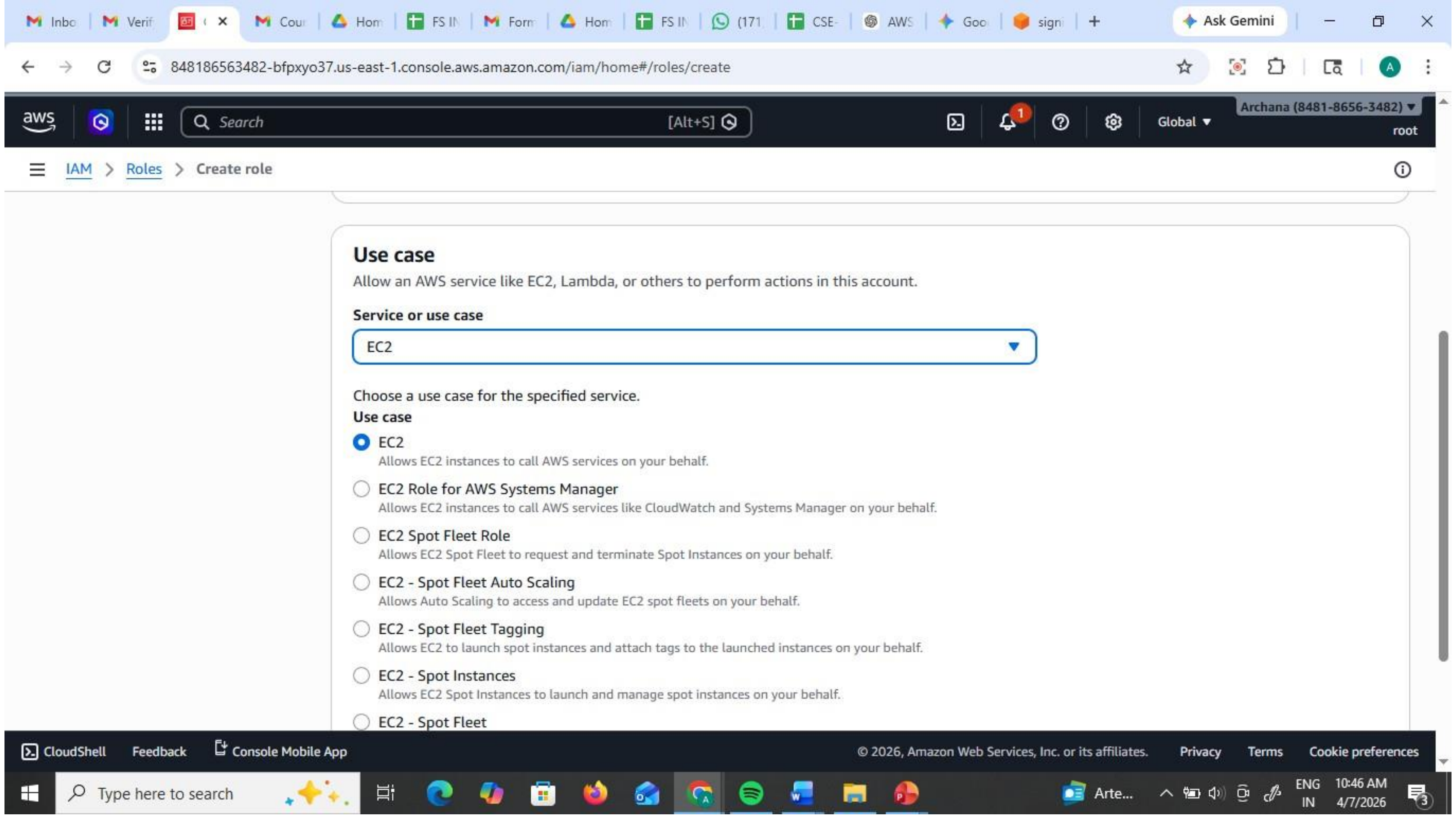
Create the Role: Go to IAM > Roles > Create role



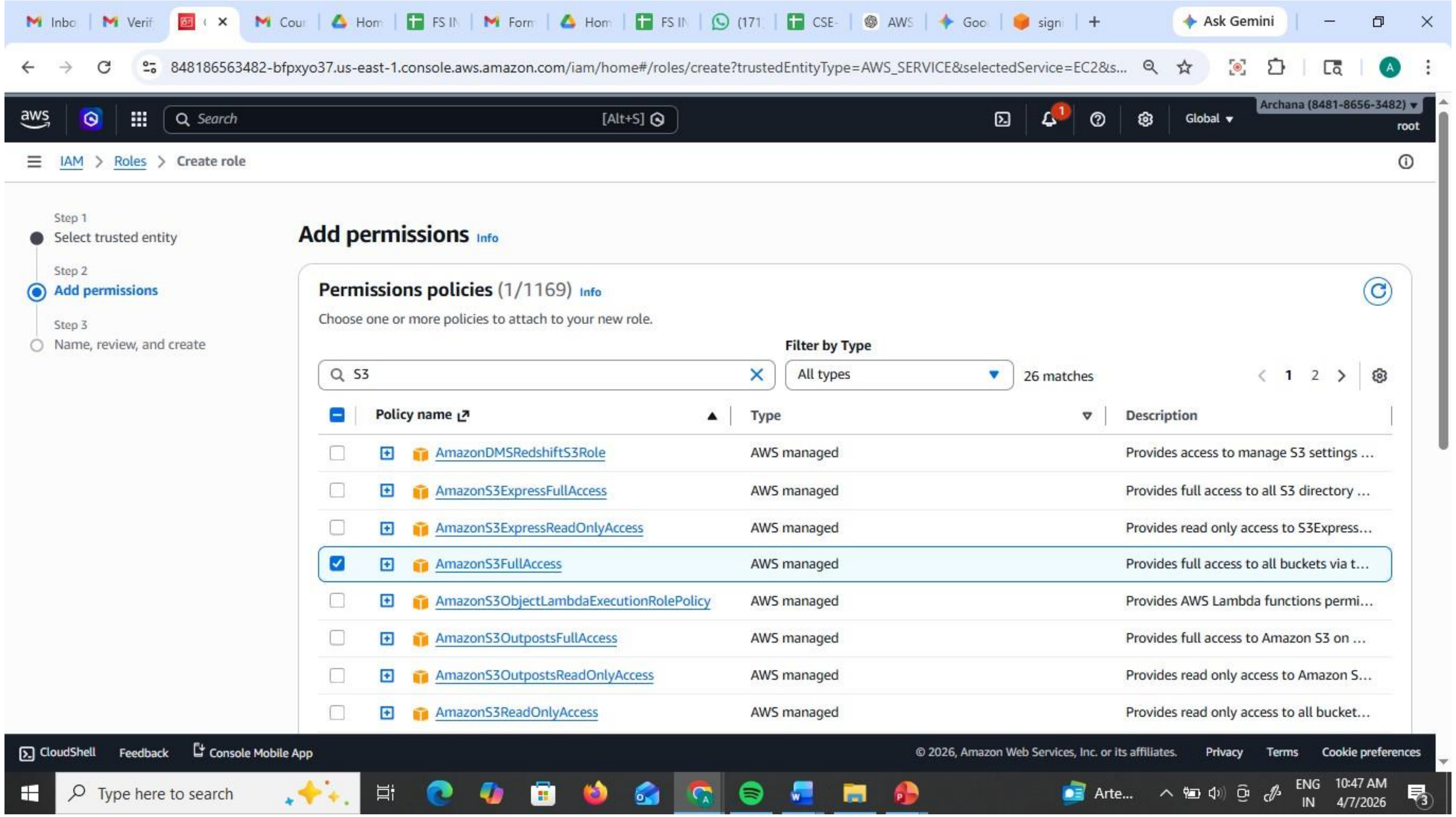
Select Trusted Entity: Choose AWS Service.



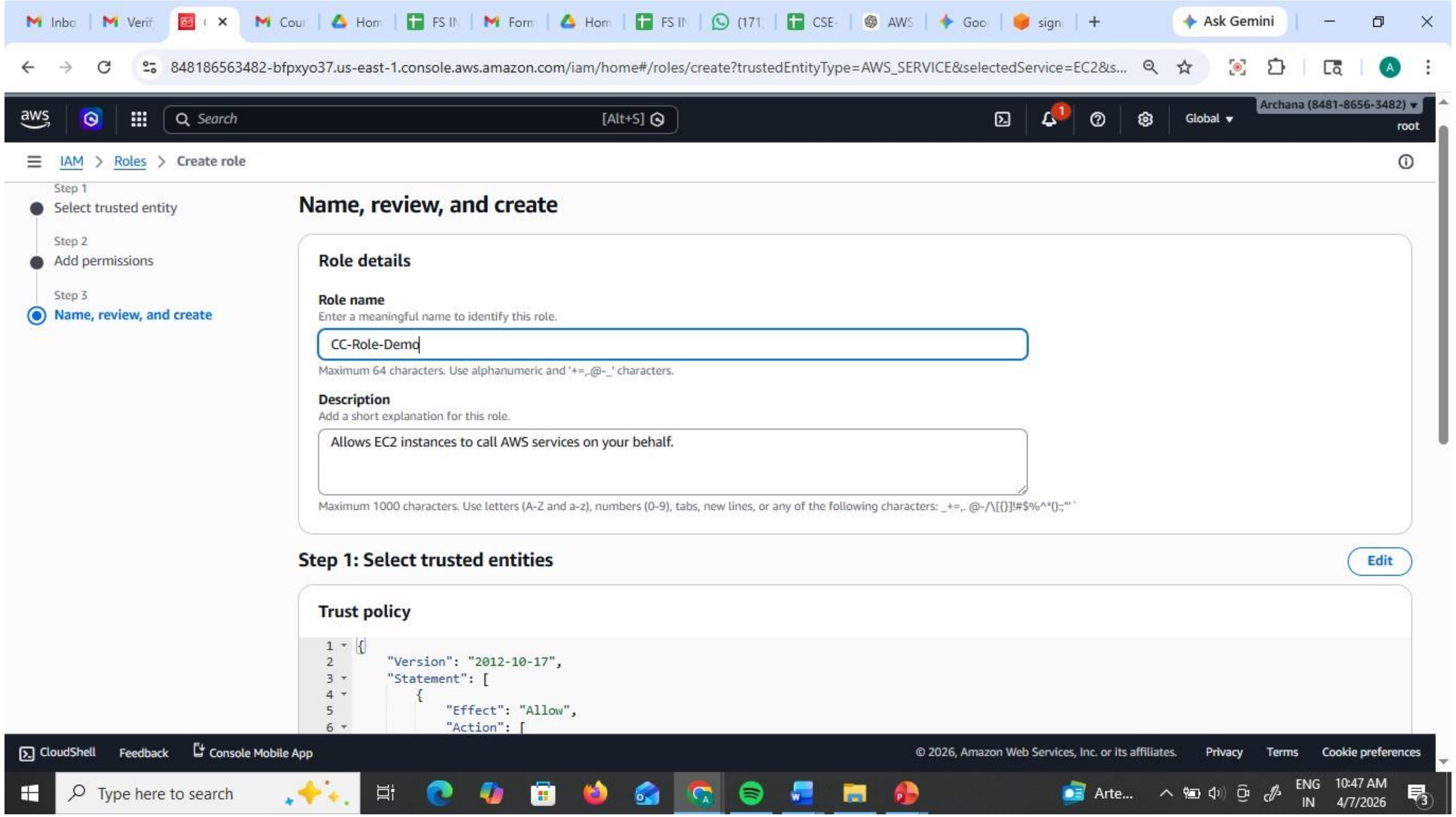
Under the "Service or use case" dropdown, select EC2. (This tells AWS: "I trust the EC2 service to use this role").



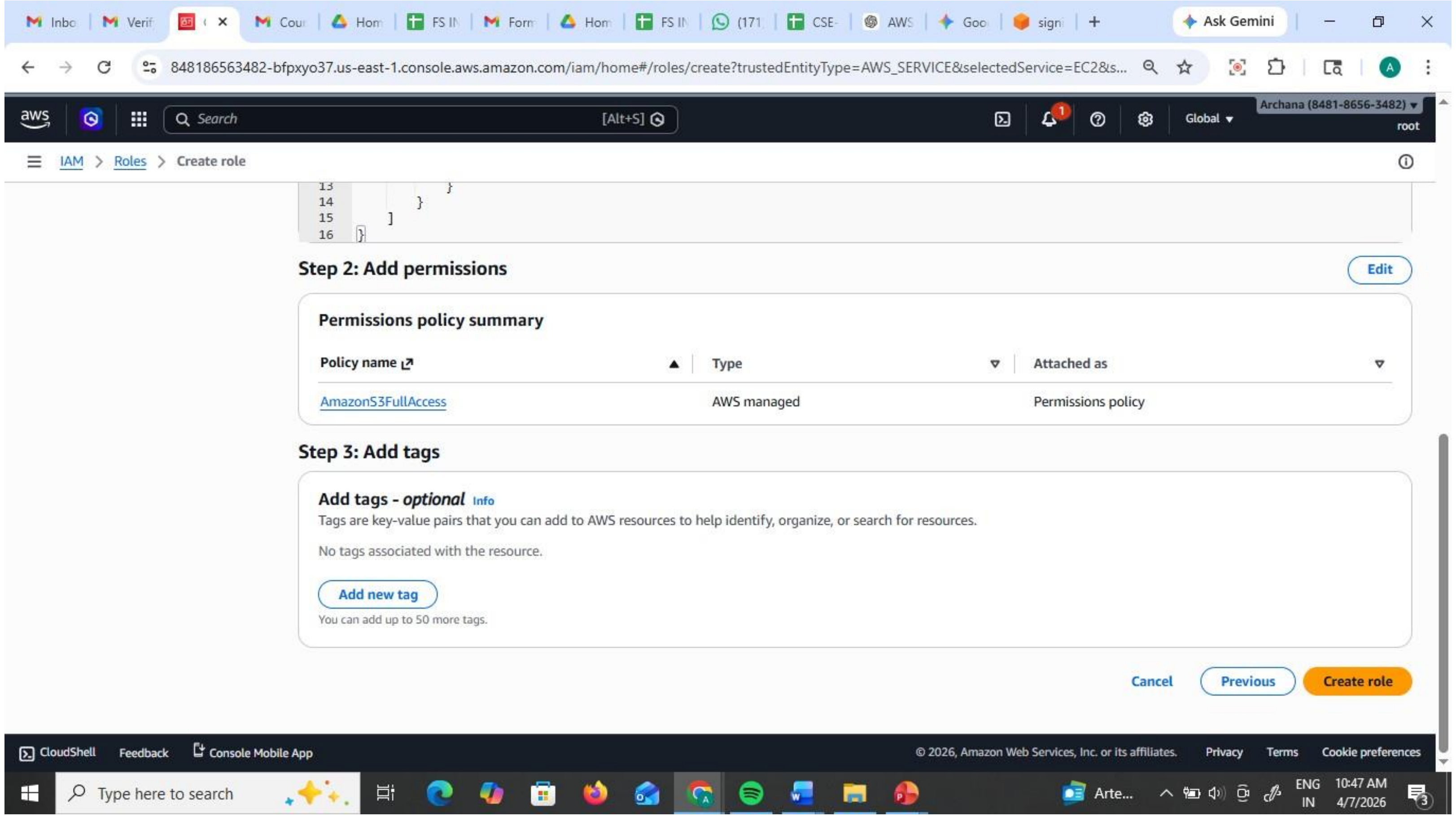
Add Permissions: Search for and check AmazonS3FullAccess.



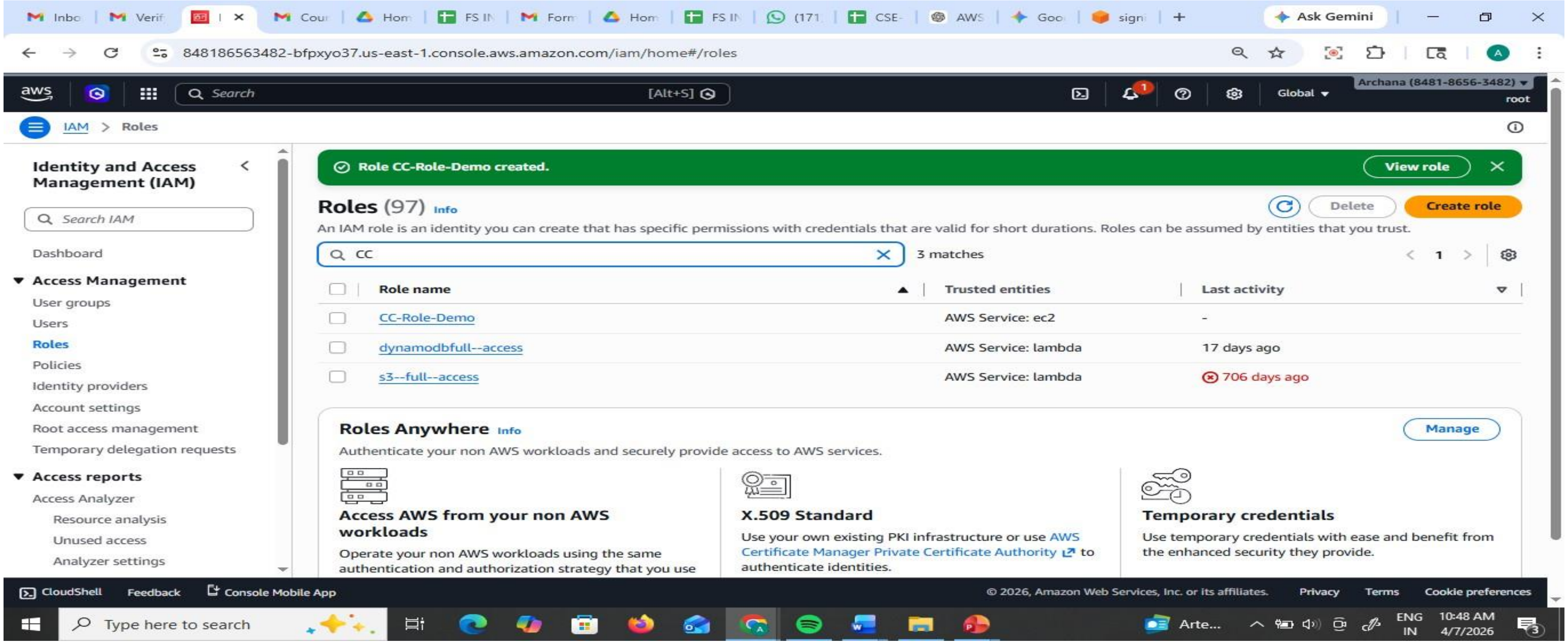
Name & Review: Name it (e.g., EC2-S3-ReadOnly-Role) and click Create role.



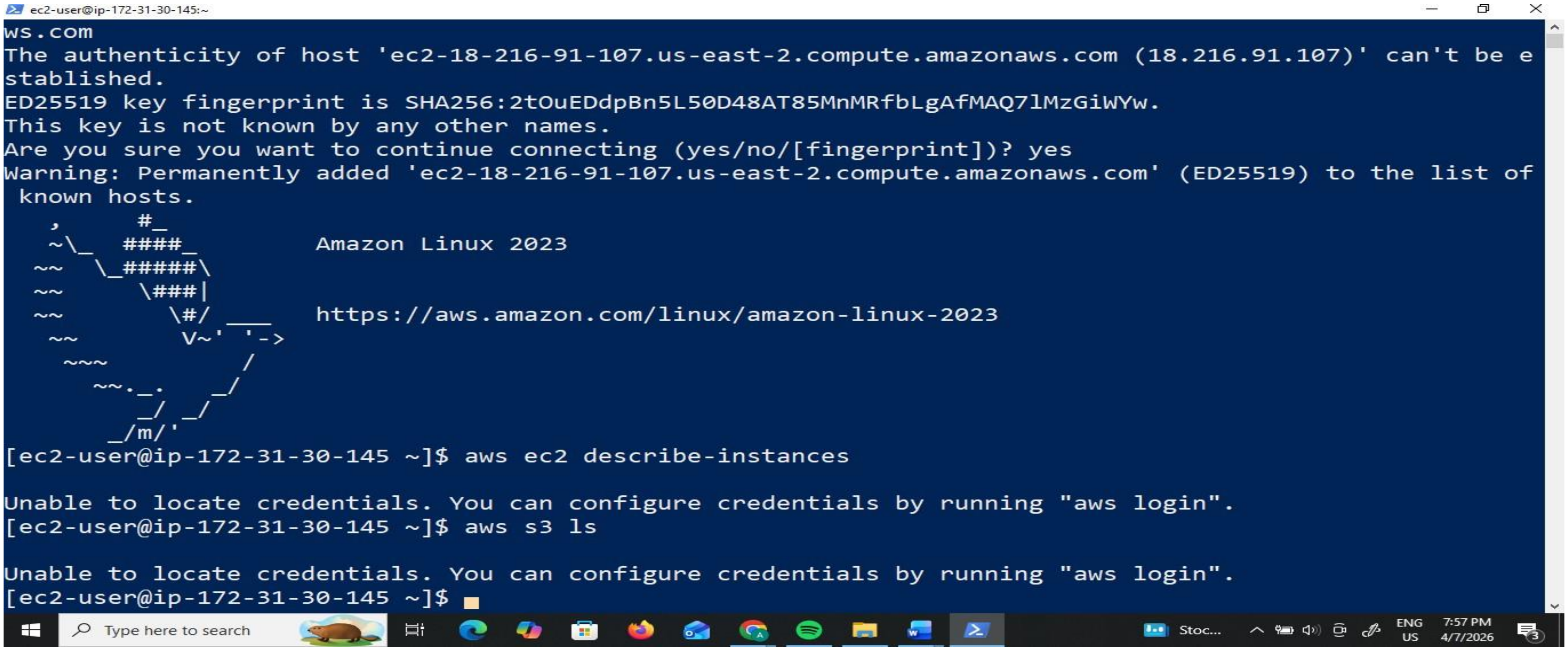
Name & Review: Name it (e.g., EC2-S3-ReadOnly-Role) and click Create role.



Role has been created



Create the ec2 instance and and connect to it and check for aws ec2 describe-instances



Attach to Instance:
Go to the EC2 Dashboard. Select your running instance.

Click Actions > Security > Modify IAM role.

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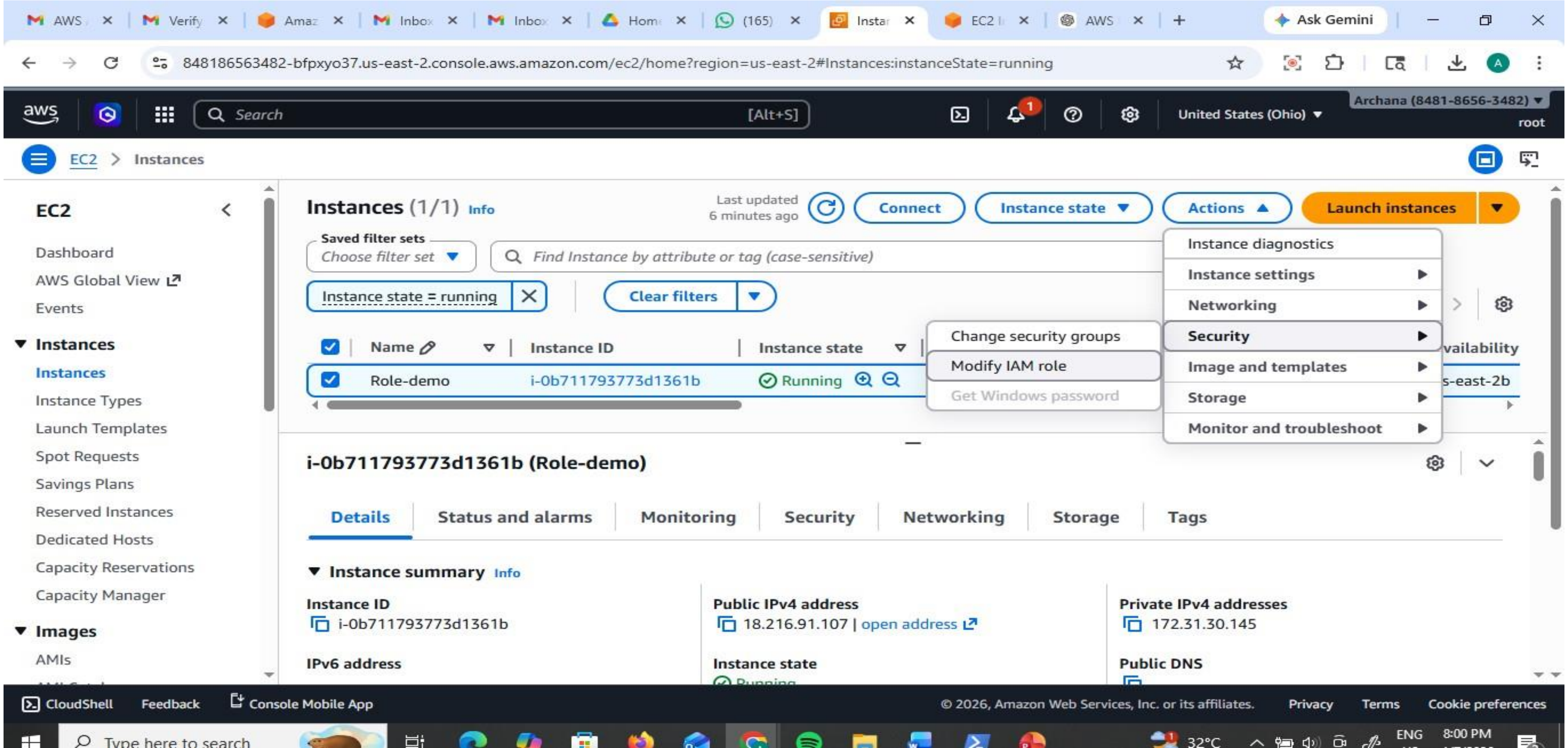
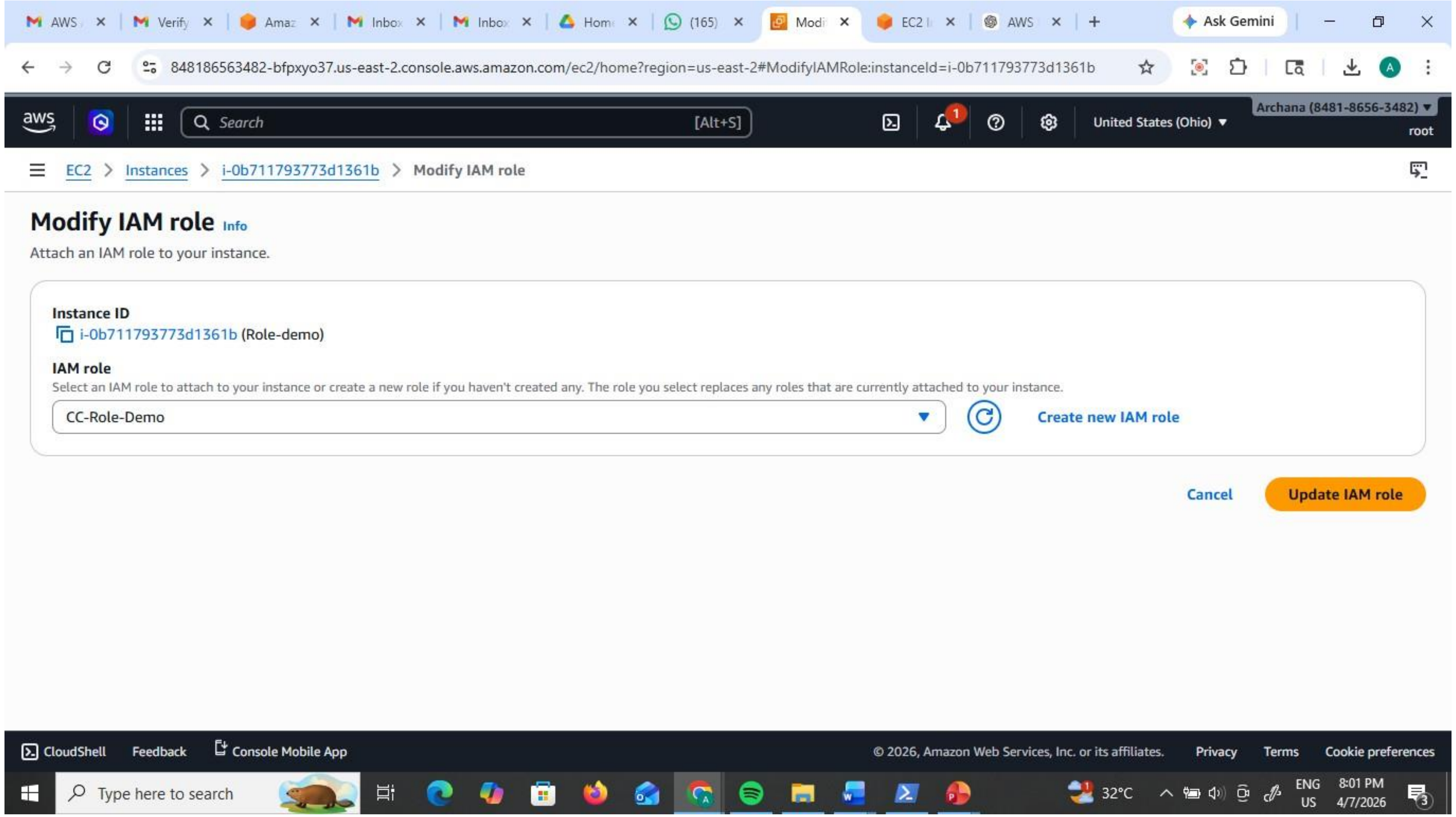
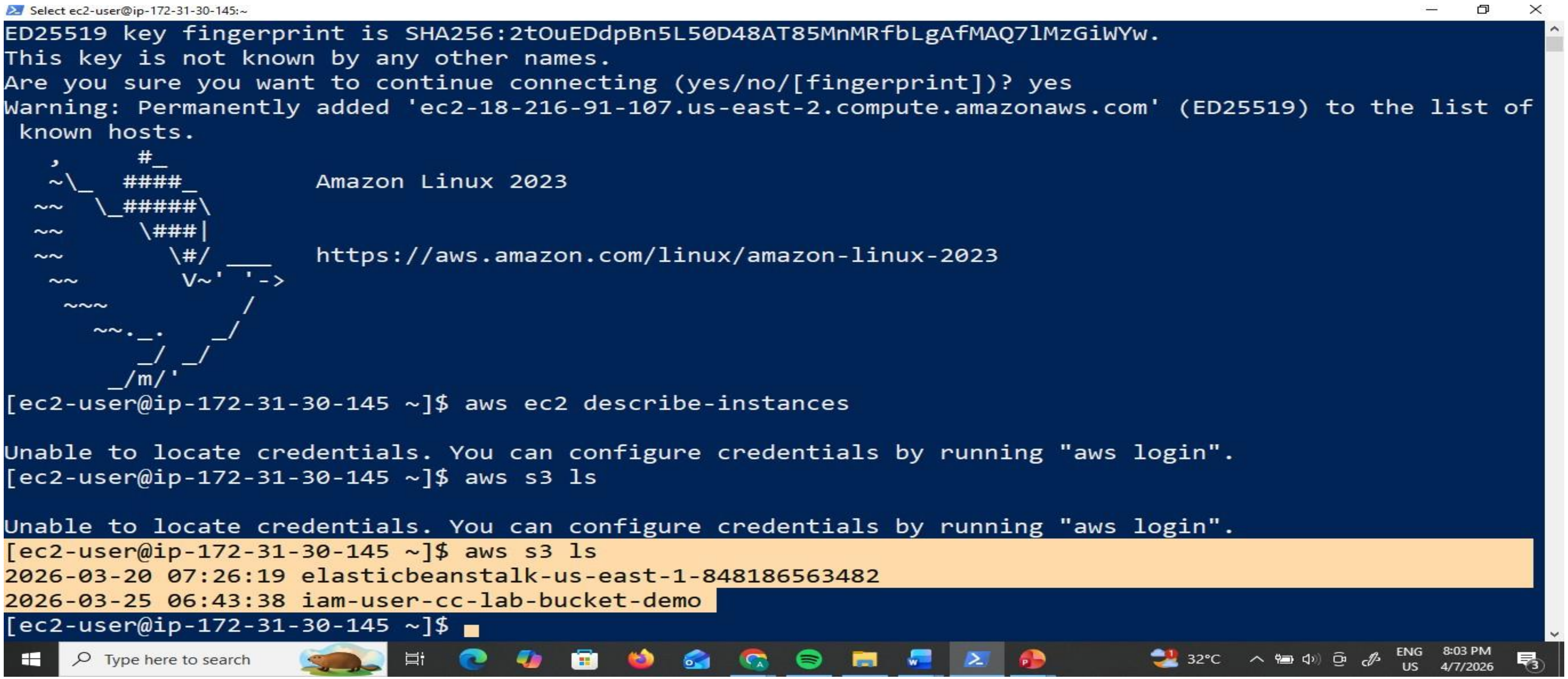


Select the role you just created and click Update IAM role



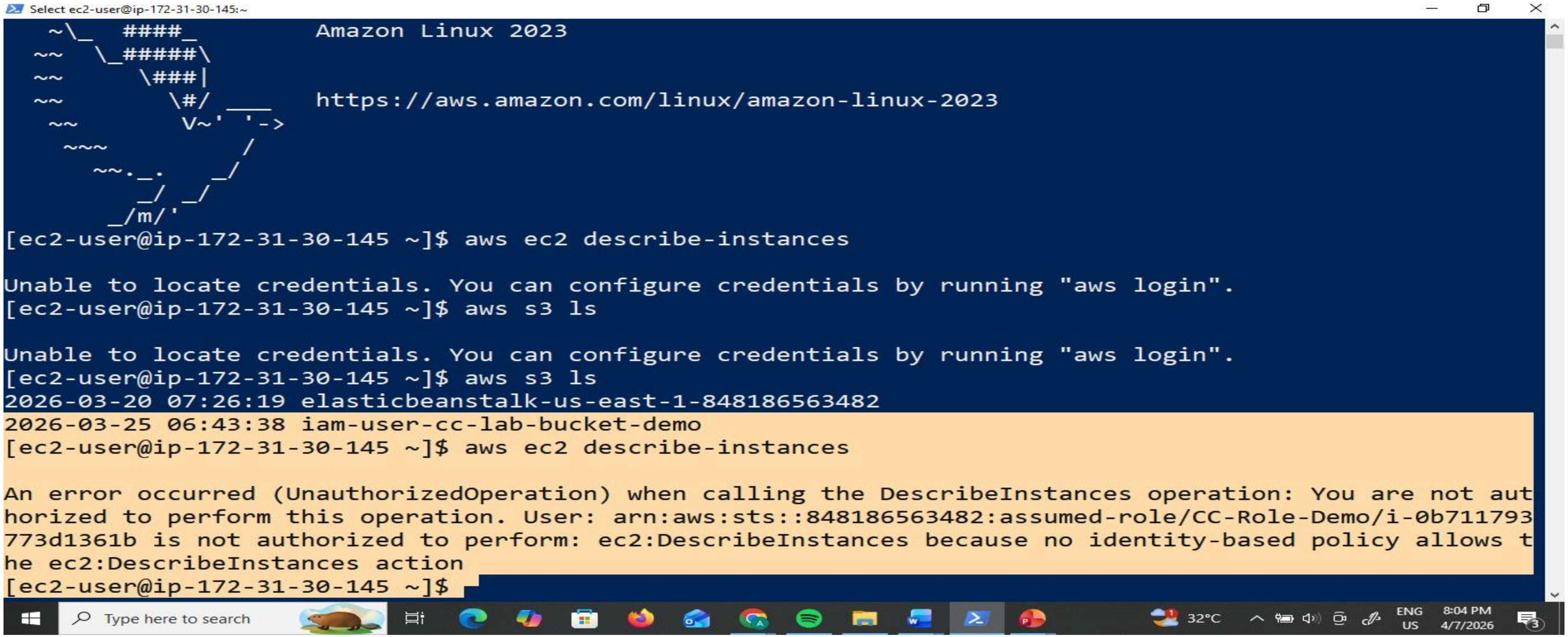
Verify via CLI:
SSH/Connect into your EC2 instance. Type aws s3 ls.

The Result: The command works immediately.



We can here role has only s3 full access only here-- aws ec2 describe-instances – unauthorized

You did not have to provide an Access Key or Secret Key. AWS automatically provided Temporary Security Credentials to the instance.



Scenario based questions:

1. Scenario: New Developer Access

You hired a new developer who needs access to EC2 and S3.

Question: How will you provide access?

2. Scenario: Temporary Access Needed

A user needs access only for 2 days. How to do this?

3. Scenario: Access Key Leak

A developer accidentally exposed access keys on GitHub.How to resolve this?

4. Scenario: CLI Access Required

User needs programmatic access. How to do this?

5. Scenario: 50 Employees Same Permissions

50 employees need same S3 access. How to do this?

6. Scenario: Change Permissions for Multiple Users

You need to revoke access from 20 users. How to do this?

7. Scenario: Different Departments

HR, Dev, Admin need different access. How to do this?

8. Scenario: EC2 Access to S3

An EC2 instance needs access to S3 without credentials.How to do this?

9. Scenario: Cross-Account Access

Account A wants to access resources in Account B. How to do this?

10. Scenario: Lambda Access to DynamoDB

Lambda needs to read/write DynamoDB. How to give this?

11. Scenario: Temporary Elevated Access

Admin access needed temporarily. How to do this?

12. Scenario: Restrict S3 Access to Specific Bucket

User should only access one bucket. How to do this?

13. Scenario: Deny Delete Actions

Users should not delete resources. Why this happens?

14. Scenario: Time-Based Access

User should access only during office hours. How to do this?

15. Scenario: IP-Based Access

Allow access only from office network. How to do?

16. Scenario: Secure Best Practice

How do you manage permissions efficiently?

17. Scenario: Root User Usage

When should root user be used?

18. Scenario: Multiple Services Access

User needs EC2 + S3 + DynamoDB. How to do this?

19. Scenario: Audit Permissions

How to track user activity?

20. Scenario: Over-Permission Issue

User has too much permission. How to solve this?